Equations and derivations for ElemCo.jl

Daniel Kats, Charlotte Rickert, Thomas Schraivogel,
Denis Usvyat, Fangcheng Wu, Kristoffer Simula

August 18, 2026

Contents

Chapter 1

Introduction

1.1 General

In this document we collect the equations and derivations for methods implemented in the ElemCo.jl package. The final goal is to have a document which can be used as a reference for the equations and derivations. The final equations should also be contained in the code as docstrings or copied to the corresponding Markdown files.

1.2 Notation

We use the following notation throughout the document.

The virtual orbitals are denoted by $a, b, c, \ldots$, the occupied orbitals by $i, j, k, \ldots$, the active (open-shell) orbitals by $t, u, v, \ldots$, and the general orbital indices are denoted by p, q, r, s. The Einstein summation convention is used for repeated indices (repeated lower and upper indices are summed over). The α and β spin orbitals are denoted by p and $\bar{p}$.

The integrals are **not antisymmetrized** and denoted by v_{pq}^{rs}, where p, q, r, s are indices of orbitals, and the lower indices correspond to the creation and the upper indices to the annihilation operators in the Hamiltonian,

$$\hat{H} = E_0 + h_p^q \hat{a}_p^\dagger \hat{a}_q + \frac{1}{2} v_{pq}^{rs} \hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_s \hat{a}_r, \tag{1.1}$$

or for the normal-ordered Hamiltonian,

$$\hat{H}_N = f_p^q \left\{ \hat{a}_p^\dagger \hat{a}_q \right\}_N + \frac{1}{2} v_{pq}^{rs} \left\{ \hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_s \hat{a}_r \right\}_N, \tag{1.2}$$

i.e., $h_p^q = \langle p|\hat{h}|q\rangle$, $f_p^q = \langle p|\hat{f}|q\rangle$ and $v_{pq}^{rs} = \langle pq|rs\rangle$.

Permutation operators:

$$\begin{aligned} \mathcal{P}\left(ab\right) X_{ab}^{ij} &= X_{ba}^{ij} \\ \mathcal{P}\left(ab \rightarrow ba\right) X_{ab}^{ij} &= X_{ba}^{ij} \end{aligned} \tag{1.3}$$

Symmetrization operators:

$$\begin{aligned} \mathcal{S}\left(ab\right) X_{ab}^{ij} &= X_{ab}^{ij} + X_{ba}^{ij} \\ \mathcal{S}\left(ab, ij\right) X_{ab}^{ij} &= X_{ab}^{ij} + X_{ba}^{ji} \end{aligned} \tag{1.4}$$

Antisymmetrization operators:

$$\begin{aligned} \mathcal{A}\left(ab\right) X_{ab}^{ij} &= X_{ab}^{ij} - X_{ba}^{ij} \\ \mathcal{A}\left(ab; ij\right) X_{ab}^{ij} &= X_{ab}^{ij} - X_{ab}^{ji} - X_{ba}^{ij} + X_{ba}^{ji} \end{aligned} \tag{1.5}$$

Chapter 2

Integrals

2.1 Density fitting and Cholesky decomposition

The electron-repulsion integrals in ElemCo.jl are obtained either from an external program through an FCIDUMP [1] interface, or are calculated using the density-fitting approximation using an interface to the `libcint`[2] library.

In the density-fitting approximation, the electron-repulsion integrals are approximated by

$$v_{pq}^{rs} \approx v_p^{rP} \left[v^{-1}\right]_{PQ} v_q^{sQ}, \tag{2.1}$$

where v_p^{rP} and v_q^{sQ} are density-fitted 3-index integrals with auxiliary basis functions P, Q,

$$v_p^{rP} = \int d\mathbf{r}_1 d\mathbf{r}_2 \frac{\phi_p^*(\mathbf{r}_1)\phi^r(\mathbf{r}_1)\phi^P(\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|}, \tag{2.2}$$

and v^{PQ} is the Coulomb metric matrix,

$$v^{PQ} = \int d\mathbf{r}_1 d\mathbf{r}_2 \frac{\phi^P(\mathbf{r}_1)\phi^Q(\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|}. \tag{2.3}$$

The Coulomb metric matrix is decomposed using the Cholesky factorization,

$$v^{PQ} = \sum_L L_L^P L_L^Q, \tag{2.4}$$

where L_L^P is a lower triangular matrix. Thereafter, a non-symmetric square root of the inverse, C_P^L, is calculated by solving the equations

$$L_{L'}^P C_P^L = \delta_{L'}^L, \tag{2.5}$$

with δ_L^K being the Kronecker delta. If L_L^P is low-rank, the equation is solved using the QR decomposition, otherwise it can be solved by simple back-substitution.

The transformed density-fitted integrals which are used throughout ElemCo.jl are then calculated by multiplying the density-fitted 3-index integrals with the non-symmetric square root of the inverse,

$$v_p^{rL} = v_p^{rP} C_P^L, \tag{2.6}$$

and the density-fitted 4-index integrals can be calculated by

$$v_{pq}^{rs} \approx \sum_L v_p^{rL} v_q^{sL}. \tag{2.7}$$

2.2 Frozen-core approximation

The frozen-core approximation is used to reduce the number of orbitals in the correlated calculation. The frozen-core approximation is implemented in ElemCo.jl by setting the corresponding integrals to zero and adding their contribution to the one-particle and zero-particle part of the Hamiltonian,

$$\begin{aligned} \tilde{h}_p^q &= h_p^q + 2v_{pI}^{qI} - v_{pI}^{Iq} \\ \tilde{E}_0 &= E_0 + 2h_I^I + 2v_{IJ}^{IJ} - v_{IJ}^{JI}, \end{aligned} \tag{2.8}$$

where $\tilde{h}$ and $\tilde{E}_0$ denote the corrected one-body and zero-body terms, I, J denote the core orbitals, and other indices do not include the core orbitals. For the UHF Hamiltonian, the frozen-core approximation is implemented as

$$\begin{aligned} \tilde{h}_p^q &= h_p^q + v_{pI}^{qI} + v_{p\bar{I}}^{q\bar{I}} - v_{pI}^{Iq} \\ \tilde{h}_{\bar{p}}^{\bar{q}} &= h_{\bar{p}}^{\bar{q}} + v_{\bar{p}\bar{I}}^{\bar{q}\bar{I}} + v_{\bar{p}I}^{\bar{q}I} - v_{\bar{p}\bar{I}}^{\bar{I}\bar{q}} \\ \tilde{E}_0 &= E_0 + h_I^I + h_{\bar{I}}^{\bar{I}} + \frac{1}{2}\left(v_{IJ}^{IJ} + v_{\bar{I}\bar{J}}^{\bar{I}\bar{J}} + 2v_{I\bar{J}}^{I\bar{J}} - v_{IJ}^{JI} - v_{\bar{I}\bar{J}}^{\bar{J}\bar{I}}\right). \end{aligned} \tag{2.9}$$

If the frozen-core approximation is used in combination with the density-fitting approximation, *JKFIT* correction terms are added to the one-body and zero-body terms of the Hamiltonian in order to ensure that the reference energy and the Fock matrix from DF-HF is not changed by the frozen-core approximation, cf. Sec. 2.3.

2.3 JKFIT/MPFIT correction

If the integrals are calculated using the density-fitting approximation using *MPFIT* fitting basis in ElemCo.jl, the correction terms are added using the *JKFIT* fitting basis to the one-body and zero-body terms of the Hamiltonian in order to ensure that the reference energy and the Fock matrix calculated using *MPFIT*-fitted integrals coincide with those from DF-HF (which employs the *JKFIT* fitting basis), i.e.,

$$\begin{aligned} \tilde{h}_p^q &= \tilde{f}_p^q - \sum_L \left(2v_p^{qL} v_i^{iL} - v_p^{iL} v_i^{qL}\right) \\ \tilde{E}_0 &= E_0 + h_i^i - \tilde{h}_i^i + \tilde{f}_I^I, \end{aligned} \tag{2.10}$$

where $\tilde{h}$ and $\tilde{E}_0$ denote the corrected one-body and zero-body terms,

$$\tilde{f}_p^q = h_p^q + \sum_{\tilde{L}} \left(2v_p^{q\tilde{L}} v_{\hat{i}}^{\hat{i}\tilde{L}} - v_p^{\hat{i}\tilde{L}} v_{\hat{i}}^{q\tilde{L}}\right), \tag{2.11}$$

and I denotes the core orbitals (cf. Sec. 2.2), $\hat{i}$ denote all occupied orbitals (including core) and other indices do not include the core orbitals. $\tilde{L}$ corresponds to the *JKFIT* density-fitting basis functions, and L corresponds to the *MPFIT* density-fitting basis functions.

2.4 Exact AO integrals: the ± supermatrix store

The exact (non-density-fitted) four-index AO integrals are stored as the ± supermatrices, the only representation from which both Hartree–Fock (Sec. 3.3) and the AO-driven coupled-cluster methods (Sec. 7.7) contract directly.

Real AO integrals possess the full eight-fold permutational symmetry, which would in principle permit an $n^4/8$ packing. The store, however, is built on the smaller symmetry group that survives for *complex* and for similarity-transformed integrals: the *joint* exchange symmetry

$$v_{\mu\nu}^{\rho\sigma} = v_{\nu\mu}^{\sigma\rho} \tag{2.12}$$

together with hermiticity, $v_{\mu\nu}^{\rho\sigma} = \left(v_{\rho\sigma}^{\mu\nu}\right)^*$. In this group neither the bra pair nor the ket pair may be packed triangularly on its own. Splitting the integrals into the parts that are symmetric and antisymmetric with respect to the exchange of the bra pair,

$$V^{\pm}_{(\mu\nu)(\rho\sigma)} = v_{\mu\nu}^{\rho\sigma} \pm v_{\nu\mu}^{\rho\sigma}, \qquad \mu \le \nu, \quad \rho \le \sigma, \tag{2.13}$$

converts the single joint symmetry Eq. (2.12) into two independent ones: together with the hermiticity of the integrals, both V^+ and V^- are symmetric (hermitian for complex integrals) matrices in the packed pair space,

$$V^{\pm}_{(\rho\sigma)(\mu\nu)} = \left(V^{\pm}_{(\mu\nu)(\rho\sigma)}\right)^*, \tag{2.14}$$

so that *both* index pairs are triangular. Only the lower block triangle has to be stored, and the store occupies $\approx n^4/4$ instead of the $n^4/2$ of the joint triangular packing.

2.4.1 Storage layout and generation

Both V^+ and V^- are kept as their lower *block* triangles over the packed pair space, written as σ-aligned column panels; the upper part is the (conjugate-)hermitian mirror, reconstructed at read time. The ± fold is fused into the integral generation: each shell-aligned ket-column block is assembled in a bounded slab of memory, folded according to Eq. (2.13) and written directly as panels, so no jointly packed $n^4/2$ intermediate exists at any point, on disk or in memory. Since the stored rows of a panel are bounded below by the panel's own block, the ERI kernel skips every bra shell pair lying entirely below that row floor (the kept-pair predicate $\max(\mu,\nu) \ge \sigma_{\text{block start}}$ is exact for shell-aligned blocks), which removes roughly half of the integral evaluations without any screening approximation.

The half-transformed store of Eq. (7.58) is kept pair-packed as well, $[(i\nu), o, (\rho\sigma)]$ with $\rho \le \sigma$ and both ket orders stored ($o = 1$: (ρ,σ); $o = 2$: (σ,ρ)) – the layout in which each output pair block owns its column range outright, so the out-of-core gather writes every block exactly once. By the exchange symmetry Eq. (2.12) a single file serves both half-transform roles of Eq. (7.60). The gather reads each ± element approximately twice (the native panel sequentially through the memory map, the mirror rows of earlier panels by positional reads with read-ahead advice), and the store is subsequently streamed by whole σ-columns for the integral blocks of the correlated methods.

Chapter 3

Hartree-Fock

3.1 Density-fitted Hartree-Fock

The density-fitted Hartree-Fock equations are given by

$$\begin{aligned} f^{\nu}_{\mu} C^{p}_{\nu} &= S^{\nu}_{\mu} C^{p}_{\nu} \epsilon_p \\ f^{\nu}_{\mu} &= h^{\nu}_{\mu} + 2\sum_L \left(v^{iL}_{\mu'} C^{\dagger\mu'}_{i} \right) C^{L}_{P} v^{\nu P}_{\mu} - v^{iL}_{\mu} v^{\dagger\nu}_{iL} \\ v^{iL}_{\mu} &= \left(v^{\nu P}_{\mu} C^{i}_{\nu} \right) C^{L}_{P}. \end{aligned} \tag{3.1}$$

Alternatively, the $v^{\nu L}_{\mu}$ integrals can be precomputed and the Fock matrix can be calculated as

$$\begin{aligned} f^{\nu}_{\mu} &= h^{\nu}_{\mu} + 2\sum_L \left(v^{iL}_{\mu'} C^{\dagger\mu'}_{i} \right) v^{\nu L}_{\mu} - v^{iL}_{\mu} v^{\dagger\nu}_{iL} \\ v^{iL}_{\mu} &= v^{\nu L}_{\mu} C^{i}_{\nu}. \end{aligned} \tag{3.2}$$

Note that if the orbitals are real, $v^{\dagger\mu}_{iL} = v^{iL}_{\mu}$, and $C^{\dagger\mu}_{i} = C^{i}_{\mu}$.

The unrestricted Hartree-Fock equations are given (for α spin) by

$$\begin{aligned} {}^{\alpha}f^{\nu}_{\mu} C^{p}_{\nu} &= S^{\nu}_{\mu} C^{p}_{\nu} \epsilon_p \\ {}^{\alpha}f^{\nu}_{\mu} &= h^{\nu}_{\mu} + \sum_L \left(v^{iL}_{\mu'} C^{\dagger\mu'}_{i} + v^{\bar{i}L}_{\mu'} C^{\dagger\mu'}_{\bar{i}} \right) C^{L}_{P} v^{\nu P}_{\mu} - v^{iL}_{\mu} v^{\dagger\nu}_{iL} \\ v^{iL}_{\mu} &= \left(v^{\nu P}_{\mu} C^{i}_{\nu} \right) C^{L}_{P}, \end{aligned} \tag{3.3}$$

and equations for β spin can be obtained by swapping the spins.

The residual of the Hartree-Fock equations, which can be used in DIIS, is given by

$$\Delta f^{\nu}_{\mu} = S^{\nu'}_{\mu} D^{\rho}_{\nu'} f^{\nu}_{\rho} - f^{\nu'}_{\mu} D^{\rho}_{\nu'} S^{\nu}_{\rho}, \quad \text{with} \quad D^{\nu}_{\mu} = C^{i}_{\mu} C^{\dagger\nu}_{i}. \tag{3.4}$$

3.2 (Bi-orthogonal) Hartree-Fock

The closed-shell Hartree-Fock on top of the FCIDUMP integrals (including the case of similarity-transformed Hamiltonians) is given by

$$\begin{aligned} f^{\tilde{q}}_{\tilde{p}} C^{p}_{\tilde{q}} &= C^{p}_{\tilde{p}} \epsilon_p, \\ f^{\tilde{q}}_{\tilde{p}} &= h^{\tilde{q}}_{\tilde{p}} + \gamma^{\tilde{r}}_{\tilde{s}} \left(V^{\tilde{q}\tilde{s}}_{\tilde{p}\tilde{r}} - \frac{1}{2} V^{\tilde{s}\tilde{q}}_{\tilde{p}\tilde{r}} \right), \\ \gamma^{\tilde{r}}_{\tilde{s}} &= 2\sum_{i\in\mathrm{occ}} \bar{C}^{\dagger\tilde{r}}_{i} C^{i}_{\tilde{s}}, \end{aligned} \tag{3.5}$$

where tilde indices correspond to the original orbitals. If the FCIDUMP is similarity-transformed, $\bar{C}_p^{\dagger\tilde{r}} \neq C_{\tilde{r}}^p$, and $\bar{C}_p^{\dagger\tilde{r}}$ are obtained as an inverse of $C_{\tilde{r}}^p$ such that $\bar{C}_p^{\dagger\tilde{r}} C_{\tilde{r}}^r = \delta_p^r$.

The unrestricted Hartree-Fock equations are given (for α case) by

$$\begin{aligned} {}^{\alpha}f_{\tilde{p}}^{\tilde{q}} C_{\tilde{q}}^{p} &= C_{\tilde{p}}^{p} \epsilon_p, \\ {}^{\alpha}f_{\tilde{p}}^{\tilde{q}} &= h_{\tilde{p}}^{\tilde{q}} + \left({}^{\alpha}\gamma_{\tilde{s}}^{\tilde{r}} + {}^{\beta}\gamma_{\tilde{s}}^{\tilde{r}} \right) V_{\tilde{p}\tilde{r}}^{\tilde{q}\tilde{s}} - {}^{\alpha}\gamma_{\tilde{s}}^{\tilde{r}} V_{\tilde{p}\tilde{r}}^{\tilde{s}\tilde{q}}, \\ {}^{\alpha}\gamma_{\tilde{s}}^{\tilde{r}} &= \sum_{i \in \mathrm{occ}} \bar{C}_i^{\dagger\tilde{r}} C_{\tilde{s}}^{i}, \\ {}^{\beta}\gamma_{\tilde{s}}^{\tilde{r}} &= \sum_{\bar{i} \in \mathrm{occ}} \bar{C}_{\bar{i}}^{\dagger\tilde{r}} C_{\tilde{s}}^{\bar{i}}, \end{aligned} \tag{3.6}$$

and equations for β spin can be obtained by swapping the spins. Note that if the FCIDUMP is of UHF type, the original indices and integrals are spin-dependent, which has to be taken into account in the equations.

The residual of the Hartree-Fock equations, which can be used in DIIS, is equivalent to Eq. (3.4), with the overlap matrix S_μ^ν removed.

3.3 Hartree–Fock from exact AO integrals

When the exact four-index AO integrals are available as the $\pm$ supermatrices of Sec. 2.4, the Fock matrix is assembled directly from the stored lower block triangle, without ever forming the full $v_{\mu\nu}^{\rho\sigma}$ tensor. With the (spin-summed) AO density $\gamma_\nu^\sigma = 2\, C_\nu^i C_i^{\dagger\sigma}$ the closed-shell Fock matrix is

$$f_\mu^\nu = h_\mu^\nu + J_\mu^\nu - \frac{1}{2} K_\mu^\nu, \qquad J_\mu^\nu = v_{\mu\rho}^{\nu\sigma}\, \gamma_\sigma^\rho, \qquad K_\mu^\nu = v_{\mu\rho}^{\sigma\nu}\, \gamma_\sigma^\rho. \tag{3.7}$$

The store is streamed once per SCF iteration, one ket-pair column $(\rho\sigma)$, $\rho \leq \sigma$, at a time: the dense slab $v_{\mu\nu}^{\rho\sigma}$ is reconstructed from $\frac{1}{2}(V^+ + V^-)$ and its hermitian mirror, and contributes the band products

$$J_\mu^\rho \mathrel{+}= v_{\mu\nu}^{\rho\sigma}\, \gamma_\nu^\sigma, \qquad J_\mu^\sigma \mathrel{+}= v_{\nu\mu}^{\rho\sigma}\, \gamma_\nu^\rho, \qquad K_\mu^\sigma \mathrel{+}= v_{\mu\nu}^{\rho\sigma}\, \gamma_\nu^\rho, \qquad K_\mu^\rho \mathrel{+}= v_{\nu\mu}^{\rho\sigma}\, \gamma_\nu^\sigma, \tag{3.8}$$

so the working memory is $\mathcal{O}(n^2)$ regardless of the basis size. For a real symmetric density (the HF case, $\gamma = \gamma^T$) the Fock matrices are symmetric and the mirror contributions of Eq. (3.8) are the transposes of the native ones; they are skipped entirely and restored at the end by the symmetrization

$$A \leftarrow A + A^T + A_\mathrm{d}, \tag{3.9}$$

where A_d collects the (self-mirroring) diagonal-tile band separately to avoid double counting – approximately halving the number of band products. The unrestricted Fock matrices are built in the *same* single sweep: the shared Coulomb term from ${}^{\alpha}\gamma + {}^{\beta}\gamma$ and the two same-spin exchange terms from ${}^{\alpha}\gamma$ and ${}^{\beta}\gamma$ per slab.

Chapter 4

Density-Fitted Multiconfigurational Self-Consistent Field

4.1 Orbital Rotation

The orbital transformed wavefunction can be expressed by

$$|\Psi\rangle = \exp(\hat{R})|0\rangle, \tag{4.1}$$

where

$$\begin{aligned} \exp(\hat{R}) &= 1 + \hat{R} + \frac{1}{2!}\hat{R}^2 + \cdots, \\ \hat{R} &= R_r^s \hat{E}_s^r, \end{aligned} \tag{4.2}$$

$\hat{E}_s^r$ is the singlet excitation operator $(\hat{a}_{r\alpha}^\dagger \hat{a}_{s\alpha} + \hat{a}_{r\beta}^\dagger \hat{a}_{s\beta})$. Mathematically if $\mathbf{R}$ is anti-symmetric, $\exp(\hat{R})$ is an unitary transformation.

$$\begin{aligned} R_r^s &= -R_s^r, \\ \hat{R} &= [R_r^s(\hat{E}_s^r - \hat{E}_r^s)]_{r>s}, \end{aligned} \tag{4.3}$$

4.2 MCSCF

The energy expectation value of the wavefunction is be given by

$$\begin{aligned} E(\mathbf{R}) &= \langle\Psi|\hat{H}|\Psi\rangle \\ &= \langle 0|\exp(-\hat{R})\hat{H}\exp(\hat{R})|0\rangle \\ &= \langle 0|\hat{H}|0\rangle + \langle 0|[\hat{H},\hat{R}]|0\rangle + \frac{1}{2!}\langle 0|[[\hat{H},\hat{R}],\hat{R}]|0\rangle + \cdots \\ &= E_0 + \mathbf{g^T x} + \mathbf{\frac{1}{2} x^T h x} + \cdots. \end{aligned} \tag{4.4}$$

In which Hamiltonian operator $\hat{H}$ is expressed as

$$\begin{aligned} \hat{H} &= E_0 + h_p^q \hat{a}_p^\dagger \hat{a}_q + \frac{1}{2} v_{pr}^{qs} \hat{a}_p^\dagger \hat{a}_r^\dagger \hat{a}_s \hat{a}_q \\ &= E_0 + h_p^q \hat{E}_q^p + \frac{1}{2} v_{pr}^{qs} \hat{e}_{qs}^{pr}, \end{aligned} \tag{4.5}$$

with $\hat{e}^{pr}_{qs}$ as the 2-electron excitation operator $\sum_{\sigma\tau} \hat{a}^{\dagger}_{p\sigma}\hat{a}^{\dagger}_{r\tau}\hat{a}_{s\tau}\hat{a}_{q\sigma}$. In the first expression, p, q, r, s denote the spin orbitals, and in the second expression, p, q, r, s denote spatial orbitals.

The orbital transformation parameters $\mathbf{R}$ is expressed as vector $\mathbf{x}$, the linear coefficients as gradient vector $\mathbf{g}$, the quadratic coefficients as matrix $\mathbf{h}$. When terms after the quadratic terms truncated, to minimize the energy expectation value, the Lagrangian equation as known as the Newton-Raphson equation is

$$\frac{\partial}{\partial \mathbf{x}}(\mathbf{g^T x} + \frac{\mathbf{1}}{\mathbf{2}}\mathbf{x^T h x}) = \mathbf{g} + \mathbf{hx} = \mathbf{0}. \tag{4.6}$$

Since the internal rotation of each of doubly occupied orbitals and virtual orbitals don't change the wavefunction, vector $\mathbf{x}$ is consisted of 4 parts, can be expressed as $[\mathbf{x^j_t}, \mathbf{x^j_a}, \mathbf{x^u_t}, \mathbf{x^u_a}]$.

Here, $\mathbf{g}$ is calculated from

$$\begin{aligned} g^k_r &= (\frac{\partial E}{\partial x^k_r})_{\mathbf{R=0}} \\ &= \langle 0|[\hat{H}, \hat{E}^r_k - \hat{E}^k_r]|0\rangle \\ &= [1 - \mathcal{P}(rk)]\langle 0|[\hat{E}^k_r, h^{q'}_{p'}\hat{E}^{p'}_{q'} + \frac{1}{2}v^{q's'}_{p'r'}\hat{e}^{p'r'}_{q's'}]|0\rangle \\ &= [1 - \mathcal{P}(rk)]\langle 0|h^{q'}_r\hat{E}^k_{q'} - h^k_{p'}\hat{E}^{p'}_r + v^{q's'}_{rr'}\hat{e}^{kr'}_{q's'} - v^{ks'}_{p'r'}\hat{e}^{p'r'}_{rs'}|0\rangle, \\ &= [1 - \mathcal{P}(rk)](h^{q'}_r\,{}^1D^k_{q'} - h^k_{p'}\,{}^1D^{p'}_r + v^{q's'}_{rr'}\,{}^2D^{kr'}_{q's'} - v^{ks'}_{p'r'}\,{}^2D^{p'r'}_{rs'}) \end{aligned} \tag{4.7}$$

let

$$A^k_r = \frac{1}{2}(h^{q'}_r\,{}^1D^k_{q'} + h^r_{p'}\,{}^1D^{p'}_k) + \frac{1}{2}(v^{q's'}_{rr'}\,{}^2D^{kr'}_{q's'} + v^{rs'}_{p'r'}\,{}^2D^{p'r'}_{ks'}), \tag{4.8}$$

then

$$g^k_r = 2(A^k_r - A^r_k). \tag{4.9}$$

In which 1-particle density matrix $\mathbf{^1D}$ and 2-particle density matrix $\mathbf{^2D}$ are defined as

$$\begin{aligned} {}^1D^t_u &= \langle 0|\hat{E}^t_u|0\rangle = c_I c_J\langle\Phi_I|\hat{E}^t_u|\Phi_J\rangle, \\ {}^2D^{tv}_{uw} &= \langle 0|\hat{e}^{tv}_{uw}|0\rangle = c_I c_J\langle\Phi_I|\hat{e}^{tv}_{uw}|\Phi_J\rangle. \end{aligned} \tag{4.10}$$

In order to simplify the expression, the ${}^1D^u_v$ is written as D^u_v, and ${}^2D^{uw}_{tv}$ is written as D^{uw}_{tv} in follow.

Fock matrix can be generalizingly defined as

$$F^s_r = {}^cF^s_r + D^t_u[v^{su}_{rt} - \frac{1}{2}v^{us}_{rt}], \tag{4.11}$$

with

$${}^cF^s_r = h^s_r + (2v^{sj}_{rj} - v^{js}_{rj}) \tag{4.12}$$

defined as the closed shell part of Fock matrix.

$\mathbf{A_r^k}$ can be calculated as

$$\begin{aligned}
A_r^i =&\frac{1}{2}(h_r^{q'}D_{q'}^i + h_{p'}^r D_i^{p'}) + \frac{1}{2}(v_{rr'}^{q's'}D_{q's'}^{ir'} + v_{p'r'}^{rs'}D_{is'}^{p'r'}),\\
=&\frac{1}{2}(h_r^i D_i^i + h_i^r D_i^i) + \frac{1}{2}(v_{rj}^{ij}D_{ij}^{ij} + v_{ij}^{rj}D_{ij}^{ij} + v_{rj}^{ji}D_{ji}^{ij} + v_{ji}^{rj}D_{ij}^{ji}\\
&+ v_{rt}^{iu}D_{iu}^{it} + v_{iu}^{rt}D_{it}^{iu} + v_{rt}^{ui}D_{ui}^{it} + v_{ui}^{rt}D_{it}^{ui})\\
=&\frac{1}{2}(2h_r^i + 2h_i^r) + \frac{1}{2}(4v_{rj}^{ij} + 4v_{ij}^{rj} - 2v_{rj}^{ji} - 2v_{ji}^{rj} + 2v_{rt}^{iu}D_u^t + 2v_{iu}^{rt}D_t^u - v_{rt}^{ui}D_u^t - v_{ui}^{rt}D_t^u)\\
=&\mathcal{S}\,(ri)\,(h_r^i + 2v_{rj}^{ij} - v_{rj}^{ji} + v_{rt}^{iu}D_u^t - \frac{1}{2}v_{rt}^{ui}D_u^t)\\
=&\mathcal{S}\,(ri)\,F_r^i,
\end{aligned} \tag{4.13}$$

$$\begin{aligned}
A_r^u =&\frac{1}{2}(h_r^{q'}D_{q'}^u + h_{p'}^r D_u^{p'}) + \frac{1}{2}(v_{rr'}^{q's'}D_{q's'}^{ur'} + v_{p'r'}^{rs'}D_{us'}^{p'r'})\\
=&\frac{1}{2}(h_r^t D_t^u + h_t^r D_u^t) + \frac{1}{2}(v_{rj}^{tj}D_{tj}^{uj} + v_{tj}^{rj}D_{uj}^{tj} + v_{rw}^{tv}D_{tv}^{uw} + v_{tv}^{rw}D_{uw}^{tv} + v_{rj}^{jt}D_{jt}^{uj} + v_{jt}^{rj}D_{uj}^{jt})\\
=&\frac{1}{2}(h_r^t D_t^u + h_t^r D_u^t) + \frac{1}{2}(2v_{rj}^{tj}D_t^u + 2v_{tj}^{rj}D_u^t + v_{rw}^{tv}D_{tv}^{uw} + v_{tv}^{rw}D_{uw}^{tv} - v_{rj}^{jt}D_t^u - v_{jt}^{rj}D_u^t)\\
=&\frac{1}{2}[(h_r^t + 2v_{rj}^{tj} - v_{rj}^{jt})D_t^u + v_{rw}^{tv}D_{tv}^{uw}] + \frac{1}{2}[(h_t^r + 2v_{tj}^{rj} - v_{tj}^{jr})D_u^t + v_{tv}^{rw}D_{uw}^{tv}]\\
=&\frac{1}{2}({}^cF_r^t D_t^u + v_{rw}^{tv}D_{tv}^{uw}) + \frac{1}{2}({}^cF_t^r D_u^t + v_{tv}^{rw}D_{uw}^{tv}),
\end{aligned} \tag{4.14}$$

$$A_r^a = 0. \tag{4.15}$$

Electronic energy E after orbital transformation is calculated with transformed orbitals and active orbital 1- and 2-particle density matrices

$$E = {}^cF_t^u D_u^t + \frac{1}{2}v_{tv}^{uw}D_{uw}^{tv} + E_c, \tag{4.16}$$

with

$$E_c = h_j^j + {}^cF_j^j, \tag{4.17}$$

E_c is the closed shell electronic energy.

The 4-index integral v_{tv}^{uw} is calculated as the density fitted integrals, as mentioned in previous chapter,

$$\begin{aligned}
v_{tv}^{uw} =&v_t^{uL}v_v^{wL},\\
v_t^{uL} =&v_\mu^{\nu L}C_t^{\dagger\mu}C_\nu^u.
\end{aligned} \tag{4.18}$$

Likewise, other four-index integrals in this chapter are calculated in the same way.

4.3 Augmented Hessian

In order to make the coefficients transforming step smaller and more robust, a level-shift ϵ is introduced to control the step $\mathbf{x}$ [3]

$$\mathbf{g} + (\mathbf{h} + \epsilon\mathbf{I})\mathbf{x} = \mathbf{0}, \tag{4.19}$$

in which

$$\epsilon = -\lambda^2 \mathbf{x}^{\mathbf{T}} \mathbf{g}, \tag{4.20}$$

$$\begin{pmatrix} 0 & \lambda \mathbf{g}^{\mathbf{T}} \\ \lambda \mathbf{g} & \mathbf{h} \end{pmatrix} \begin{pmatrix} \frac{1}{\lambda} \\ \mathbf{x} \end{pmatrix} = \nu \begin{pmatrix} \frac{1}{\lambda} \\ \mathbf{x} \end{pmatrix}. \tag{4.21}$$

We search for the value of λ with a combination method of linear search and logarithmic bisection search:

$$\begin{aligned} \lambda =& \frac{(\|\mathbf{x_{small\lambda}}\| - trust) + (trust - \|\mathbf{x_{large\lambda}}\|)}{\|\mathbf{x_{small\lambda}}\| - \|\mathbf{x_{large\lambda}}\|}, \\ \lambda =& \exp\left(\ln small\lambda + (\ln big\lambda - \ln small\lambda) * bisecdamp\right). \end{aligned} \tag{4.22}$$

If there are both tested smallest and biggest limits in one iteration search for λ, we adopt the linear search with the norm of both limit $\mathbf{x}$, otherwise, we use either the upper and lower boundaries value(by default set to be 1000 and 1) and the tested λ to do the logarithmic bisection search.

4.4 Hessian Approximation

4.4.1 First Order Approximation

We adopt the Super-CI optimization approximation[4], in which

$$\hat{H}^{eff} = \sum_{pq} F_p^q \hat{E}_p^q, \tag{4.23}$$

$$E^{(0)} = \langle 0 | \hat{H}^{eff} | 0 \rangle, \tag{4.24}$$

$$^{SCI}H_{rs}^{kl} = 2\langle rk | \hat{H}^{eff} - E^{(0)} | sl \rangle. \tag{4.25}$$

More specifically,

$$\begin{aligned} ^{SCI}H_{ab}^{ij} =& 4(\delta_i^j F_a^b - \delta_a^b F_i^j), \\ ^{SCI}H_{ab}^{iu} =& -2\delta_a^b F_i^v D_v^u, \\ ^{SCI}H_{au}^{ij} =& \delta_i^j (4F_a^u - 2F_a^v D_v^u), \\ ^{SCI}H_{tb}^{iu} =& 0, \\ ^{SCI}H_{tu}^{ij} =& (2D_t^u - 4\delta_t^u) F_i^j + 2\delta_i^j [2F_t^u - (D_{uw}^{tv} - D_u^t D_w^v) F_v^w - D_t^v F_v^u - F_t^v D_u^v], \\ ^{SCI}H_{ab}^{tu} =& 2\delta_b^a (D_{uw}^{tv} - D_u^t D_w^v) F_v^w + 2D_u^t F_a^b. \end{aligned} \tag{4.26}$$

4.4.2 Second Order Approximation

In general, the second-order Hessian matrix elements are calculated as

$$
\begin{aligned}
{}^{SO}H^{kl}_{rs} =& (\frac{\partial E}{\partial x^k_r \partial x^l_s})_{\mathbf{R=0}} \\
=& \frac{1}{2}\mathcal{S}\,(rs,kl)\,\langle 0|[[\hat{H},\hat{E}^k_r-\hat{E}^r_k],\hat{E}^l_s-\hat{E}^s_l]|0\rangle \\
=& \frac{1}{2}\mathcal{S}\,(rs,kl)\,\mathcal{A}\,(rk;sl)\,\langle 0|[[\hat{H},\hat{E}^k_r],\hat{E}^l_s]|0\rangle \\
=& \frac{1}{2}\mathcal{S}\,(rs,kl)\,\mathcal{A}\,(rk;sl)\,\langle 0|[\hat{E}^l_s,[\hat{E}^k_r,h^{q'}_{p'}\hat{E}^{p'}_{q'}+\frac{1}{2}v^{q's'}_{p'r'}\hat{e}^{p'r'}_{q's'}]]|0\rangle \\
=& \frac{1}{2}\mathcal{S}\,(rs,kl)\,\mathcal{A}\,(rk;sl)\,(-h^l_rD^k_s-h^k_sD^l_r+h^{q'}_rD^l_{q'}\delta^k_s+h^k_{p'}D^{p'}_s\delta^l_r \\
&+\delta^k_sv^{q's'}_{rr'}D^{lr'}_{q's'}+\delta^l_rv^{ks'}_{p'r'}D^{p'r'}_{ss'}-v^{ls'}_{rr'}D^{kr'}_{ss'}-v^{ks'}_{sr'}D^{lr'}_{rs'} \\
&+v^{q's'}_{rs}D^{kl}_{q's'}+v^{kl}_{p'r'}D^{p'r'}_{rs}-v^{q'l}_{rr'}D^{kr'}_{q's}-v^{ks'}_{p's}D^{p'l}_{rs'}) \\
=& \mathcal{A}\,(rk;sl)\,(-h^l_rD^k_s-h^k_sD^l_r-v^{ls'}_{rr'}D^{kr'}_{ss'}-v^{ks'}_{sr'}D^{lr'}_{rs'}+v^{q's'}_{rs}D^{kl}_{q's'}+v^{kl}_{p'r'}D^{p'r'}_{rs}-v^{q'l}_{rr'}D^{kr'}_{q's}-v^{ks'}_{p's}D^{p'l}_{rs'}) \\
&+\frac{1}{2}\mathcal{S}\,(rs,kl)\,\mathcal{A}\,(rk;sl)\,\delta^k_s(h^{q'}_rD^l_{q'}+h^r_{p'}D^{p'}_l+v^{q's'}_{rr'}D^{lr'}_{q's'}+v^{rs'}_{p'r'}D^{p'r'}_{ls'}) \\
=& \mathcal{A}\,(rk;sl)\,(h^s_rD^k_l+h^r_sD^l_k+v^{ss'}_{rr'}D^{kr'}_{ls'}+v^{rs'}_{sr'}D^{lr'}_{ks'}+v^{q's'}_{rs}D^{kl}_{q's'}+v^{rs}_{p'r'}D^{p'r'}_{kl}+v^{q's}_{rr'}D^{kr'}_{q'l}+v^{rs'}_{p's}D^{p'l}_{ks'}) \\
&+\mathcal{A}\,(rk;sl)\,(\delta^k_sA^l_r+\delta^l_rA^k_s) \\
=& \mathcal{A}\,(rk;sl)\,[2G^{kl}_{rs}+\delta^k_s(A^l_r+A^r_l)] \\
=& \mathcal{A}\,(rk;sl)\,[2G^{kl}_{rs}-\delta^k_l(A^s_r+A^r_s)],
\end{aligned}
\tag{4.27}
$$

matrices $\mathbf{G}$ are defined as

$$
G^{kl}_{rs}=\frac{1}{2}(h^s_rD^k_l+h^r_sD^l_k+v^{ss'}_{rr'}D^{kr'}_{ls'}+v^{rs'}_{sr'}D^{lr'}_{ks'}+v^{q's'}_{rs}D^{kl}_{q's'}+v^{rs}_{p'r'}D^{p'r'}_{kl}+v^{q's}_{rr'}D^{kr'}_{q'l}+v^{rs'}_{p's}D^{p'l}_{ks'})
\tag{4.28}
$$

$$
G^{ij}_{rs}=\delta^j_i(F^r_s+F^s_r)+L^{ij}_{rs}+L^{rs}_{ij},
\tag{4.29}
$$

$$
G^{tj}_{rs}=\frac{1}{2}(D^t_uL^{uj}_{rs}+D^u_tL^{rs}_{uj})=G^{jt}_{sr},
\tag{4.30}
$$

$$
G^{tu}_{rs}=\frac{1}{2}({}^cF^s_rD^u_t+{}^cF^r_sD^t_u+v^{sw}_{rv}D^{tv}_{uw}+v^{rv}_{sw}D^{uw}_{tv})+v^{vw}_{rs}D^{tu}_{vw}+v^{rs}_{vw}D^{vw}_{tu},
\tag{4.31}
$$

where

$$
L^{pq}_{rs}=2v^{pq}_{rs}+2^{ps}_{rq}-v^{sp}_{rq}-v^{qp}_{rs}.
\tag{4.32}
$$

4.4.3 Combined Second-Order and Super-CI Hessian Approximation

By default, the SO-SCI Hessian matrix [5] is approximated as below:

$$\begin{aligned}
{}^{SO-SCI}H_{ab}^{ij} &= {}^{SCI}H_{ab}^{ij},\\
{}^{SO-SCI}H_{ab}^{iu} &= {}^{SCI}H_{ab}^{iu},\\
{}^{SO-SCI}H_{au}^{ij} &= {}^{SCI}H_{au}^{ij},\\
{}^{SO-SCI}H_{tb}^{iu} &= {}^{SO}H_{tb}^{iu},\\
{}^{SO-SCI}H_{tu}^{ij} &= {}^{SO}H_{tu}^{ij},\\
{}^{SO-SCI}H_{ab}^{tu} &= {}^{SO}H_{ab}^{tu}.
\end{aligned} \tag{4.33}$$

If the **SO_SCI_origin** option is set to be $false$,

$$ {}^{SO-SCI}H_{ab}^{tu} = {}^{SCI}H_{ab}^{tu}, \tag{4.34}$$

and the rest blocks of Hessian matrix remain the same.

Chapter 5

MP2: Second-Order Møller–Plesset Perturbation Theory

5.1 Spin-Orbital Formulation

In the spin-orbital basis, the MP2 correlation energy is

$$E^{(2)} = \frac{1}{4} \sum_{IJAB} \frac{(v_{IJ}^{AB} - v_{IJ}^{BA})(v_{AB}^{IJ} - v_{BA}^{IJ})}{\varepsilon_I + \varepsilon_J - \varepsilon_A - \varepsilon_B}, \tag{5.1}$$

where I, J, A, B denote *spin-orbitals*. The factor $\frac{1}{4}$ compensates for unrestricted summation over equivalent excitations ($IJ \leftrightarrow JI, AB \leftrightarrow BA$).

5.2 Closed-Shell RHF–MP2

For a closed-shell RHF reference, α and β electrons occupy the same spatial orbitals with identical orbital energies. We sum over spin explicitly.

5.2.1 Spin Cases

The integral $v_{i_{\sigma_1} j_{\sigma_2}}^{a_{\sigma_3} b_{\sigma_4}}$ is nonzero only when $\sigma_1 = \sigma_3$ and $\sigma_2 = \sigma_4$ (spin conservation).

Same-spin ($\alpha\alpha$ and $\beta\beta$). Both direct and exchange terms survive:

$$v_{i_\alpha j_\alpha}^{a_\alpha b_\alpha} - v_{i_\alpha j_\alpha}^{b_\alpha a_\alpha} = v_{ij}^{ab} - v_{ij}^{ba}. \tag{5.2}$$

Opposite-spin ($\alpha\beta$ and $\beta\alpha$). The exchange term vanishes by spin orthogonality:

$$v_{i_\alpha j_\beta}^{a_\alpha b_\beta} = v_{ij}^{ab}. \tag{5.3}$$

5.2.2 Spin Summation

Denoting $\Delta_{IJAB} = \varepsilon_I + \varepsilon_J - \varepsilon_A - \varepsilon_B$ and converting to spatial orbitals, we must account for overcounting.

Same-spin ($\alpha\alpha + \beta\beta$). Each spatial configuration (i, j, a, b) appears 4 times in the spin-orbital sum (swapping $i \leftrightarrow j$ and $a \leftrightarrow b$). With two identical spin cases and the $\frac{1}{4}$ factor:

$$E^{(2)}_{\alpha\alpha} + E^{(2)}_{\beta\beta} = 2 \times \frac{1}{4} \sum_{ijab} \frac{(v^{ab}_{ij} - v^{ba}_{ij})(v^{ij}_{ab} - v^{ij}_{ba})}{\Delta_{ijab}}. \tag{5.4}$$

Opposite-spin ($\alpha\beta + \beta\alpha$). Each spatial configuration appears twice (swapping both pairs simultaneously). With two identical spin cases:

$$E^{(2)}_{\alpha\beta} + E^{(2)}_{\beta\alpha} = 2 \times \frac{1}{2} \sum_{ijab} \frac{v^{ab}_{ij} v^{ij}_{ab}}{\Delta_{ijab}}. \tag{5.5}$$

5.2.3 Expanding the Same-Spin Term

Expand the squared antisymmetric integral:

$$(v^{ab}_{ij} - v^{ba}_{ij})(v^{ij}_{ab} - v^{ij}_{ba}) = v^{ab}_{ij} v^{ij}_{ab} - v^{ab}_{ij} v^{ij}_{ba} - v^{ba}_{ij} v^{ij}_{ab} + v^{ba}_{ij} v^{ij}_{ba}. \tag{5.6}$$

Since $\sum_{ijab} v^{ba}_{ij} v^{ij}_{ba} = \sum_{ijab} v^{ab}_{ij} v^{ij}_{ab}$ by relabeling $a \leftrightarrow b$:

$$\sum_{ijab} (v^{ab}_{ij} - v^{ba}_{ij})(v^{ij}_{ab} - v^{ij}_{ba}) = 2 \sum_{ijab} \big(v^{ab}_{ij} v^{ij}_{ab} - v^{ab}_{ij} v^{ij}_{ba}\big). \tag{5.7}$$

5.2.4 SS and OS Contributions

Substituting the expansion into the same-spin term:

$$E^{(2)}_{\text{SS}} = \frac{1}{2} \cdot 2 \sum_{ijab} \frac{v^{ab}_{ij} v^{ij}_{ab} - v^{ab}_{ij} v^{ij}_{ba}}{\Delta_{ijab}} = \sum_{ijab} \frac{v^{ab}_{ij} (v^{ij}_{ab} - v^{ij}_{ba})}{\Delta_{ijab}}. \tag{5.8}$$

The opposite-spin contribution simplifies to:

$$E^{(2)}_{\text{OS}} = \sum_{ijab} \frac{v^{ab}_{ij} v^{ij}_{ab}}{\Delta_{ijab}}. \tag{5.9}$$

5.2.5 Final RHF–MP2 Formula

Adding SS and OS:

$$\begin{aligned} E^{(2)}_{\text{RHF}} &= E^{(2)}_{\text{SS}} + E^{(2)}_{\text{OS}} \\ &= \sum_{ijab} \frac{v^{ab}_{ij} (v^{ij}_{ab} - v^{ij}_{ba}) + v^{ab}_{ij} v^{ij}_{ab}}{\Delta_{ijab}}. \end{aligned} \tag{5.10}$$

Factoring:

$$\boxed{E^{(2)}_{\text{RHF}} = \sum_{ijab} \frac{v^{ab}_{ij} (2 v^{ij}_{ab} - v^{ij}_{ba})}{\Delta_{ijab}}.} \tag{5.11}$$

5.2.6 Amplitude Form

Using spatial-orbital amplitudes $T^{ij}_{ab} = v^{ij}_{ab}/\Delta_{ijab}$:

$$\begin{aligned} E^{(2)}_{\mathrm{SS}} &= \sum_{ijab} v^{ab}_{ij}(T^{ij}_{ab} - T^{ij}_{ba}), \\ E^{(2)}_{\mathrm{OS}} &= \sum_{ijab} v^{ab}_{ij}\, T^{ij}_{ab}, \\ E^{(2)}_{\mathrm{RHF}} &= \sum_{ijab} v^{ab}_{ij}(2T^{ij}_{ab} - T^{ij}_{ba}). \end{aligned} \tag{5.12}$$

5.3 Unrestricted UHF–MP2

For UHF, α and β electrons occupy *different* spatial orbitals with different orbital energies. We use (i, j, a, b) for α and $(\bar{i}, \bar{j}, \bar{a}, \bar{b})$ for β.

5.3.1 Three Spin Contributions

$$E^{(2)}_{\mathrm{UHF}} = E^{(2)}_{\alpha\alpha} + E^{(2)}_{\beta\beta} + E^{(2)}_{\alpha\beta}. \tag{5.13}$$

Same-spin $\alpha\alpha$. With antisymmetrized integrals and antisymmetric amplitudes:

$$E^{(2)}_{\alpha\alpha} = \frac{1}{4}\sum_{ijab} \frac{(v^{ab}_{ij} - v^{ba}_{ij})(v^{ij}_{ab} - v^{ij}_{ba})}{\Delta_{ijab}} = \frac{1}{2}\sum_{ijab} v^{ab}_{ij}\, T^{ij}_{ab}. \tag{5.14}$$

The $\frac{1}{4} \to \frac{1}{2}$ follows from amplitude antisymmetry.

Same-spin $\beta\beta$.

$$E^{(2)}_{\beta\beta} = \frac{1}{2}\sum_{\bar{i}\bar{j}\bar{a}\bar{b}} v^{\bar{a}\bar{b}}_{\bar{i}\bar{j}}\, T^{\bar{i}\bar{j}}_{\bar{a}\bar{b}}. \tag{5.15}$$

Opposite-spin $\alpha\beta$. No exchange, no antisymmetry, no overcounting:

$$E^{(2)}_{\alpha\beta} = \sum_{i\bar{j}a\bar{b}} \frac{v^{a\bar{b}}_{i\bar{j}} v^{i\bar{j}}_{a\bar{b}}}{\Delta_{i\bar{j}a\bar{b}}} = \sum_{i\bar{j}a\bar{b}} v^{a\bar{b}}_{i\bar{j}}\, T^{i\bar{j}}_{a\bar{b}}. \tag{5.16}$$

5.3.2 Final UHF–MP2 Formula

$$\boxed{E^{(2)}_{\mathrm{UHF}} = \frac{1}{2}\sum_{ijab} v^{ab}_{ij}\, T^{ij}_{ab} + \frac{1}{2}\sum_{\bar{i}\bar{j}\bar{a}\bar{b}} v^{\bar{a}\bar{b}}_{\bar{i}\bar{j}}\, T^{\bar{i}\bar{j}}_{\bar{a}\bar{b}} + \sum_{i\bar{j}a\bar{b}} v^{a\bar{b}}_{i\bar{j}}\, T^{i\bar{j}}_{a\bar{b}},} \tag{5.17}$$

with amplitudes

$$\begin{aligned} T^{ij}_{ab} &= \frac{v^{ij}_{ab} - v^{ij}_{ba}}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}, \\ T^{\bar{i}\bar{j}}_{\bar{a}\bar{b}} &= \frac{v^{\bar{i}\bar{j}}_{\bar{a}\bar{b}} - v^{\bar{i}\bar{j}}_{\bar{b}\bar{a}}}{\varepsilon_{\bar{i}} + \varepsilon_{\bar{j}} - \varepsilon_{\bar{a}} - \varepsilon_{\bar{b}}}, \\ T^{i\bar{j}}_{a\bar{b}} &= \frac{v^{i\bar{j}}_{a\bar{b}}}{\varepsilon_i + \varepsilon_{\bar{j}} - \varepsilon_a - \varepsilon_{\bar{b}}}. \end{aligned} \tag{5.18}$$

5.3.3 Key Differences from RHF

- No factor of 2 on SS terms (α and β orbitals differ).
- Separate integrals and orbital energies for each spin.
- No exchange in the OS contribution.

Chapter 6

Positron

We consider systems with electrons and a single positron. Positron orbital indices are marked with a superscript plus: occupied positron orbitals $i^+, j^+, k^+, \ldots$, virtual positron orbitals $a^+, b^+, c^+, \ldots$, and general positron indices p^+, q^+, r^+, s^+. Electron indices remain undecorated as defined in Sec. 1.2. We use $\hat{b}^\dagger, \hat{b}$ for positron creation and annihilation operators to distinguish them from the electronic operators $\hat{a}^\dagger, \hat{a}$.

The full electron–positron Hamiltonian in second quantization reads

$$\hat{H} = E_0 + h_p^q \hat{a}_p^\dagger \hat{a}_q + h_{p^+}^{q^+(+)} \hat{b}_{p^+}^\dagger \hat{b}_{q^+} + \frac{1}{2} v_{pq}^{rs} \hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_s \hat{a}_r - v_{pq^+}^{rs^+} \hat{a}_p^\dagger \hat{b}_{q^+}^\dagger \hat{b}_{s^+} \hat{a}_r, \tag{6.1}$$

where $h_{p^+}^{q^+(+)}$ is the positron one–body Hamiltonian (differing from the electronic one by the sign of the nuclear attraction), $v_{pq^+}^{rs^+} = \langle pq^+ | rs^+ \rangle$ are the electron–positron Coulomb integrals, and the minus sign reflects the attractive electron–positron interaction. With a single positron, any positron–positron repulsion term vanishes.

6.1 Density fitting with positron

With positrons, the density-fitted correction equations (2.10) and (2.11) are modified to include the positron contributions:

$$\begin{aligned}
\tilde{h}_p^q &= \tilde{f}_p^q - \sum_L \left(2 v_p^{qL} v_i^{iL} - v_p^{iL} v_i^{qL} - v_p^{qL} v_{i^+}^{i^+L} \right) \\
\tilde{f}_p^q &= h_p^q + \sum_{\tilde{L}} \left(2 v_p^{q\tilde{L}} v_{\hat{i}}^{\hat{i}\tilde{L}} - v_p^{\hat{i}\tilde{L}} v_{\hat{i}}^{q\tilde{L}} - v_p^{q\tilde{L}} v_{i^+}^{i^+\tilde{L}} \right),
\end{aligned} \tag{6.2}$$

where $i^+ = 1$ denote the single occupied positron orbital, $\hat{i}$ denote all occupied orbitals (including core), and $\tilde{L}$ denote the auxiliary basis functions. The positron one-body terms get corrections according to

$$\begin{aligned}
\tilde{h}_{p^+}^{q^+(+)} &= \tilde{f}_{p^+}^{q^+(+)} + \sum_L \left(2 v_i^{iL} v_{p^+}^{q^+L} \right) \\
\tilde{f}_{p^+}^{q^+(+)} &= h_{p^+}^{q^+(+)} - \sum_L \left(2 v_{\hat{i}}^{\hat{i}\tilde{L}} v_{p^+}^{q^+\tilde{L}} \right),
\end{aligned} \tag{6.3}$$

and the core energy is saved as

$$\tilde{E}_0 = E_0 + h_i^i + \tfrac{1}{2} h_{i^+}^{i^+(+)} - \left(\tilde{h}_i^i + \tfrac{1}{2} \tilde{h}_{i^+}^{i^+(+)} \right). \tag{6.4}$$

At the moment the positron integral evaluation does not support frozen-core approximation.

6.2 Positron Hartree–Fock

A closed–shell electron reference plus a single positron is formally analogous to an *unrestricted* HF problem with two spin blocks: the α electrons and the positron (e^+). The key difference is that the positron block contains *exactly one* occupied orbital, so there is no exchange term between two positrons. Also there is an attractive Coulomb interaction between the positron and the electrons instead of the repulsive one between α and β electrons.

6.2.1 Density–fitted working equations

The coupled Roothaan equations

$$f^{\nu(e)}_{\mu} C^{p}_{\nu} = S^{\nu}_{\mu} C^{p}_{\nu} \epsilon_p, \qquad f^{\nu(p)}_{\mu} C^{p^+}_{\nu} = S^{\nu}_{\mu} C^{p^+}_{\nu} \epsilon_{p^+}, \tag{6.5}$$

are solved self–consistently.

With density fitting [cf. Sec. 2.1] we reuse the three–index intermediates $v^{\nu L}_{\mu}$ and the metric factors C^{L}_{P}. The AO density matrices are

$$D^{\nu}_{\mu} = 2 C^{i}_{\mu} C^{\dagger\nu}_{i}, \qquad P^{\nu}_{\mu} = C^{i^+}_{\mu} C^{\dagger\nu}_{i^+}. \tag{6.6}$$

Electron block. Including the attractive Coulomb term with the positron the electronic Fock matrix becomes

$$\begin{aligned} f^{\nu(e)}_{\mu} &= h^{\nu}_{\mu} + J^{\nu}_{\mu}[D] - K^{\nu}_{\mu}[D] - J^{\nu}_{\mu}[P] \\ J^{\nu}_{\mu}[D] &= \sum_{L} v^{\mathrm{o}L}_{\xi} D^{\xi}_{\mathrm{o}} v^{\nu L}_{\mu}, \quad J^{\nu}_{\mu}[P] = \sum_{L} v^{\mathrm{o}L}_{\xi} P^{\xi}_{\mathrm{o}} v^{\nu L}_{\mu}, \\ K^{\nu}_{\mu}[D] &= v^{iL}_{\mu} v^{\nu L}_{i}, \qquad v^{iL}_{\mu} = v^{\rho L}_{\mu} C^{i}_{\rho} \end{aligned} \tag{6.7}$$

Positron block. Exchange vanishes for a single particle; the Fock matrix for the positron therefore contains only the one–body term and the *attractive* Coulomb interaction with the electrons,

$$f^{\nu(p)}_{\mu} = h^{\nu(p)}_{\mu} - J^{\nu}_{\mu}[D], \tag{6.8}$$

where $h^{(p)}$ differs from the electronic one–electron matrix only by the sign of the nuclear attraction operator.

6.2.2 HF energy with a positron

The Hartree–Fock total energy is

$$E^{e/e^+}_{\mathrm{HF}} = D^{\nu}_{\mu} h^{\mu}_{\nu} + \tfrac{1}{2} D^{\nu}_{\mu} \big(f^{\mu(e)}_{\nu} + J^{\mu}_{\nu}[P] \big) + P^{\nu}_{\mu} f^{\mu(p)}_{\nu} + E_0, \tag{6.9}$$

which reduces to the ordinary closed–shell expression when $P = 0$. In practice the Coulomb terms are accumulated on the fly from the three–index intermediates as described above.

6.3 RHF–MP2 with a Single Positron

Consider a closed-shell RHF system with an additional single positron. We use indices i, j, a, b for electrons and i^+, a^+, p^+, q^+ for positrons, where i, j, i^+ denote occupied and a, b, a^+ denote virtual orbitals.

6.3.1 Contributions

The total MP2 energy has two parts:

$$E^{(2)} = E^{(2)}_{\mathrm{ee}} + E^{(2)}_{\mathrm{ep}}, \tag{6.10}$$

where $E^{(2)}_{\mathrm{ee}}$ is the standard RHF electron–electron correlation from Eq. (5.11), and $E^{(2)}_{\mathrm{ep}}$ is the electron–positron correlation. With only one positron, there is no positron–positron contribution.

6.3.2 Electron–Positron Correlation

Since electrons and positrons are mutually distinguishable, there is *no exchange* between them—only the direct (Coulomb) term contributes. For RHF electrons, the α and β spatial orbitals are identical, and the single positron occupies one spatial orbital i^+. Both electron spin channels (α and β) contribute equally, giving a factor of 2 after spin integration:

$$E^{(2)}_{\mathrm{ep}} = 2 \sum_{ii^+aa^+} \frac{v^{aa^+}_{ii^+} v^{ii^+}_{aa^+}}{\Delta_{ii^+aa^+}}, \tag{6.11}$$

where $\Delta_{ii^+aa^+} = \varepsilon_i + \varepsilon_{i^+} - \varepsilon_a - \varepsilon_{a^+}$, with i, a running over spatial electron orbitals and i^+, a^+ over spatial positron orbitals (one occupied, several virtual).

6.3.3 Amplitude Form

Defining the electron–positron amplitude:

$$T^{ii^+}_{aa^+} = \frac{v^{ii^+}_{aa^+}}{\Delta_{ii^+aa^+}}, \tag{6.12}$$

the electron–positron correlation energy becomes:

$$\boxed{E^{(2)}_{\mathrm{ep}} = 2 \sum_{ii^+aa^+} v^{aa^+}_{ii^+} T^{ii^+}_{aa^+}.} \tag{6.13}$$

Chapter 7

CCSD and DCSD amplitude and Λ equations

7.1 Closed-shell CCSD/DCSD Lagrangian

The singles-dressed factorization of the closed-shell CCSD and DCSD amplitude equations roughly follows the factorization from Ref. [6]. The closed-shell CCSD and DCSD Lagrangian is given by

$$\begin{aligned}
\mathcal{L} = {} & v^{cd}_{kl}\tilde{T}^{kl}_{cd} + \left(\hat{f}^c_k + f^c_k\right) T^k_c + \Lambda^{ab}_{ij}\hat{v}^{ij}_{ab} + \Lambda^{ab}_{ij}\left(\hat{v}^{ij}_{kl} + v^{cd}_{kl}T^{ij}_{cd}\right) T^{kl}_{ab} + \Lambda^{ab}_{ij}\hat{v}^{cd}_{ab}T^{ij}_{cd} \\
& + \Lambda^{ab}_{ij} v^{cd}_{kl} T^{kj}_{ad} T^{il}_{cb} \\
& + \Lambda^{ab}_{ij}\mathcal{S}\left(ab, ij\right)\left\{\left(\hat{f}^c_a - 2\times\frac{1}{2}v^{cd}_{kl}\tilde{T}^{kl}_{ad}\right) T^{ij}_{cb} - \left(\hat{f}^i_k + 2\times\frac{1}{2}v^{cd}_{kl}\tilde{T}^{il}_{cd}\right) T^{kj}_{ab}\right. \\
& \left. + \left(\hat{v}^{id}_{al} + \frac{1}{2}v^{cd}_{kl}\tilde{T}^{ik}_{ac}\right)\tilde{T}^{lj}_{db} - \hat{v}^{ic}_{ka}T^{kj}_{cb} - \hat{v}^{ic}_{kb}T^{kj}_{ac} - v^{cd}_{kl}T^{ki}_{da}\left(T^{lj}_{cb} - T^{lj}_{bc}\right)\right\} \\
& + \Lambda^a_i\hat{f}^i_a + \Lambda^a_i\hat{f}^b_j\tilde{T}^{ij}_{ab} + \Lambda^a_i\hat{v}^{bc}_{ak}\tilde{T}^{ki}_{cb} - \Lambda^a_i\hat{v}^{ic}_{jk}\tilde{T}^{kj}_{ca}.
\end{aligned} \tag{7.1}$$

The DCSD Lagrangian is obtained by removing terms in red.

$\tilde{T}^{ij}_{ab}$ are the contravariant amplitudes,

$$\tilde{T}^{ij}_{ab} = 2T^{ij}_{ab} - T^{ij}_{ba}. \tag{7.2}$$

Integrals with hats are dressed integrals, i.e. they are obtained by dressing the integrals with the singles amplitudes, and the Fock matrix is internally dressed, too, e.g.,

$$\begin{aligned}
\hat{v}^{id}_{kl} &= v^{id}_{kl} + v^{cd}_{kl}T^i_c \\
\hat{v}^{cd}_{al} &= v^{cd}_{al} - v^{cd}_{kl}T^k_a \\
\hat{f}^c_k &= h^c_k + 2\hat{v}^{cl}_{kl} - \hat{v}^{cl}_{lk} = f^c_k + \left(2v^{cd}_{kl} - v^{cd}_{lk}\right) T^l_d.
\end{aligned} \tag{7.3}$$

Note that only the **lower virtual** and **upper occupied** indices are dressed.

The amplitude equations can be obtained by taking the derivative of the Lagrangian with respect to the Lagrange multipliers Λ and setting the result to zero.

The most efficient version of CCSD/DCSD in ElemCo.jl combines the dressed factorization from above with the cckext type of factorization from Ref. [7] and is given by

$$\begin{aligned}\mathcal{L} = {} & v_{kl}^{cd}\tilde{T}_{cd}^{kl} + \left(\hat{f}_k^c + f_k^c\right) T_c^k + \Lambda_{ij}^{ab}\left(\hat{v}_{kl}^{ij} + v_{kl}^{cd}T_{cd}^{ij}\right) T_{ab}^{kl} + \Lambda_{ij}^{ab} K_{pq}^{ij}\delta_a^p\delta_b^q \\ & + \Lambda_{ij}^{ab} v_{kl}^{cd} T_{ad}^{kj} T_{cb}^{il} \\ & + \Lambda_{ij}^{ab}\mathcal{S}\left(ab,ij\right)\left\{\left(\hat{f}_a^c - 2\times\frac{1}{2}v_{kl}^{cd}\tilde{T}_{ad}^{kl}\right) T_{cb}^{ij} - \left(\hat{f}_k^i + 2\times\frac{1}{2}v_{kl}^{cd}\tilde{T}_{cd}^{il}\right) T_{ab}^{kj}\right. \\ & + \left(\hat{v}_{al}^{id} + \frac{1}{2}v_{kl}^{cd}\tilde{T}_{ac}^{ik}\right)\tilde{T}_{db}^{lj} - \hat{v}_{ka}^{ic}T_{cb}^{kj} - \hat{v}_{kb}^{ic}T_{ac}^{kj} - v_{kl}^{cd}T_{da}^{ki}\left(T_{cb}^{lj} - T_{bc}^{lj}\right) \\ & \left. - K_{pq}^{ij}\left(\delta_k^p\delta_b^q - \frac{1}{2}\delta_k^p\delta_l^q T_b^l\right) T_a^k\right\} + \Lambda_i^a K_{pq}^{ij}\left(2\delta_a^p\delta_j^q - \delta_j^p\delta_a^q\right) \\ & - \Lambda_i^a T_a^k K_{pq}^{ij}\left(2\delta_k^p\delta_j^q - \delta_j^p\delta_k^q\right) + \Lambda_i^a\hat{h}_a^i + \Lambda_i^a\hat{f}_j^b\tilde{T}_{ab}^{ij} - \Lambda_i^a\hat{v}_{jk}^{ic}\tilde{T}_{ca}^{kj},\end{aligned} \tag{7.4}$$

where

$$K_{pq}^{ij} = v_{pq}^{rs}\left(\left(T_{ab}^{ij} + T_a^i T_b^j\right)\delta_r^a\delta_s^b + \delta_r^i T_b^j\delta_s^b + T_a^i\delta_r^a\delta_s^j + \delta_r^i\delta_s^j\right) \tag{7.5}$$

and h is the one-particle part of the Hamiltonian.

7.2 Closed-shell CCSD/DSCD Lagrangian multipliers equations

The Λ equations are obtained by taking the derivative of the Lagrangian Eq. (7.1) with respect to the amplitudes and setting the result to zero, i.e.,

$$\begin{aligned}\frac{\partial\mathcal{L}}{\partial T_e^m} = {} & 2\left(2v_{jm}^{be} - v_{jm}^{eb}\right) T_b^j + 2f_m^e - \Lambda_{ij}^{eb}\hat{v}_{mb}^{ij} - \Lambda_{ij}^{ae}\hat{v}_{am}^{ij} + \Lambda_{mj}^{ab}\hat{v}_{ab}^{ej} + \Lambda_{im}^{ab}\hat{v}_{ab}^{ie} \\ & + \Lambda_{mj}^{ab}\hat{v}_{kl}^{ej}T_{ab}^{kl} + \Lambda_{im}^{ab}\hat{v}_{kl}^{ie}T_{ab}^{kl} - \Lambda_{ij}^{eb}\hat{v}_{mb}^{cd}T_{cd}^{ij} - \Lambda_{ij}^{ae}\hat{v}_{am}^{cd}T_{cd}^{ij} \\ & - \Lambda_{ij}^{eb}\hat{f}_m^d T_{db}^{ij} - \Lambda_{ij}^{ae}\hat{f}_m^d T_{ad}^{ij} - \Lambda_{mj}^{ab}\hat{f}_k^e T_{ab}^{kj} - \Lambda_{im}^{ab}\hat{f}_k^e T_{ab}^{ik} \\ & + \Lambda_{ij}^{ab}\mathcal{S}\left(ab,ij\right)\left\{\left(2\hat{v}_{am}^{de} - \hat{v}_{am}^{ed}\right) T_{db}^{ij} - \left(2\hat{v}_{km}^{ie} - \hat{v}_{km}^{ei}\right) T_{ab}^{kj}\right\} \\ & - \Lambda_{ij}^{eb}\hat{v}_{ml}^{id}\tilde{T}_{db}^{lj} - \Lambda_{ij}^{ae}\hat{v}_{ml}^{jd}\tilde{T}_{ad}^{il} + \Lambda_{mj}^{ab}\hat{v}_{al}^{ed}\tilde{T}_{db}^{lj} + \Lambda_{im}^{ab}\hat{v}_{bl}^{ed}\tilde{T}_{ad}^{il} \\ & + \Lambda_{ij}^{eb}\hat{v}_{km}^{ic}T_{cb}^{kj} - \Lambda_{mj}^{ab}\hat{v}_{ka}^{ec}T_{cb}^{kj} + \Lambda_{ij}^{ae}\hat{v}_{km}^{jc}T_{ac}^{ik} - \Lambda_{im}^{ab}\hat{v}_{kb}^{ec}T_{ac}^{ik} \\ & + \Lambda_{ij}^{ae}\hat{v}_{km}^{ic}T_{ac}^{kj} - \Lambda_{mj}^{ab}\hat{v}_{kb}^{ec}T_{ac}^{kj} + \Lambda_{ij}^{eb}\hat{v}_{km}^{jc}T_{cb}^{ik} - \Lambda_{im}^{ab}\hat{v}_{ka}^{ec}T_{cb}^{ik} \\ & - \Lambda_i^e\hat{f}_m^i + \Lambda_m^a\hat{f}_a^e + \Lambda_i^a\left(2\hat{v}_{am}^{ie} - \hat{v}_{am}^{ei}\right) \\ & + \Lambda_i^a\left(2v_{jm}^{be} - v_{jm}^{eb}\right)\tilde{T}_{ab}^{ij} - \Lambda_i^e v_{mk}^{bc}\tilde{T}_{cb}^{ki} - \Lambda_m^a v_{jk}^{ec}\tilde{T}_{ca}^{kj}.\end{aligned} \tag{7.6}$$

$$
\begin{aligned}
\frac{\partial \mathcal{L}}{\partial T_{ef}^{mn}} = &\frac{1}{2}\mathcal{S}\left(ef,mn\right)\Big[\tilde{v}_{mn}^{ef} + \Lambda_{ij}^{ef}\left(\hat{v}_{mn}^{ij} + v_{mn}^{cd}T_{cd}^{ij}\right) + \Lambda_{mn}^{ab}v_{kl}^{ef}T_{ab}^{kl} + \Lambda_{mn}^{ab}\hat{v}_{ab}^{ef} \\
&+\Lambda_{in}^{eb}v_{ml}^{cf}T_{cb}^{il} + \Lambda_{mj}^{af}v_{kn}^{ed}T_{ad}^{kj} \\
&+ \Lambda_{mn}^{af}\mathcal{S}\left(af,mn\right)\left(\hat{f}_{a}^{e} - 2\times\frac{1}{2}v_{kl}^{ed}\tilde{T}_{ad}^{kl}\right) - \Lambda_{ij}^{eb}\mathcal{S}\left(eb,ij\right)2\times\frac{1}{2}\tilde{v}_{mn}^{cf}T_{cb}^{ij} \\
&- \Lambda_{in}^{ef}\mathcal{S}\left(ef,in\right)\left(\hat{f}_{m}^{i} + 2\times\frac{1}{2}v_{ml}^{cd}\tilde{T}_{cd}^{il}\right) - \Lambda_{mj}^{ab}\mathcal{S}\left(ab,mj\right)2\times\frac{1}{2}\tilde{v}_{kn}^{ef}T_{ab}^{kj} \\
&+ 2\Lambda_{in}^{af}\mathcal{S}\left(af,in\right)\left(\hat{v}_{am}^{ie} + \frac{1}{2}v_{km}^{ce}\tilde{T}_{ac}^{ik}\right) - \Lambda_{im}^{af}\mathcal{S}\left(af,im\right)\left(\hat{v}_{an}^{ie} + \frac{1}{2}v_{kn}^{ce}\tilde{T}_{ac}^{ik}\right) \\
&+ 2\Lambda_{mj}^{eb}\mathcal{S}\left(eb,mj\right)\frac{1}{2}v_{nl}^{fd}\tilde{T}_{db}^{lj} - \Lambda_{nj}^{eb}\mathcal{S}\left(eb,nj\right)\frac{1}{2}v_{ml}^{fd}\tilde{T}_{db}^{lj} \\
&- \Lambda_{in}^{af}\mathcal{S}\left(af,in\right)\hat{v}_{ma}^{ie} - \Lambda_{in}^{eb}\mathcal{S}\left(eb,in\right)\hat{v}_{mb}^{if} \\
&-\Lambda_{nj}^{fb}\mathcal{S}\left(fb,nj\right)v_{ml}^{ce}\left(T_{cb}^{lj} - T_{bc}^{lj}\right) - \Lambda_{in}^{af}\mathcal{S}\left(af,in\right)v_{km}^{ed}T_{da}^{ki} \\
&+\Lambda_{in}^{ae}\mathcal{S}\left(ae,in\right)v_{km}^{fd}T_{da}^{ki} \\
&+ \mathcal{T}(mn)\left\{\Lambda_{m}^{e}\hat{f}_{n}^{f} + \Lambda_{n}^{a}\hat{v}_{am}^{fe} - \Lambda_{i}^{f}\hat{v}_{nm}^{ie}\right\}\Big],
\end{aligned}
\tag{7.7}
$$

with a "contravariation" operator,

$$\mathcal{T}(mn)X_{mn}^{ef} = 2X_{mn}^{ef} - X_{nm}^{ef}, \tag{7.8}$$

and

$$\tilde{v}_{mn}^{ef} = 2v_{mn}^{ef} - v_{nm}^{ef}. \tag{7.9}$$

Now we can introduce useful intermediate quantities, related to the density matrices. The one-body reduced density matrices can be written as

$$
\begin{aligned}
D_{i}^{j} &= -2\Lambda_{ik}^{cd}T_{cd}^{jk}, \\
D_{a}^{b} &= 2\Lambda_{kl}^{bc}T_{ac}^{kl}, \\
D_{i}^{a} &= \Lambda_{i}^{a}, \\
D_{a}^{i} &= \Lambda_{k}^{c}\tilde{T}_{ac}^{ik}.
\end{aligned}
\tag{7.10}
$$

Note that we have excluded here terms coming from the singles amplitudes. Thus, if this density matrix is used to calculate properties, the corresponding integrals should be dressed. Alternatively, one can define "dressed" density matrices which include the singles contributions,

$$
\begin{aligned}
\hat{D}_{i}^{j} &= D_{i}^{j} - D_{i}^{c}T_{c}^{j}, \\
\hat{D}_{a}^{b} &= D_{a}^{b} + D_{k}^{b}T_{a}^{k}, \\
\hat{D}_{i}^{a} &= D_{i}^{a}, \\
\hat{D}_{a}^{i} &= D_{a}^{i} + 2T_{a}^{i} - D_{a}^{c}T_{c}^{i} + \hat{D}_{k}^{i}T_{a}^{k}.
\end{aligned}
\tag{7.11}
$$

Some parts of the two-body reduced density matrices can be written as

$$
\begin{aligned}
D_{ij}^{kl} &= \Lambda_{ij}^{cd}T_{cd}^{kl} \\
D_{ib}^{aj} &= \Lambda_{ik}^{ac}\tilde{T}_{cb}^{kj} \\
\bar{D}_{ib}^{aj} &= \Lambda_{ik}^{ac}T_{cb}^{kj} + \Lambda_{ik}^{ca}T_{bc}^{kj}
\end{aligned}
\tag{7.12}
$$

Finally, we define the following quantities which correspond to the cckext factorization and a doubles-dressing of the Fock matrix,

$$
\begin{aligned}
K^{rs}_{mn} &= \hat{\Lambda}^{pq}_{mn} v^{rs}_{pq} \\
\hat{\Lambda}^{pq}_{mn} &= \Lambda^{ab}_{mn}\delta^p_a\delta^q_b - \Lambda^{ab}_{mn}T^i_a\delta^p_i\delta^q_b - \Lambda^{ab}_{mn}\delta^p_a T^j_b\delta^q_j + \Lambda^{ab}_{mn}T^i_a T^j_b\delta^p_i\delta^q_j \\
x^i_m &= \tilde{T}^{il}_{cd}v^{cd}_{ml} \qquad\qquad x^e_a = \tilde{T}^{kl}_{ac}v^{ec}_{kl}
\end{aligned}
\tag{7.13}
$$

With these definitions, the Λ equations can be written as

$$
\begin{aligned}
\frac{\partial\mathcal{L}}{\partial T^m_e} = {}& \left(2v^{pe}_{qm} - v^{ep}_{qm}\right)\hat{D}^q_p + 2f^e_m - 2\Lambda^{eb}_{ij}\hat{v}^{ij}_{mb} + 2K^{rs}_{mj}\delta^e_r\left(\delta^j_s + \delta^b_s T^j_b\right) \\
& + 2D^{kl}_{mj}\hat{v}^{ej}_{kl} - 2\Lambda^{eb}_{ij}\left(\hat{v}^{cd}_{mb}T^{ij}_{cd}\right) - D^e_d\hat{f}^d_m + D^k_m\hat{f}^e_k - 2D^{el}_{id}\hat{v}^{id}_{ml} + 2D^{al}_{md}\hat{v}^{ed}_{al} \\
& + 2\bar{D}^{ek}_{ic}\hat{v}^{ic}_{km} - 2\bar{D}^{ak}_{mc}\hat{v}^{ec}_{ka} - \Lambda^e_i\hat{f}^i_m + \Lambda^a_m\hat{f}^e_a - \Lambda^e_i x^i_m - \Lambda^a_m x^e_a .
\end{aligned}
\tag{7.14}
$$

$$
\begin{aligned}
\frac{\partial\mathcal{L}}{\partial T^{mn}_{ef}} = {}& \tilde{v}^{ef}_{mn} + \Lambda^{ef}_{ij}\left(\hat{v}^{ij}_{mn} + v^{cd}_{mn}T^{ij}_{cd}\right) + D^{kl}_{mn}v^{ef}_{kl} + K^{rs}_{mn}\delta^e_r\delta^f_s \\
& + \mathcal{S}(ef,mn)\Bigg\{\Lambda^{af}_{mn}\left(\hat{f}^e_a - 2\times\frac{1}{2}x^e_a\right) - \Lambda^{ef}_{in}\left(\hat{f}^i_m + 2\times\frac{1}{2}x^i_m\right) \\
& + \mathcal{T}(mn)\Bigg[2\times\frac{1}{4}v^{ef}_{kn}D^k_m - 2\times\frac{1}{4}v^{cf}_{mn}D^e_c + \Lambda^{af}_{in}\left(\hat{v}^{ie}_{am} + v^{ce}_{km}\tilde{T}^{ik}_{ac}\right) \\
& + \frac{1}{2}\left(\Lambda^e_m\hat{f}^f_n + \Lambda^a_n\hat{v}^{fe}_{am} - \Lambda^f_i\hat{v}^{ie}_{nm}\right)\Bigg] \\
& - \Lambda^{af}_{in}\hat{v}^{ie}_{ma} - \Lambda^{eb}_{in}\hat{v}^{if}_{mb} - D^{fl}_{nc}v^{ce}_{ml} + \bar{D}^{ek}_{nd}v^{fd}_{km}\Bigg\} .
\end{aligned}
\tag{7.15}
$$

7.3 Perturbative triples for closed-shell CCSD

The perturbative triples equations for CCSD are given by

$$
\begin{aligned}
E_{[T]} &= \sum_{i\le j\le k} p(i,j,k)K^{abc}_{ijk}X^{ijk}_{abc} \\
p(i,j,k) &= \begin{bmatrix} 2 & i\neq j\neq k \\ 1 & i=j\oplus j=k \\ 0 & i=j=k \end{bmatrix}
\end{aligned}
\tag{7.16}
$$

X^{ijk}_{abc} and K^{abc}_{ijk} are calculated for the triangular set of indices $i\le j\le k$ (with $k = 1 : n_{occ}$),

$$
\begin{aligned}
K^{abc}_{ijk} = K^{ijk}_{abc} &= v^{dk}_{bc}T^{ij}_{ad} + v^{dk}_{ac}T^{ij}_{db} + v^{dj}_{cb}T^{ik}_{ad} + v^{dj}_{ab}T^{ik}_{dc} + v^{di}_{ca}T^{jk}_{bd} + v^{di}_{ba}T^{jk}_{dc} \\
&\quad - v^{jk}_{lc}T^{li}_{ba} - v^{ik}_{lc}T^{lj}_{ab} - v^{kj}_{lb}T^{li}_{ca} - v^{ij}_{lb}T^{lk}_{ac} - v^{ki}_{la}T^{lj}_{cb} - v^{ji}_{la}T^{lk}_{bc} \\
X^{ijk}_{abc} &= \frac{4K^{ijk}_{abc} - 2K^{ijk}_{acb} - 2K^{ijk}_{cba} - 2K^{ijk}_{bac} + K^{ijk}_{cab} + K^{ijk}_{bca}}{\epsilon_i + \epsilon_j + \epsilon_k - \epsilon_a - \epsilon_b - \epsilon_c}
\end{aligned}
\tag{7.17}
$$

The (T) correction contains additionally the following terms,

$$
\begin{aligned}
E_{(T)} = E_{[T]} + \sum_{i\le j\le k} p(i,j,k)\Big[& v^{bc}_{jk}X^{ijk}_{abc}T^{\dagger a}_i + v^{ac}_{ik}X^{ijk}_{abc}T^{\dagger b}_j + v^{ab}_{ij}X^{ijk}_{abc}T^{\dagger c}_k \\
& + T^{\dagger bc}_{jk}X^{ijk}_{abc}f^a_i + T^{\dagger ac}_{ik}X^{ijk}_{abc}f^b_j + T^{\dagger ab}_{ij}X^{ijk}_{abc}f^c_k\Big] .
\end{aligned}
\tag{7.18}
$$

In case of **ΛCCSD(T)** K^{abc}_{ijk} is different from K^{ijk}_{abc} and is calculated using the Lagrange multipliers,

$$\begin{aligned} K^{abc}_{ijk} &= v^{bc}_{dk}\bar{\Lambda}^{ad}_{ij} + v^{ac}_{dk}\bar{\Lambda}^{db}_{ij} + v^{cb}_{dj}\bar{\Lambda}^{ad}_{ik} + v^{ab}_{dj}\bar{\Lambda}^{dc}_{ik} + v^{ca}_{di}\bar{\Lambda}^{bd}_{jk} + v^{ba}_{di}\bar{\Lambda}^{dc}_{jk} \\ &- v^{lc}_{jk}\bar{\Lambda}^{ba}_{li} - v^{lc}_{ik}\bar{\Lambda}^{ab}_{lj} - v^{lb}_{kj}\bar{\Lambda}^{ca}_{li} - v^{lb}_{ij}\bar{\Lambda}^{ac}_{lk} - v^{la}_{ki}\bar{\Lambda}^{cb}_{lj} - v^{la}_{ji}\bar{\Lambda}^{bc}_{lk}, \end{aligned} \tag{7.19}$$

where $\bar{\Lambda}^{ab}_{ij}$ are the covariant Lagrange multipliers,

$$\bar{\Lambda}^{ab}_{ij} = \frac{2}{3}\Lambda^{ab}_{ij} + \frac{1}{3}\Lambda^{ba}_{ij}. \tag{7.20}$$

Additionally, the conjugate-transposed amplitudes in Eq. (7.18) are replaced by the covariant Lagrange multipliers $\bar{\Lambda}^{ab}_{ij}$ and $\bar{\Lambda}^{a}_{i} = \frac{1}{2}\Lambda^{a}_{i}$.

7.4 Open-shell CCSD/DSCD Lagrangian

The factorization of the open-shell CCSD/DCSD amplitude equations roughly follows the factorization of the closed-shell equations, Sec. 7.1. The open-shell CCSD and DCSD Lagrangian – i.e., spin dependent – is given by

$$\mathcal{L} = \mathcal{L}_\alpha + \mathcal{L}_\beta + \mathcal{L}_{\alpha\beta}, \tag{7.21}$$

$$\begin{aligned} \mathcal{L}_\alpha = \frac{1}{2}&\left[v^{cd}_{kl}T^{kl}_{cd} + \left(\hat{f}^c_k + f^c_k\right)T^k_c\right] + \frac{1}{4}\Lambda^{ab}_{ij}\left(\hat{v}^{ij}_{ab} - \hat{v}^{ji}_{ab}\right) + \frac{1}{4}\Lambda^{ab}_{ij}\left(\hat{v}^{ij}_{kl} + \frac{1}{2}v^{cd}_{kl}T^{ij}_{cd}\right)T^{kl}_{ab} \\ &+ \frac{1}{4}\Lambda^{ab}_{ij}\hat{v}^{cd}_{ab}T^{ij}_{cd} + \frac{1}{4}\Lambda^{ab}_{ij}\mathcal{S}\left(ab,ij\right)\left\{\hat{x}^c_a T^{ij}_{cb} - \hat{x}^i_k T^{kj}_{ab}\right\} \\ &+ \frac{1}{4}\Lambda^{ab}_{ij}\mathcal{A}\left(ab;ij\right)\left\{\left(\hat{v}^{id}_{al} - \hat{v}^{di}_{al} + \bar{x}^{id}_{al}\right)T^{lj}_{db} + \left(\hat{v}^{i\bar{d}}_{a\bar{l}} + x^{i\bar{d}}_{a\bar{l}}\right)T^{j\bar{l}}_{b\bar{d}}\right\} \\ &+ \Lambda^a_i\hat{v}^{cd}_{al}T^{il}_{cd} + \Lambda^a_i\hat{v}^{c\bar{d}}_{a\bar{l}}T^{i\bar{l}}_{c\bar{d}} - \Lambda^a_i\hat{v}^{ic}_{jk}T^{jk}_{ac} - \Lambda^a_i\hat{v}^{i\bar{c}}_{j\bar{k}}T^{j\bar{k}}_{a\bar{c}} \\ &+ \Lambda^a_i\hat{f}^i_a + \Lambda^a_i\hat{f}^b_j T^{ij}_{ab} + \Lambda^a_i\hat{f}^{\bar{b}}_{\bar{j}}T^{i\bar{j}}_{a\bar{b}}, \end{aligned} \tag{7.22}$$

or using the cckext factorization,

$$\begin{aligned} \mathcal{L}_\alpha = \frac{1}{2}&\left[v^{cd}_{kl}T^{kl}_{cd} + \left(\hat{f}^c_k + f^c_k\right)T^k_c\right] + \frac{1}{4}\Lambda^{ab}_{ij}\left(\hat{v}^{ij}_{kl} + \frac{1}{2}v^{cd}_{kl}T^{ij}_{cd}\right)T^{kl}_{ab} \\ &+ \frac{1}{4}\Lambda^{ab}_{ij}K^{ij}_{pq}\left(\delta^p_a - \delta^p_k T^k_a\right)\left(\delta^q_b - \delta^q_l T^l_b\right) + \frac{1}{4}\Lambda^{ab}_{ij}\mathcal{S}\left(ab,ij\right)\left\{\hat{x}^c_a T^{ij}_{cb} - \hat{x}^i_k T^{kj}_{ab}\right\} \\ &+ \frac{1}{4}\Lambda^{ab}_{ij}\mathcal{A}\left(ab;ij\right)\left\{\left(\hat{v}^{id}_{al} - \hat{v}^{di}_{al} + \bar{x}^{id}_{al}\right)T^{lj}_{db} + \left(\hat{v}^{i\bar{d}}_{a\bar{l}} + x^{i\bar{d}}_{a\bar{l}}\right)T^{j\bar{l}}_{b\bar{d}}\right\} \\ &+ \Lambda^a_i\left(K^{ij}_{pq}\delta^q_j + K^{i\bar{j}}_{p\bar{q}}\delta^{\bar{q}}_{\bar{j}}\right)\left(\delta^p_a - \delta^p_k T^k_a\right) - \Lambda^a_i\hat{v}^{ic}_{jk}T^{jk}_{ac} - \Lambda^a_i\hat{v}^{i\bar{c}}_{j\bar{k}}T^{j\bar{k}}_{a\bar{c}} \\ &+ \Lambda^a_i\hat{h}^i_a + \Lambda^a_i\hat{f}^b_j T^{ij}_{ab} + \Lambda^a_i\hat{f}^{\bar{b}}_{\bar{j}}T^{i\bar{j}}_{a\bar{b}}; \end{aligned} \tag{7.23}$$

$\mathcal{L}_\beta$ is obtained from $\mathcal{L}_\alpha$ by flipping the spins;

$$\begin{aligned} \mathcal{L}_{\alpha\beta} = v^{c\bar{d}}_{k\bar{l}}T^{k\bar{l}}_{c\bar{d}} &+ \Lambda^{a\bar{b}}_{i\bar{j}}\hat{v}^{i\bar{j}}_{a\bar{b}} + \Lambda^{a\bar{b}}_{i\bar{j}}\left(\hat{v}^{i\bar{j}}_{k\bar{l}} + v^{c\bar{d}}_{k\bar{l}}T^{i\bar{j}}_{c\bar{d}}\right)T^{k\bar{l}}_{a\bar{b}} + \Lambda^{a\bar{b}}_{i\bar{j}}\hat{v}^{c\bar{d}}_{a\bar{b}}T^{i\bar{j}}_{c\bar{d}} \\ &+ \Lambda^{a\bar{b}}_{i\bar{j}}\left\{\hat{x}^c_a T^{i\bar{j}}_{c\bar{b}} + \hat{x}^{\bar{d}}_{\bar{b}}T^{i\bar{j}}_{a\bar{d}} - \hat{x}^i_k T^{kj}_{ab} - \hat{x}^{\bar{j}}_{\bar{l}}T^{i\bar{l}}_{a\bar{b}}\right\} \\ &+ \Lambda^{a\bar{b}}_{i\bar{j}}\left\{\left(\hat{v}^{id}_{al} - \hat{v}^{di}_{al} + x^{id}_{al} + \bar{x}^{id}_{al}\right)T^{l\bar{j}}_{d\bar{b}} + \left(\hat{v}^{\bar{j}\bar{d}}_{\bar{b}\bar{l}} - \hat{v}^{\bar{d}\bar{j}}_{\bar{b}\bar{l}} + 2x^{\bar{j}\bar{d}}_{\bar{b}\bar{l}}\right)T^{i\bar{l}}_{a\bar{d}}\right. \\ &+ \left.\left(\hat{v}^{i\bar{d}}_{a\bar{l}} + v^{c\bar{d}}_{k\bar{l}}T^{ik}_{ac}\right)T^{\bar{l}\bar{j}}_{\bar{d}\bar{b}} + \hat{v}^{d\bar{j}}_{l\bar{b}}T^{il}_{ad} - \hat{v}^{c\bar{j}}_{a\bar{k}}T^{i\bar{k}}_{c\bar{b}} - \left(\hat{v}^{i\bar{d}}_{k\bar{b}} - v^{c\bar{d}}_{k\bar{l}}T^{i\bar{l}}_{c\bar{b}}\right)T^{k\bar{j}}_{a\bar{d}}\right\}, \end{aligned} \tag{7.24}$$

or using the cckext factorization,

$$\begin{aligned}
\mathcal{L}_{\alpha\beta} &= v^{c\bar{d}}_{k\bar{l}} T^{k\bar{l}}_{c\bar{d}} + \Lambda^{a\bar{b}}_{i\bar{j}} \left(\hat{v}^{i\bar{j}}_{k\bar{l}} + v^{c\bar{d}}_{k\bar{l}} T^{i\bar{j}}_{c\bar{d}} \right) T^{k\bar{l}}_{a\bar{b}} + \Lambda^{a\bar{b}}_{i\bar{j}} K^{i\bar{j}}_{p\bar{q}} \left(\delta^p_a - \delta^p_k T^k_a \right) \left(\delta^{\bar{q}}_{\bar{b}} - \delta^{\bar{q}}_{\bar{l}} T^{\bar{l}}_{\bar{b}} \right) \\
&+ \Lambda^{a\bar{b}}_{i\bar{j}} \left\{ \hat{x}^c_a T^{i\bar{j}}_{c\bar{b}} + \hat{x}^{\bar{d}}_{\bar{b}} T^{i\bar{j}}_{a\bar{d}} - \hat{x}^i_k T^{k\bar{j}}_{a\bar{b}} - \hat{x}^{\bar{j}}_{\bar{l}} T^{i\bar{l}}_{a\bar{b}} \right\} \\
&+ \Lambda^{a\bar{b}}_{i\bar{j}} \left\{ \left(\hat{v}^{id}_{al} - \hat{v}^{di}_{al} + x^{id}_{al} + \bar{x}^{id}_{al} \right) T^{l\bar{j}}_{d\bar{b}} + \left(\hat{v}^{\bar{j}\bar{d}}_{\bar{b}\bar{l}} - \hat{v}^{\bar{d}\bar{j}}_{\bar{b}\bar{l}} + 2x^{\bar{j}\bar{d}}_{\bar{b}\bar{l}} \right) T^{i\bar{l}}_{a\bar{d}} \right. \\
&\left. + \left(\hat{v}^{i\bar{d}}_{a\bar{l}} + v^{c\bar{d}}_{k\bar{l}} T^{ik}_{ac} \right) T^{\bar{l}\bar{j}}_{\bar{d}\bar{b}} + \hat{v}^{d\bar{j}}_{l\bar{b}} T^{il}_{ad} - \hat{v}^{c\bar{j}}_{a\bar{k}} T^{i\bar{k}}_{c\bar{b}} - \left(\hat{v}^{i\bar{d}}_{k\bar{b}} - v^{c\bar{d}}_{k\bar{l}} T^{i\bar{l}}_{c\bar{b}} \right) T^{k\bar{j}}_{a\bar{d}} \right\} .
\end{aligned} \tag{7.25}$$

The intermediate quantities are defined as follows,

$$\begin{aligned}
K^{ij}_{pq} &= v^{rs}_{pq} D^{ij}_{rs} \qquad\qquad K^{i\bar{j}}_{p\bar{q}} = v^{r\bar{s}}_{p\bar{q}} D^{i\bar{j}}_{r\bar{s}} \\
D^{ij}_{rs} &= \left(T^{ij}_{ab} + T^i_a T^j_b - T^i_b T^j_a \right) \delta^a_r \delta^b_s + \mathcal{A}\left(ij; rs\right) \delta^i_r T^j_b \delta^b_s + \delta^i_r \delta^j_s - \delta^i_s \delta^j_r \\
D^{i\bar{j}}_{r\bar{s}} &= \left(T^{i\bar{j}}_{a\bar{b}} + T^i_a T^{\bar{j}}_{\bar{b}} \right) \delta^a_r \delta^{\bar{b}}_{\bar{s}} + \delta^i_r T^{\bar{j}}_{\bar{b}} \delta^{\bar{b}}_{\bar{s}} + T^i_a \delta^a_s \delta^{\bar{j}}_{\bar{s}} + \delta^i_r \delta^{\bar{j}}_{\bar{s}} \\
x^{id}_{al} &= \frac{1}{2} T^{ik}_{ac} \left(v^{cd}_{kl} - v^{dc}_{kl} \right) \\
\bar{x}^{id}_{al} &= x^{id}_{al} + T^{i\bar{k}}_{a\bar{c}} v^{d\bar{c}}_{l\bar{k}} \\
x^{i\bar{d}}_{a\bar{l}} &= \frac{1}{2} T^{i\bar{k}}_{a\bar{c}} \left(v^{\bar{c}\bar{d}}_{\bar{k}\bar{l}} - v^{\bar{d}\bar{c}}_{\bar{k}\bar{l}} \right) \\
\hat{x}^i_k &= \hat{f}^i_k + 2\times \frac{1}{2} \left(v^{cd}_{kl} T^{il}_{cd} + v^{c\bar{d}}_{k\bar{l}} T^{i\bar{l}}_{c\bar{d}} \right) \\
\hat{x}^c_a &= \hat{f}^c_a - 2\times \frac{1}{2} \left(v^{cd}_{kl} T^{kl}_{ad} + v^{c\bar{d}}_{k\bar{l}} T^{k\bar{l}}_{a\bar{d}} \right)
\end{aligned} \tag{7.26}$$

7.4.1 Spin-restricted open-shell CCSD/DSCD

The spin-restricted versions `rccsd` and `rdcsd` are obtained through spin-projection of the residuals and amplitudes from the spin-dependent equations in each iteration. [8, 9]

In this section we use the following notation:

$$\begin{aligned}
{}^{\alpha\alpha}T^{ij}_{ab} &= T^{ij}_{ab}, \\
{}^{\beta\beta}T^{ij}_{ab} &= T^{\bar{i}\bar{j}}_{\bar{a}\bar{b}}, \\
{}^{\alpha\beta}T^{ij}_{ab} &= T^{i\bar{j}}_{a\bar{b}},
\end{aligned} \tag{7.27}$$

and the spin-projected amplitudes are denoted by a bar, e.g., ${}^{\alpha\beta}\bar{T}^{ij}_{ab}$. Moreover, the indices $i, j, \ldots$ run in the following part of the section over the closed-shell part of occupied orbitals, $a, b, \ldots$ over the (doubly) virtual orbitals, and $t, u, \ldots$ over the singly occupied (or singly-virtual) orbitals.

The "closed-shell" part of spin-projected $\alpha\beta$ amplitudes is given by

$${}^{\alpha\beta}\bar{T}^{ij}_{ab} = \frac{1}{6}({}^{\alpha\alpha}T^{ij}_{ab} + {}^{\beta\beta}T^{ij}_{ab} + 2\,{}^{\alpha\beta}T^{ij}_{ab} + {}^{\alpha\beta}T^{ij}_{ba} + 2\,{}^{\alpha\beta}T^{ji}_{ba} + {}^{\alpha\beta}T^{ji}_{ab}) \tag{7.28}$$

The "open-shell" part of spin-projected $\alpha\beta$ amplitudes is given by

$$\begin{aligned}
{}^{\alpha\beta}\bar{T}^{ij}_{at} &= \frac{1}{3}({}^{\beta\beta}T^{ij}_{at} + 2\,{}^{\alpha\beta}T^{ij}_{at} + {}^{\alpha\beta}T^{ji}_{at}) \\
{}^{\alpha\beta}\bar{T}^{tj}_{ab} &= \frac{1}{3}({}^{\alpha\alpha}T^{tj}_{ab} + 2\,{}^{\alpha\beta}T^{tj}_{ab} + {}^{\alpha\beta}T^{tj}_{ba}) \\
{}^{\alpha\beta}\bar{T}^{tj}_{au} &= {}^{\alpha\beta}T^{tj}_{au} + \frac{\delta^t_u}{2m_s + 2} \left({}^{\beta}T^j_a - {}^{\alpha}T^j_a - {}^{\alpha\beta}T^{vj}_{av} \right)
\end{aligned} \tag{7.29}$$

The projection corrections for the remaining amplitudes are defined in terms of the new spin-projected $\alpha\beta$ amplitudes as follows. For singles amplitudes,

$$\begin{aligned}
{}^{\alpha}\bar{T}^{i}_{a} &= \frac{1}{2}\left({}^{\alpha}T^{i}_{a} + {}^{\beta}T^{i}_{a} - {}^{\alpha\beta}\bar{T}^{vi}_{av}\right), \\
{}^{\beta}\bar{T}^{i}_{a} &= \frac{1}{2}\left({}^{\alpha}T^{i}_{a} + {}^{\beta}T^{i}_{a} + {}^{\alpha\beta}\bar{T}^{vi}_{av}\right), \\
{}^{\alpha}\bar{T}^{t}_{a} &= {}^{\alpha}T^{t}_{a} \qquad \text{and} \qquad {}^{\beta}\bar{T}^{i}_{t} = {}^{\beta}T^{i}_{t}.
\end{aligned} \tag{7.30}$$

For the $\alpha\alpha$ and $\beta\beta$ amplitudes,

$$\begin{aligned}
{}^{\sigma\sigma}\bar{T}^{ij}_{ab} &= {}^{\alpha\beta}\bar{T}^{ij}_{ab} - {}^{\alpha\beta}\bar{T}^{ij}_{ba}, \\
{}^{\alpha\alpha}\bar{T}^{tj}_{ab} &= {}^{\alpha\alpha}\bar{T}^{jt}_{ba} = {}^{\alpha\beta}\bar{T}^{tj}_{ab} - {}^{\alpha\beta}\bar{T}^{tj}_{ba}, \\
{}^{\beta\beta}\bar{T}^{ij}_{at} &= {}^{\beta\beta}\bar{T}^{ji}_{ta} = {}^{\alpha\beta}\bar{T}^{ij}_{at} - {}^{\alpha\beta}\bar{T}^{ji}_{at}, \\
{}^{\alpha\alpha}\bar{T}^{tu}_{ab} &= {}^{\alpha\alpha}T^{tu}_{ab} \qquad \text{and} \qquad {}^{\beta\beta}\bar{T}^{ij}_{tu} = {}^{\beta\beta}T^{ij}_{tu}.
\end{aligned} \tag{7.31}$$

7.5 Open-shell CCSD/DSCD Lagrangian multipliers equations

The Lagrange multipliers equations for the open-shell CCSD/DSCD Lagrangian can be obtained by taking the derivatives with respect to the amplitudes and setting them to zero.

$$\begin{aligned}
\frac{\partial\mathcal{L}_{\alpha}}{\partial T^{m}_{e}} = {} & \left(v^{ce}_{km} - v^{ec}_{km}\right) T^{k}_{c} + \frac{1}{2} v^{e\bar{c}}_{m\bar{k}} T^{\bar{k}}_{\bar{c}} + f^{e}_{m} - \Lambda^{eb}_{ij}\hat{v}^{ij}_{mb} + \Lambda^{ab}_{mj}\hat{v}^{ej}_{ab} + \frac{1}{4}\Lambda^{ab}_{mj}\hat{v}^{ej}_{kl}T^{kl}_{ab} \\
& + \frac{1}{4}\Lambda^{ab}_{im}\hat{v}^{ie}_{kl}T^{kl}_{ab} - \frac{1}{4}\Lambda^{eb}_{ij}\hat{v}^{cd}_{mb}T^{ij}_{cd} - \frac{1}{4}\Lambda^{ae}_{ij}\hat{v}^{cd}_{am}T^{ij}_{cd} - \frac{1}{2}\Lambda^{eb}_{ij}\hat{f}^{c}_{m}T^{ij}_{cb} - \frac{1}{2}\Lambda^{ab}_{mj}\hat{f}^{e}_{k}T^{kj}_{ab} \\
& + \frac{1}{2}\Lambda^{ab}_{ij}\left\{\left(\hat{v}^{ce}_{am} - \hat{v}^{ec}_{am}\right)T^{ij}_{cb} - \left(\hat{v}^{ie}_{km} - \hat{v}^{ie}_{mk}\right)T^{kj}_{ab}\right\} \\
& - \Lambda^{eb}_{ij}\hat{v}^{id}_{ml}T^{lj}_{db} + \Lambda^{ab}_{mj}\hat{v}^{ed}_{al}T^{lj}_{db} + \Lambda^{eb}_{ij}\hat{v}^{di}_{ml}T^{lj}_{db} \\
& - \Lambda^{ab}_{mj}\hat{v}^{de}_{al}T^{lj}_{db} - \Lambda^{eb}_{ij}\hat{v}^{i\bar{d}}_{m\bar{l}}T^{j\bar{l}}_{b\bar{d}} + \Lambda^{ab}_{mj}\hat{v}^{e\bar{d}}_{a\bar{l}}T^{j\bar{l}}_{b\bar{d}} \\
& - \Lambda^{e}_{i}v^{cd}_{ml}T^{il}_{cd} - \Lambda^{e}_{i}v^{c\bar{d}}_{m\bar{l}}T^{i\bar{l}}_{c\bar{d}} - \Lambda^{a}_{m}\hat{v}^{ec}_{jk}T^{jk}_{ac} - \Lambda^{a}_{m}\hat{v}^{e\bar{c}}_{j\bar{k}}T^{j\bar{k}}_{a\bar{c}} \\
& - \Lambda^{e}_{i}\hat{f}^{i}_{m} + \Lambda^{a}_{m}\hat{f}^{e}_{a} + \Lambda^{a}_{i}\left(\hat{v}^{ie}_{am} - \hat{v}^{ei}_{am}\right) + \Lambda^{a}_{i}\left(v^{be}_{jm} - v^{eb}_{jm}\right)T^{ij}_{ab} + \Lambda^{a}_{i}v^{e\bar{b}}_{m\bar{j}}T^{i\bar{j}}_{a\bar{b}}
\end{aligned} \tag{7.32}$$

$$\begin{aligned}
\frac{\partial\mathcal{L}_{\beta}}{\partial T^{m}_{e}} = {} & \frac{1}{2}v^{e\bar{c}}_{m\bar{k}}T^{\bar{k}}_{\bar{c}} + \frac{1}{2}\Lambda^{\bar{a}\bar{b}}_{\bar{i}\bar{j}}\left\{\hat{v}^{e\bar{c}}_{m\bar{a}}T^{\bar{i}\bar{j}}_{\bar{c}\bar{b}} - \hat{v}^{e\bar{i}}_{m\bar{k}}T^{\bar{k}\bar{j}}_{\bar{a}\bar{b}}\right\} \\
& + \Lambda^{\bar{a}}_{\bar{i}}\hat{v}^{e\bar{i}}_{m\bar{a}} + \Lambda^{\bar{a}}_{\bar{i}}v^{e\bar{b}}_{m\bar{j}}T^{\bar{i}\bar{j}}_{\bar{a}\bar{b}} + \Lambda^{\bar{a}}_{\bar{i}}\left(v^{eb}_{mj} - v^{be}_{mj}\right)T^{j\bar{i}}_{b\bar{a}},
\end{aligned} \tag{7.33}$$

$$\begin{aligned}
\frac{\partial\mathcal{L}_{\alpha\beta}}{\partial T^{m}_{e}} = {} & -\Lambda^{e\bar{b}}_{i\bar{j}}\hat{v}^{i\bar{j}}_{m\bar{b}} + \Lambda^{a\bar{b}}_{m\bar{j}}\hat{v}^{e\bar{j}}_{a\bar{b}} + \Lambda^{a\bar{b}}_{m\bar{j}}\hat{v}^{e\bar{j}}_{k\bar{l}}T^{k\bar{l}}_{a\bar{b}} - \Lambda^{e\bar{b}}_{i\bar{j}}\hat{v}^{c\bar{d}}_{m\bar{b}}T^{i\bar{j}}_{c\bar{d}} \\
& - \Lambda^{e\bar{b}}_{i\bar{j}}\hat{f}^{c}_{m}T^{i\bar{j}}_{c\bar{b}} - \Lambda^{a\bar{b}}_{m\bar{j}}\hat{f}^{e}_{k}T^{k\bar{j}}_{a\bar{b}} \\
& + \Lambda^{a\bar{b}}_{i\bar{j}}\left\{\left(\hat{v}^{ce}_{am} - \hat{v}^{ec}_{am}\right)T^{i\bar{j}}_{c\bar{b}} + \hat{v}^{e\bar{d}}_{m\bar{b}}T^{i\bar{j}}_{a\bar{d}} - \left(\hat{v}^{ie}_{km} - \hat{v}^{ie}_{mk}\right)T^{k\bar{j}}_{a\bar{b}} - \hat{v}^{e\bar{j}}_{m\bar{l}}T^{i\bar{l}}_{a\bar{b}}\right\} \\
& - \Lambda^{e\bar{b}}_{i\bar{j}}\hat{v}^{id}_{ml}T^{l\bar{j}}_{d\bar{b}} + \Lambda^{a\bar{b}}_{m\bar{j}}\hat{v}^{ed}_{al}T^{l\bar{j}}_{d\bar{b}} + \Lambda^{e\bar{b}}_{i\bar{j}}\hat{v}^{di}_{ml}T^{l\bar{j}}_{d\bar{b}} - \Lambda^{a\bar{b}}_{m\bar{j}}\hat{v}^{de}_{al}T^{l\bar{j}}_{d\bar{b}} \\
& - \Lambda^{e\bar{b}}_{i\bar{j}}\hat{v}^{i\bar{d}}_{m\bar{l}}T^{\bar{l}\bar{j}}_{\bar{d}\bar{b}} + \Lambda^{a\bar{b}}_{m\bar{j}}\hat{v}^{e\bar{d}}_{a\bar{l}}T^{\bar{l}\bar{j}}_{\bar{d}\bar{b}} + \Lambda^{e\bar{b}}_{i\bar{j}}\hat{v}^{c\bar{j}}_{m\bar{k}}T^{i\bar{k}}_{c\bar{b}} - \Lambda^{a\bar{b}}_{m\bar{j}}\hat{v}^{e\bar{d}}_{k\bar{b}}T^{k\bar{j}}_{a\bar{d}},
\end{aligned} \tag{7.34}$$

The corresponding equations for the derivatives with respect to the β amplitudes are obtained by flipping the spins.

Derivatives with respect to doubles amplitudes are given by

$$\begin{aligned}4\frac{\partial\mathcal{L}_\alpha}{\partial T^{mn}_{ef}} = \mathcal{A}(ef;mn)\Bigg[&\frac{1}{2}v^{ef}_{mn}+\frac{1}{4}\Lambda^{ef}_{ij}\left(\hat{v}^{ij}_{mn}+\frac{1}{2}v^{cd}_{mn}T^{ij}_{cd}\right)+\frac{1}{8}\Lambda^{ab}_{mn}v^{ef}_{kl}T^{kl}_{ab}\\&+\frac{1}{4}\Lambda^{ab}_{mn}\hat{v}^{ef}_{ab}+\frac{1}{2}\Lambda^{af}_{mn}\hat{x}^{e}_{a}-\frac{1}{2}\Lambda^{ef}_{in}\hat{x}^{i}_{m}-2\times\frac{1}{4}\Lambda^{eb}_{ij}v^{cf}_{mn}T^{ij}_{cb}-2\times\frac{1}{4}\Lambda^{ab}_{mj}v^{ef}_{kn}T^{kj}_{ab}\\&+\Lambda^{af}_{in}\left(\hat{v}^{ie}_{am}-\hat{v}^{ei}_{am}+\bar{x}^{ie}_{am}\right)+\frac{1}{2}\Lambda^{eb}_{mj}\left(v^{fd}_{nl}-v^{df}_{nl}\right)T^{lj}_{db}\\&+\Lambda^{a}_{m}\hat{v}^{ef}_{an}-\Lambda^{e}_{i}\hat{v}^{if}_{mn}+\Lambda^{e}_{m}\hat{f}^{f}_{n}\Bigg],\end{aligned}\tag{7.35}$$

$$\frac{\partial\mathcal{L}_\beta}{\partial T^{mn}_{ef}}=0,\tag{7.36}$$

$$\begin{aligned}4\frac{\partial\mathcal{L}_{\alpha\beta}}{\partial T^{mn}_{ef}} = \mathcal{A}(ef;mn)\Bigg[&-2\times\frac{1}{2}\Lambda^{e\bar{b}}_{i\bar{j}}v^{cf}_{mn}T^{i\bar{j}}_{c\bar{b}}-2\times\frac{1}{2}\Lambda^{a\bar{b}}_{m\bar{j}}v^{ef}_{kn}T^{k\bar{j}}_{a\bar{b}}\\&+\Lambda^{e\bar{b}}_{m\bar{j}}\left(v^{fd}_{nl}-v^{df}_{nl}\right)T^{l\bar{j}}_{d\bar{b}}+\Lambda^{e\bar{b}}_{m\bar{j}}v^{f\bar{d}}_{n\bar{l}}T^{\bar{l}\bar{j}}_{\bar{d}\bar{b}}+\Lambda^{e\bar{b}}_{m\bar{j}}\hat{v}^{f\bar{j}}_{n\bar{b}}\Bigg],\end{aligned}\tag{7.37}$$

$$\begin{aligned}\frac{\partial\mathcal{L}_\alpha}{\partial T^{m\bar{n}}_{e\bar{f}}} = &-2\times\frac{1}{4}\Lambda^{eb}_{ij}v^{c\bar{f}}_{m\bar{n}}T^{ij}_{cb}-2\times\frac{1}{4}\Lambda^{ab}_{mj}v^{e\bar{f}}_{k\bar{n}}T^{kj}_{ab}\\&+\Lambda^{eb}_{mj}v^{d\bar{f}}_{l\bar{n}}T^{lj}_{db}+\Lambda^{ae}_{im}\left(\hat{v}^{i\bar{f}}_{a\bar{n}}+x^{i\bar{f}}_{a\bar{n}}\right)+\frac{1}{2}\Lambda^{eb}_{mj}\left(v^{\bar{f}\bar{d}}_{\bar{n}\bar{l}}-v^{\bar{d}\bar{f}}_{\bar{n}\bar{l}}\right)T^{j\bar{l}}_{b\bar{d}}\\&+\Lambda^{a}_{m}\hat{v}^{e\bar{f}}_{a\bar{n}}-\Lambda^{e}_{i}\hat{v}^{i\bar{f}}_{m\bar{n}}+\Lambda^{e}_{m}\hat{f}^{\bar{f}}_{\bar{n}},\end{aligned}\tag{7.38}$$

$$\begin{aligned}\frac{\partial\mathcal{L}_\beta}{\partial T^{m\bar{n}}_{e\bar{f}}} = &-2\times\frac{1}{4}\Lambda^{\bar{f}\bar{b}}_{\bar{i}\bar{j}}v^{e\bar{c}}_{m\bar{n}}T^{\bar{i}\bar{j}}_{\bar{c}\bar{b}}-2\times\frac{1}{4}\Lambda^{\bar{a}\bar{b}}_{\bar{n}\bar{j}}v^{e\bar{f}}_{m\bar{k}}T^{\bar{k}\bar{j}}_{\bar{a}\bar{b}}\\&+\Lambda^{\bar{f}\bar{b}}_{\bar{n}\bar{j}}v^{e\bar{d}}_{m\bar{l}}T^{\bar{l}\bar{j}}_{\bar{d}\bar{b}}+\Lambda^{\bar{a}\bar{f}}_{\bar{i}\bar{n}}\left(\hat{v}^{e\bar{i}}_{m\bar{a}}+x^{\bar{i}e}_{\bar{a}m}\right)+\frac{1}{2}\Lambda^{\bar{f}\bar{b}}_{\bar{n}\bar{j}}\left(v^{ed}_{ml}-v^{de}_{ml}\right)T^{l\bar{j}}_{d\bar{b}}\\&+\Lambda^{\bar{a}}_{\bar{n}}\hat{v}^{\bar{f}e}_{\bar{a}m}-\Lambda^{\bar{f}}_{\bar{i}}\hat{v}^{\bar{i}e}_{\bar{n}m}+\Lambda^{\bar{f}}_{\bar{n}}\hat{f}^{e}_{m},\end{aligned}\tag{7.39}$$

$$\begin{aligned}\frac{\partial\mathcal{L}_{\alpha\beta}}{\partial T^{m\bar{n}}_{e\bar{f}}} = &v^{e\bar{f}}_{m\bar{n}}+\Lambda^{e\bar{f}}_{i\bar{j}}\left(\hat{v}^{i\bar{j}}_{m\bar{n}}+v^{c\bar{d}}_{m\bar{n}}T^{i\bar{j}}_{c\bar{d}}\right)+\Lambda^{a\bar{b}}_{m\bar{n}}v^{e\bar{f}}_{k\bar{l}}T^{k\bar{l}}_{a\bar{b}}+\Lambda^{a\bar{b}}_{m\bar{n}}\hat{v}^{e\bar{f}}_{a\bar{b}}\\&+\Lambda^{a\bar{f}}_{m\bar{n}}\hat{x}^{e}_{a}-2\times\frac{1}{2}\Lambda^{e\bar{b}}_{i\bar{j}}v^{c\bar{f}}_{m\bar{n}}T^{i\bar{j}}_{c\bar{b}}+\Lambda^{e\bar{b}}_{m\bar{n}}\hat{x}^{\bar{f}}_{\bar{b}}-2\times\frac{1}{2}\Lambda^{a\bar{f}}_{i\bar{j}}v^{e\bar{d}}_{m\bar{n}}T^{i\bar{j}}_{a\bar{d}}\\&-\Lambda^{e\bar{f}}_{i\bar{n}}\hat{x}^{i}_{m}-2\times\frac{1}{2}\Lambda^{a\bar{b}}_{m\bar{j}}v^{e\bar{f}}_{k\bar{n}}T^{k\bar{j}}_{a\bar{b}}-\Lambda^{e\bar{f}}_{m\bar{j}}\hat{x}^{\bar{j}}_{\bar{n}}-2\times\frac{1}{2}\Lambda^{a\bar{b}}_{i\bar{n}}v^{e\bar{f}}_{m\bar{l}}T^{i\bar{l}}_{a\bar{b}}\\&+\Lambda^{a\bar{f}}_{i\bar{n}}\left(\hat{v}^{ie}_{am}-\hat{v}^{ei}_{am}+x^{ie}_{am}+\bar{x}^{ie}_{am}\right)+\Lambda^{e\bar{b}}_{m\bar{j}}v^{d\bar{f}}_{l\bar{n}}T^{l\bar{j}}_{d\bar{b}}+\Lambda^{e\bar{b}}_{m\bar{j}}\left(\hat{v}^{\bar{j}\bar{f}}_{\bar{b}\bar{n}}-\hat{v}^{\bar{f}\bar{j}}_{\bar{b}\bar{n}}+2x^{\bar{j}\bar{f}}_{\bar{b}\bar{n}}\right)\\&-\Lambda^{a\bar{f}}_{m\bar{j}}\hat{v}^{e\bar{j}}_{a\bar{n}}-\Lambda^{e\bar{b}}_{i\bar{n}}\left(\hat{v}^{i\bar{f}}_{m\bar{b}}-v^{c\bar{f}}_{m\bar{l}}T^{i\bar{l}}_{c\bar{b}}\right)+\Lambda^{a\bar{f}}_{m\bar{j}}v^{ed}_{kn}T^{k\bar{j}}_{a\bar{d}},\end{aligned}\tag{7.40}$$

and the derivatives with respect to the $\beta\beta$ amplitudes are obtained by flipping the spins.

The one-body reduced density matrix (without singles contributions) is given by

$$\begin{aligned}D^{j}_{i} &= -\frac{1}{2}\Lambda^{cd}_{ik}T^{jk}_{cd}-\Lambda^{c\bar{d}}_{i\bar{k}}T^{j\bar{k}}_{c\bar{d}}\\D^{b}_{a} &= \frac{1}{2}\Lambda^{bc}_{kl}T^{kl}_{ac}+\Lambda^{b\bar{c}}_{k\bar{l}}T^{k\bar{l}}_{a\bar{c}}\\D^{a}_{i} &= \Lambda^{a}_{i}\\D^{i}_{a} &= \Lambda^{c}_{k}T^{ik}_{ac}+\Lambda^{\bar{c}}_{\bar{k}}T^{i\bar{k}}_{a\bar{c}}\end{aligned}\tag{7.41}$$

and the β 1RDM is obtained by flipping the spins.

The full (dressed) one-body reduced density matrix is given by

$$\begin{aligned} \hat{D}^j_i &= D^j_i - D^c_i T^j_c, \\ \hat{D}^b_a &= D^b_a + D^b_k T^k_a, \\ \hat{D}^a_i &= D^a_i, \\ \hat{D}^i_a &= D^i_a + T^i_a - D^c_a T^i_c + \hat{D}^i_k T^k_a. \end{aligned} \tag{7.42}$$

Additionally, we define intermediates related to the two-body reduced density matrix,

$$\begin{aligned} D^{kl}_{ij} &= \frac{1}{2}\Lambda^{cd}_{ij} T^{kl}_{cd} \\ D^{k\bar{l}}_{i\bar{j}} &= \Lambda^{c\bar{d}}_{i\bar{j}} T^{k\bar{l}}_{c\bar{d}} \\ D^{aj}_{ib} &= \Lambda^{ac}_{ik} T^{jk}_{bc} + \Lambda^{a\bar{c}}_{i\bar{k}} T^{j\bar{k}}_{b\bar{c}} \\ \bar{D}^{a\bar{j}}_{i\bar{b}} &= \Lambda^{ac}_{ik} T^{k\bar{j}}_{c\bar{b}} + \Lambda^{a\bar{c}}_{i\bar{k}} T^{\bar{j}\bar{k}}_{\bar{b}\bar{c}} \end{aligned} \tag{7.43}$$

and doubles-dressed Fock matrix,

$$\begin{aligned} x^i_k &= v^{cd}_{kl} T^{il}_{cd} + v^{c\bar{d}}_{k\bar{l}} T^{i\bar{l}}_{c\bar{d}} \\ x^c_a &= v^{cd}_{kl} T^{kl}_{ad} + v^{c\bar{d}}_{k\bar{l}} T^{k\bar{l}}_{a\bar{d}}. \end{aligned} \tag{7.44}$$

The intermediates for the cckext factorization are given by

$$\begin{aligned} K^{rs}_{mn} &= \hat{\Lambda}^{pq}_{mn} v^{rs}_{pq} \\ \hat{\Lambda}^{pq}_{mn} &= \Lambda^{ab}_{mn} \delta^p_a \delta^q_b - \Lambda^{ab}_{mn} T^i_a \delta^p_i \delta^q_b - \Lambda^{ab}_{mn} \delta^p_a T^j_b \delta^q_j + \Lambda^{ab}_{mn} T^i_a T^j_b \delta^p_i \delta^q_j. \end{aligned} \tag{7.45}$$

$K^{r\bar{s}}_{m\bar{n}}$ and $K^{\bar{r}\bar{s}}_{\bar{m}\bar{n}}$ are obtained by flipping the spins.

Finally, we define useful intermediates which can be precalculated and reused in the equations,

$$\begin{aligned} \hat{y}^{ie}_{am} &= \hat{v}^{ie}_{am} - \hat{v}^{ei}_{am} + \bar{x}^{ie}_{am} + x^{ie}_{am} \\ \hat{y}^{\bar{j}f}_{\bar{b}n} &= v^{f\bar{d}}_{n\bar{l}} T^{l\bar{j}}_{d\bar{b}} + \hat{v}^{f\bar{j}}_{n\bar{b}} + 2x^{\bar{j}f}_{\bar{b}n} \end{aligned} \tag{7.46}$$

With these intermediates the equations for the α Lagrange multipliers are given by

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial T^m_e} = {} & \left(v^{qe}_{pm} - v^{eq}_{pm}\right) \hat{D}^p_q + v^{e\bar{q}}_{m\bar{p}} \hat{D}^{\bar{p}}_{\bar{q}} + f^e_m - \Lambda^{eb}_{ij} \hat{v}^{ij}_{mb} - \Lambda^{e\bar{b}}_{i\bar{j}} \hat{v}^{i\bar{j}}_{m\bar{b}} \\ & + K^{rs}_{mj} \delta^e_r \left(\delta^j_s + \delta^b_s T^j_b\right) + K^{r\bar{s}}_{m\bar{j}} \delta^e_r \left(\delta^{\bar{j}}_{\bar{s}} + \delta^{\bar{b}}_{\bar{s}} T^{\bar{j}}_{\bar{b}}\right) + D^{kl}_{mj} \hat{v}^{ej}_{kl} + D^{k\bar{l}}_{m\bar{j}} \hat{v}^{e\bar{j}}_{k\bar{l}} \\ & - \frac{1}{2}\Lambda^{eb}_{ij} \left(\hat{v}^{cd}_{mb} T^{ij}_{cd}\right) - \Lambda^{e\bar{b}}_{i\bar{j}} \left(\hat{v}^{c\bar{d}}_{m\bar{b}} T^{i\bar{j}}_{c\bar{d}}\right) - D^e_c \hat{f}^c_m + D^k_m \hat{f}^e_k \\ & + D^{el}_{id} \left(\hat{v}^{di}_{ml} - \hat{v}^{di}_{lm}\right) + D^{al}_{md} \left(\hat{v}^{ed}_{al} - \hat{v}^{de}_{al}\right) - \bar{D}^{e\bar{l}}_{i\bar{d}} \hat{v}^{i\bar{d}}_{m\bar{l}} + \bar{D}^{a\bar{l}}_{m\bar{d}} \hat{v}^{e\bar{d}}_{a\bar{l}} \\ & + \Lambda^{e\bar{b}}_{i\bar{j}} \hat{v}^{c\bar{j}}_{m\bar{k}} T^{i\bar{k}}_{c\bar{b}} - \Lambda^{a\bar{b}}_{m\bar{j}} \hat{v}^{e\bar{d}}_{k\bar{b}} T^{k\bar{j}}_{a\bar{d}} \\ & - \Lambda^e_i \hat{f}^i_m + \Lambda^a_m \hat{f}^e_a - \Lambda^e_i x^i_m - \Lambda^a_m x^e_a \end{aligned} \tag{7.47}$$

$$
\begin{aligned}
4\frac{\partial \mathcal{L}}{\partial T^{mn}_{ef}} &= v^{ef}_{mn} - v^{ef}_{nm} + \Lambda^{ef}_{ij}\left(\hat{v}^{ij}_{mn} + \frac{1}{2}v^{cd}_{mn}T^{ij}_{cd}\right) + D^{kl}_{mn}v^{ef}_{kl} + K^{rs}_{mn}\delta^e_r\delta^f_s \\
&+ \mathcal{S}\left(ef, mn\right)\left\{\Lambda^{af}_{mn}\hat{x}^e_a - \Lambda^{ef}_{in}\hat{x}^i_m\right\} \\
&+ \mathcal{A}\left(ef; mn\right)\left\{2\times\frac{1}{2}D^k_m v^{ef}_{kn} - 2\times\frac{1}{2}D^e_c v^{cf}_{mn} + \Lambda^{af}_{in}\hat{y}^{ie}_{am} + \Lambda^{e\bar{b}}_{m\bar{j}}\hat{y}^{\bar{j}f}_{\bar{b}n}\right. \\
&\left.+\Lambda^a_m\hat{v}^{ef}_{an} - \Lambda^e_i\hat{v}^{if}_{mn} + \Lambda^e_m\hat{f}^f_n\right\}
\end{aligned}
\tag{7.48}
$$

The equations for the β Lagrange multipliers are obtained by flipping the spins. The equations for the $\alpha\beta$ Lagrange multipliers are given by

$$
\begin{aligned}
\frac{\partial \mathcal{L}}{\partial T^{m\bar{n}}_{e\bar{f}}} &= v^{e\bar{f}}_{m\bar{n}} + \Lambda^{e\bar{f}}_{i\bar{j}}\left(\hat{v}^{i\bar{j}}_{m\bar{n}} + v^{c\bar{d}}_{m\bar{n}}T^{i\bar{j}}_{c\bar{d}}\right) + D^{k\bar{l}}_{m\bar{n}}v^{e\bar{f}}_{k\bar{l}} + K^{r\bar{s}}_{m\bar{n}}\delta^e_r\delta^{\bar{f}}_{\bar{s}} \\
&+ \Lambda^{a\bar{f}}_{m\bar{n}}\hat{x}^e_a + \Lambda^{e\bar{b}}_{m\bar{n}}\hat{x}^{\bar{f}}_{\bar{b}} - \Lambda^{e\bar{f}}_{i\bar{n}}\hat{x}^i_m - \Lambda^{e\bar{f}}_{m\bar{j}}\hat{x}^{\bar{j}}_{\bar{n}} \\
&+ 2\times\frac{1}{2}\left(D^k_m v^{e\bar{f}}_{k\bar{n}} + D^{\bar{k}}_{\bar{n}}v^{e\bar{f}}_{m\bar{k}} - D^e_c v^{c\bar{f}}_{m\bar{n}} - D^{\bar{f}}_{\bar{c}}v^{e\bar{c}}_{m\bar{n}}\right) \\
&+ \Lambda^{a\bar{f}}_{i\bar{n}}\hat{y}^{ie}_{am} + \Lambda^{e\bar{b}}_{m\bar{j}}\hat{y}^{\bar{j}\bar{f}}_{\bar{b}\bar{n}} + \Lambda^{ae}_{im}\hat{y}^{i\bar{f}}_{a\bar{n}} + \Lambda^{\bar{a}\bar{f}}_{\bar{i}\bar{n}}\hat{y}^{\bar{i}e}_{\bar{a}m} \\
&- \Lambda^{a\bar{f}}_{m\bar{j}}\left(\hat{v}^{e\bar{j}}_{a\bar{n}} - v^{e\bar{d}}_{k\bar{n}}T^{k\bar{j}}_{a\bar{d}}\right) - \Lambda^{e\bar{b}}_{i\bar{n}}\left(\hat{v}^{i\bar{f}}_{m\bar{b}} - v^{c\bar{f}}_{m\bar{l}}T^{i\bar{l}}_{c\bar{b}}\right) \\
&+ \Lambda^a_m\hat{v}^{e\bar{f}}_{a\bar{n}} - \Lambda^e_i\hat{v}^{i\bar{f}}_{m\bar{n}} + \Lambda^e_m\hat{f}^{\bar{f}}_{\bar{n}} + \Lambda^{\bar{a}}_{\bar{n}}\hat{v}^{\bar{f}e}_{\bar{a}m} - \Lambda^{\bar{f}}_{\bar{i}}\hat{v}^{\bar{i}e}_{\bar{n}m} + \Lambda^{\bar{f}}_{\bar{n}}\hat{f}^e_m.
\end{aligned}
\tag{7.49}
$$

7.6 Perturbative triples for unrestricted CCSD

The perturbative triples equations for unrestricted CCSD are given by

$$
E_{[T]} = \frac{1}{6}\sum_{i<j<k} K^{abc}_{ijk}T^{ijk}_{abc} + \frac{1}{6}\sum_{\bar{i}<\bar{j}<\bar{k}} K^{\bar{a}\bar{b}\bar{c}}_{\bar{i}\bar{j}\bar{k}}T^{\bar{i}\bar{j}\bar{k}}_{\bar{a}\bar{b}\bar{c}} + \frac{1}{2}\sum_{i<j;\bar{k}} K^{ab\bar{c}}_{ij\bar{k}}X^{ij\bar{k}}_{ab\bar{c}} + \frac{1}{2}\sum_{\bar{i}<\bar{j};k} K^{\bar{a}\bar{b}c}_{\bar{i}\bar{j}k}X^{\bar{i}\bar{j}k}_{\bar{a}\bar{b}c}.
\tag{7.50}
$$

K^{abc}_{ijk} and T^{ijk}_{abc} (and the all-β-counterparts) are calculated for a triangular set of indices $i<j<k$ (with $k = 3 : n_\alpha$),

$$
\begin{aligned}
K^{abc}_{ijk} &= K^{ijk}_{abc} = \mathcal{A}\left(abc\right)\left\{v^{dk}_{bc}T^{ij}_{ad} + v^{dj}_{cb}T^{ik}_{ad} + v^{di}_{ba}T^{jk}_{dc} - \frac{1}{2}\left(\bar{v}^{kj}_{cl}T^{il}_{ab} + \bar{v}^{ki}_{cl}T^{lj}_{ab} + \bar{v}^{ij}_{al}T^{kl}_{cb}\right)\right\}, \\
\bar{v}^{ij}_{al} &= v^{ij}_{al} - v^{ji}_{al}, \\
T^{ijk}_{abc} &= \frac{K^{ijk}_{abc}}{\epsilon_i + \epsilon_j + \epsilon_k - \epsilon_a - \epsilon_b - \epsilon_c}
\end{aligned}
\tag{7.51}
$$

$K^{ab\bar{c}}_{ij\bar{k}}$ and $T^{ij\bar{k}}_{ab\bar{c}}$ (and the spin-flipped counterparts) are calculated for a triangular set of first two indices $i<j$ (with $j = 2 : n_\alpha$),

$$
\begin{aligned}
K^{ab\bar{c}}_{ij\bar{k}} &= K^{ij\bar{k}}_{ab\bar{c}} = \mathcal{A}\left(ab\right)\left\{v^{d\bar{k}}_{b\bar{c}}T^{ij}_{ad} + v^{di}_{ba}T^{j\bar{k}}_{d\bar{c}} + v^{dj}_{ab}T^{i\bar{k}}_{d\bar{c}} + v^{i\bar{d}}_{a\bar{c}}T^{j\bar{k}}_{b\bar{d}} + v^{j\bar{d}}_{b\bar{c}}T^{i\bar{k}}_{a\bar{d}} - \bar{v}^{ij}_{al}T^{l\bar{k}}_{b\bar{c}}\right. \\
&\left.-v^{j\bar{k}}_{b\bar{l}}T^{i\bar{l}}_{a\bar{c}} - v^{i\bar{k}}_{a\bar{l}}T^{j\bar{l}}_{b\bar{c}}\right\} - v^{j\bar{k}}_{l\bar{c}}T^{il}_{ab} - v^{i\bar{k}}_{l\bar{c}}T^{lj}_{ab}, \\
T^{ij\bar{k}}_{ab\bar{c}} &= \frac{K^{ij\bar{k}}_{ab\bar{c}}}{\epsilon_i + \epsilon_j + \epsilon_{\bar{k}} - \epsilon_a - \epsilon_b - \epsilon_{\bar{c}}}
\end{aligned}
\tag{7.52}
$$

The (T) correction contains additionally the following terms,

$$\begin{aligned}
E_{(T)} = E_{[T]} + \sum_{i<j<k} &\left[v^{bc}_{jk} T^{ijk}_{abc} T^{\dagger a}_{i} + v^{ac}_{ik} T^{ijk}_{abc} T^{\dagger b}_{j} + v^{ab}_{ij} T^{ijk}_{abc} T^{\dagger c}_{k} \right. \\
&\left. + \frac{1}{2} \left(T^{\dagger bc}_{jk} T^{ijk}_{abc} f^{a}_{i} + T^{\dagger ac}_{ik} T^{ijk}_{abc} f^{b}_{j} + T^{\dagger ab}_{ij} T^{ijk}_{abc} f^{c}_{k} \right) \right] \\
&+ \sum_{i<j;\bar{k}} \left[v^{b\bar{c}}_{j\bar{k}} T^{ij\bar{k}}_{ab\bar{c}} T^{\dagger a}_{i} + v^{a\bar{c}}_{i\bar{k}} T^{ij\bar{k}}_{ab\bar{c}} T^{\dagger b}_{j} + v^{ab}_{ij} T^{ij\bar{k}}_{ab\bar{c}} T^{\dagger \bar{c}}_{\bar{k}} \right. \\
&\left. + T^{\dagger b\bar{c}}_{j\bar{k}} T^{ij\bar{k}}_{ab\bar{c}} f^{a}_{i} + T^{\dagger a\bar{c}}_{i\bar{k}} T^{ij\bar{k}}_{ab\bar{c}} f^{b}_{j} + \frac{1}{2} T^{\dagger ab}_{ij} T^{ij\bar{k}}_{ab\bar{c}} f^{\bar{c}}_{\bar{k}} \right] \\
&+ \text{spin-flipped terms.}
\end{aligned} \tag{7.53}$$

In case of **ΛUCCSD(T)**, K^{abc}_{ijk} and T^{ijk}_{abc} etc are different from K^{abc}_{ijk} and T^{ijk}_{abc} and can be calculated by replacing amplitudes with Lagrange multipliers (and integrals with transpose integrals) in the above equations,

$$\begin{aligned}
K^{abc}_{ijk} &= \mathcal{A}\left(abc\right) \left\{ v^{bc}_{dk} \Lambda^{ad}_{ij} + v^{cb}_{dj} \Lambda^{ad}_{ik} + v^{ba}_{di} \Lambda^{dc}_{jk} - \frac{1}{2} \left(\bar{v}^{cl}_{kj} \Lambda^{ab}_{il} + \bar{v}^{cl}_{ki} \Lambda^{ab}_{lj} + \bar{v}^{al}_{ij} \Lambda^{cb}_{kl} \right) \right\}, \\
K^{ab\bar{c}}_{ij\bar{k}} &= \mathcal{A}\left(ab\right) \left\{ v^{b\bar{c}}_{d\bar{k}} \Lambda^{ad}_{ij} + v^{ba}_{di} \Lambda^{d\bar{c}}_{j\bar{k}} + v^{ab}_{dj} \Lambda^{d\bar{c}}_{i\bar{k}} + v^{a\bar{c}}_{i\bar{d}} \Lambda^{b\bar{d}}_{j\bar{k}} + v^{b\bar{c}}_{j\bar{d}} \Lambda^{a\bar{d}}_{i\bar{k}} - \bar{v}^{al}_{ij} \Lambda^{b\bar{c}}_{l\bar{k}} \right. \\
&\quad \left. - v^{b\bar{l}}_{j\bar{k}} \Lambda^{a\bar{c}}_{i\bar{l}} - v^{a\bar{l}}_{i\bar{k}} \Lambda^{b\bar{c}}_{j\bar{l}} \right\} - v^{l\bar{c}}_{j\bar{k}} \Lambda^{ab}_{il} - v^{l\bar{c}}_{i\bar{k}} \Lambda^{ab}_{lj},
\end{aligned} \tag{7.54}$$

and the conjugate-transpose of the amplitudes in Eq. (7.53) are replaced with the Lagrange multipliers.

7.7 AO-driven coupled cluster

In the `cckext` factorization of Sec. 7.1 the only quantity that requires integrals with two general indices in the bra is K^{ij}_{pq}, Eq. (7.5); every other integral block appearing in the amplitude, Λ, perturbative-triples and EOM equations carries at least one occupied index. This makes it possible to evaluate the equations *directly* from the AO integrals $v^{\rho\sigma}_{\mu\nu}$, without ever constructing the MO integrals, and thus without the $\mathcal{O}(N^5)$ four-index transformation and without storing an $\mathcal{O}(N^4)$ MO integral set. Below μ, ν, ρ, σ denote AO indices, n the number of AOs and o the number of correlated occupied orbitals.

7.7.1 Singles dressing as a coefficient transformation

Only lower virtual and upper occupied indices are dressed (Sec. 7.1), and the dressing is linear in each index separately. It can therefore be absorbed entirely into two sets of coefficients, one for the lower (bra) and one for the upper (ket) indices,

$$L^{\mu}_{p} = \begin{cases} C^{\mu}_{i}, & p = i \\ C^{\mu}_{a} - C^{\mu}_{k} T^{k}_{a}, & p = a \end{cases} \qquad R^{p}_{\mu} = \begin{cases} C^{i}_{\mu} + C^{c}_{\mu} T^{i}_{c}, & p = i \\ C^{a}_{\mu}, & p = a, \end{cases} \tag{7.55}$$

so that every dressed integral is an *undressed* AO integral transformed with L on the lower and R on the upper indices,

$$\hat{v}^{rs}_{pq} = L^{\mu}_{p} L^{\nu}_{q}\, v^{\rho\sigma}_{\mu\nu}\, R^{r}_{\rho} R^{s}_{\sigma}. \tag{7.56}$$

The two sets are biorthogonal with respect to the AO overlap S_μ^ν,

$$L_p^\mu\, S_\mu^\nu\, R_\nu^q = \delta_p^q, \tag{7.57}$$

which follows from the orthonormality of the MOs, $C_p^\mu S_\mu^\nu C_\nu^q = \delta_p^q$, together with the fact that the occupied–virtual blocks of Eq. (7.55) cancel: $-T_a^k\delta_k^j$ from L against $+\delta_a^c T_c^j$ from R. Eq. (7.57) is the statement that the dressing is a similarity transformation. Consequently the singles never enter as a separate dressing step: each iteration simply rebuilds Eq. (7.55) and re-contracts the AO integrals.

7.7.2 Half-transformed integrals

The lower occupied coefficients in Eq. (7.55) are *undressed*, $L_i^\mu = C_i^\mu$. The half-transformed integrals

$$v_{i\nu}^{\rho\sigma} = C_i^\mu\, v_{\mu\nu}^{\rho\sigma} \tag{7.58}$$

are therefore independent of T_1 and are constructed *once* per set of orbitals – by the out-of-core gather from the ± store described in Sec. 2.4, which keeps them pair-packed in both ket orders. Every block with at least one occupied index is then obtained from Eq. (7.58) by matrix multiplications alone, at a cost of $\mathcal{O}(o\,n^3)$ per block; the per-iteration data volume drops accordingly, from the $\approx n^4/4$ elements of the ± triangle to the $\approx o\,n^3$ elements of the half-transformed store, and the ± store itself is streamed per iteration only by the four-external term (Sec. 7.7.4). Blocks whose only occupied index sits on an upper (ket) slot are reached through

$$v_{pq}^{rs} = \left(v_{rs}^{pq}\right)^*, \tag{7.59}$$

which for real integrals is the plain bra–ket exchange.

7.7.3 One-pass occupied-early dressing

The T$_1$-dressed integral blocks that the CCSD/DCSD residual reads (all carrying at least one occupied index) are assembled once per iteration in a *single* pass over the half-transformed store: for every ket column σ the slabs

$$h_{i\nu}^{(\rho\sigma)} = C_i^\mu\, v_{\mu\nu}^{\rho\sigma}, \qquad \bar{h}_{i\mu}^{(\rho\sigma)} = C_i^\nu\, v_{\mu\nu}^{\rho\sigma} \tag{7.60}$$

are *read* – both roles from the same file, by the exchange symmetry Eq. (2.12) – and the closed-shell sweep accumulates the three occupied-early intermediates

$$v_{ij}^{\rho\sigma} = h_{i\nu}^{(\rho\sigma)} C_j^\nu, \qquad v_{i\nu}^{j\sigma} = h_{i\nu}^{(\rho\sigma)}\, R_\rho^j, \qquad v_{\mu i}^{j\sigma} = \bar{h}_{i\mu}^{(\rho\sigma)}\, R_\rho^j, \tag{7.61}$$

with the dressed occupied-ket coefficients R_ρ^j of Eq. (7.55); the open-shell sweep carries two coefficient sets and five such intermediates. Every dressed block of the residual (and the dressed Fock matrix) then follows from Eq. (7.61) by contracting the remaining AO indices with L and R, and the ket contractions over ρ are batched into level-3 BLAS products over accumulated slabs rather than performed as per-slab rank-one updates. Each element of the half-transformed store is read exactly once per iteration.

When the half-transformed store has not been built (a single-shot doubles-only calculation, or its per-macro-iteration rebuild in orbital-optimized methods), the same intermediates are formed in one pass over the ± triangle instead, with the band half-transforms Eq. (7.60) computed on the fly from each reconstructed slab – the identical sweep at $\approx n^4/4$ streamed elements.

7.7.4 Four-external term from the ± supermatrices

With the ± supermatrices of Sec. 2.4 both index pairs of the AO integrals are packed triangularly and only the lower block triangle is stored.

With the effective pair density of Eq. (7.5) transformed to the AO basis, $D^{ij}_{\rho\sigma}$, and folded in the same way,

$$D^{\pm,ij}_{(\rho\sigma)} = \frac{1}{2}\left(D^{ij}_{\rho\sigma} \pm D^{ij}_{\sigma\rho}\right) \quad (\rho < \sigma), \qquad D^{+,ij}_{(\rho\rho)} = \frac{1}{2}D^{ij}_{\rho\rho}, \qquad D^{-,ij}_{(\rho\rho)} = 0, \tag{7.62}$$

the four-external term reduces to two matrix products in the packed pair space,

$$\begin{aligned} K^{ij}_{\mu\nu} &= \left(V^{+}D^{+,ij} + V^{-}D^{-,ij}\right)_{(\mu\nu)}, \\ K^{ij}_{\nu\mu} &= \left(V^{+}D^{+,ij} - V^{-}D^{-,ij}\right)_{(\mu\nu)}, \end{aligned} \tag{7.63}$$

i.e. *both* orders of the bra pair are obtained from the same pair of products. The MO-basis K^{ij}_{pq} of Eq. (7.5) follows by contracting the remaining pair with L.

7.7.5 Occupied-pair packing and operation count

The effective pair density satisfies $D^{ij}_{\rho\sigma} = D^{ji}_{\sigma\rho}$, which together with Eq. (2.12) gives

$$K^{ij}_{pq} = K^{ji}_{qp}. \tag{7.64}$$

Because Eq. (7.63) delivers both bra orders simultaneously, Eq. (7.64) allows the occupied pair to be restricted to $i \le j$ as well. With $n_{pp} = n(n+1)/2$ and $n_{oo} = o(o+1)/2$ the ± form therefore requires

$$2\, n_{pp}^2\, n_{oo} \tag{7.65}$$

multiplications, whereas the contraction over the jointly packed integrals requires $n^2 n_{pp} o^2$: there the ket pair alone is packed, the result is recovered as $K^{ij}_{pq} = A^{ij}_{pq} + A^{ji}_{qp}$ from a partial contraction A, and this symmetrization consumes *both* occupied orders, so the occupied triangle cannot be exploited. The ratio is

$$\frac{n^2 n_{pp} o^2}{2\, n_{pp}^2\, n_{oo}} = \frac{2no}{(n+1)(o+1)} \xrightarrow[n,o \gg 1]{} 2, \tag{7.66}$$

the product of the bra-pair packing ($n^2 \to n_{pp}$), the two products instead of one, and the occupied-pair packing ($o^2 \to n_{oo}$). Both the operation count and the amount of data streamed are thus halved. Note that the second factor in Eq. (7.66) approaches unity only for a sufficiently large occupied space: n_{oo} is the width of the right-hand side of the matrix products Eq. (7.63), and for small o the products are too narrow to reach the asymptotic performance of the underlying matrix multiplication.

The same supermatrices are used for the closed- and open-shell Fock-like intermediates, which are built by streaming $V^{\pm}$ once, and — through Eq. (7.58) — for the Λ, perturbative-triples and EOM integral blocks, so that the AO integrals are the only integral set that is ever stored.

Chapter 8

Two determinant coupled cluster

Amplitudes are normal ordered with respect to the formal reference with two active orbitals t and $\bar{u}$. The occupied $(i, j, \ldots)$ and virtual $(a, b, \ldots)$ spaces do not contain the active orbitals. The equations follow the equations presented in Ref.[10]. Differences because of fixed typos or other reasons are coloured blue. Terms we have added to ensure energy invariance with respect to the reference choice and which are not explicitly listed in Ref.[10] are coloured magenta. Terms we have added to ensure proper antisymmetry and which are not explicitly listed in Ref.[10] are coloured green. IAS terms, which are terms including the all internal singles, are coloured brown.

$$\begin{aligned} R_a^i &= \langle\, {}^A\Phi_a^i | \hat{H}_N e^{\hat{T}_A} |\, {}^A\Phi\rangle_C - \Big(\langle\, {}^A\Phi_a^i | e^{\hat{T}_B} |\, {}^B\Phi\rangle \langle\, {}^B\Phi | \hat{H}_N e^{\hat{T}_A} |\, {}^A\Phi\rangle \Big)_C \\ &\equiv \langle\, {}^A\Phi_a^i | \hat{H}_N e^{\hat{T}_A} |\, {}^A\Phi\rangle_C + M_a^i W = 0, \end{aligned} \tag{8.1}$$

$$\begin{aligned} R_{ab}^{ij} &= \langle\, {}^A\Phi_{ab}^{ij} | \hat{H}_N e^{\hat{T}_A} |\, {}^A\Phi\rangle_C - \Big(\langle\, {}^A\Phi_{ab}^{ij} | e^{\hat{T}_B} |\, {}^B\Phi\rangle \langle\, {}^B\Phi | \hat{H}_N e^{\hat{T}_A} |\, {}^A\Phi\rangle \Big)_C \\ &\quad - \mathcal{A}\,(ij;ab) \Big[\langle\, {}^A\Phi_a^i | e^{\hat{T}_A} |\, {}^A\Phi\rangle \Big(\langle\, {}^A\Phi_b^j | e^{\hat{T}_B} |\, {}^B\Phi\rangle \langle\, {}^B\Phi | \hat{H}_N e^{\hat{T}_A} |\, {}^A\Phi\rangle \Big)_C \\ &\quad - \hat{R}(ia) \langle\, {}^B\Phi_a^i | e^{\hat{T}_B} |\, {}^B\Phi\rangle \Big(\langle\, {}^A\Phi_b^j | e^{\hat{T}_B} |\, {}^B\Phi\rangle \langle\, {}^B\Phi | \hat{H}_N e^{\hat{T}_A} |\, {}^A\Phi\rangle \Big)_C \Big] \\ &\equiv \langle\, {}^A\Phi_{ab}^{ij} | \hat{H}_N e^{\hat{T}_A} |\, {}^A\Phi\rangle_C + M_{ab}^{ij} W = 0, \end{aligned} \tag{8.2}$$

The operator $\hat{R}(ia)$ excludes the active orbitals from the corresponding orbital spaces. The following intermediates are used:

$$\tau_a^i = T_a^i - T_{\bar{a}}^{\bar{i}}, \tag{8.3}$$

$$\tau_{\bar{a}}^{\bar{i}} = T_{\bar{a}}^{\bar{i}} - T_a^i, \tag{8.4}$$

$$\tau_{ab}^{ij} = T_{ab}^{ij} + T_a^i T_b^j. \tag{8.5}$$

The singles M tensor is built as follows,

$$M^{i}_{u} = T^{t\bar{i}}_{u\bar{t}}, \tag{8.6}$$
$$M^{i}_{u} = T^{t}_{u}T^{\bar{i}}_{\bar{t}}, \tag{8.7}$$
$$M^{t}_{a} = -T^{t\bar{u}}_{u\bar{a}}, \tag{8.8}$$
$$M^{t}_{a} = -T^{t}_{u}T^{\bar{u}}_{\bar{a}}, \tag{8.9}$$
$$M^{i}_{a} = T^{\bar{u}}_{\bar{a}}T^{t\bar{i}}_{u\bar{t}} + T^{\bar{i}}_{\bar{t}}T^{t\bar{u}}_{u\bar{a}}, \tag{8.10}$$
$$M^{i}_{a} = T^{t}_{u}T^{\bar{u}\bar{i}}_{\bar{a}\bar{t}}, \tag{8.11}$$
$$M^{i}_{a} = T^{t}_{u}T^{\bar{u}}_{\bar{a}}T^{\bar{i}}_{\bar{t}}, \tag{8.12}$$
$$M^{t}_{u} = T^{t}_{u}. \tag{8.13}$$

$$M^{\bar{i}}_{\bar{t}} = T^{i\bar{u}}_{u\bar{t}}, \tag{8.14}$$
$$M^{\bar{i}}_{\bar{t}} = T^{\bar{u}}_{\bar{t}}T^{i}_{u}, \tag{8.15}$$
$$M^{\bar{u}}_{\bar{a}} = -T^{t\bar{u}}_{a\bar{t}}, \tag{8.16}$$
$$M^{\bar{u}}_{\bar{a}} = -T^{\bar{u}}_{\bar{t}}T^{t}_{a}, \tag{8.17}$$
$$M^{\bar{i}}_{\bar{a}} = T^{t}_{a}T^{i\bar{u}}_{u\bar{t}} + T^{i}_{u}T^{t\bar{u}}_{a\bar{t}}, \tag{8.18}$$
$$M^{\bar{i}}_{\bar{a}} = T^{\bar{u}}_{\bar{t}}T^{ti}_{au}, \tag{8.19}$$
$$M^{\bar{i}}_{\bar{a}} = T^{\bar{u}}_{\bar{t}}T^{t}_{a}T^{i}_{u}, \tag{8.20}$$
$$M^{\bar{u}}_{\bar{t}} = T^{\bar{u}}_{\bar{t}}. \tag{8.21}$$

The all alpha part of the doubles part is built as follows,

$$
\begin{aligned}
M^{ij}_{ua} =& \mathcal{A}\left(ij\right)\tau^{i}_{a}T^{t\bar{j}}_{u\bar{t}} + \mathcal{A}\left(ij\right)T^{\bar{i}}_{\bar{t}}T^{t\bar{j}}_{u\bar{a}}, && (8.22)\\
M^{ij}_{ua} =& T^{t}_{u}T^{\bar{i}\bar{j}}_{\bar{t}\bar{a}}, && (8.23)\\
M^{ij}_{au} =& -\mathcal{A}\left(ij\right)\tau^{i}_{a}T^{t\bar{j}}_{u\bar{t}} - \mathcal{A}\left(ij\right)T^{\bar{i}}_{\bar{t}}T^{t\bar{j}}_{u\bar{a}}, && (8.24)\\
M^{ij}_{au} =& -T^{t}_{u}T^{\bar{i}\bar{j}}_{\bar{t}\bar{a}}, && (8.25)\\
M^{ti}_{ab} =& \mathcal{A}\left(ab\right)\tau^{i}_{b}T^{t\bar{u}}_{u\bar{a}} + \mathcal{A}\left(ab\right)T^{\bar{u}}_{\bar{b}}T^{t\bar{i}}_{u\bar{a}}, && (8.26)\\
M^{ti}_{ab} =& T^{t}_{u}T^{\bar{u}\bar{i}}_{\bar{b}\bar{a}}, && (8.27)\\
M^{it}_{ab} =& -\mathcal{A}\left(ab\right)\tau^{i}_{b}T^{t\bar{u}}_{u\bar{a}} - \mathcal{A}\left(ab\right)T^{\bar{u}}_{\bar{b}}T^{t\bar{i}}_{u\bar{a}}, && (8.28)\\
M^{it}_{ab} =& -T^{t}_{u}T^{\bar{u}\bar{i}}_{\bar{b}\bar{a}}, && (8.29)\\
M^{ij}_{ab} =& -\mathcal{A}\left(ij;ab\right)\tau^{j}_{b}\left(T^{\bar{u}}_{\bar{a}}T^{t\bar{i}}_{u\bar{t}} + T^{\bar{i}}_{\bar{t}}T^{t\bar{u}}_{u\bar{a}}\right) - \mathcal{A}\left(\bar{i}\bar{j};\bar{a}\bar{b}\right)\left(T^{\bar{u}\bar{i}}_{\bar{t}\bar{a}}T^{t\bar{j}}_{u\bar{b}}\right) \\
& - \mathcal{A}\left(\bar{i}\bar{j}\right)T^{\bar{u}\bar{i}}_{\bar{a}\bar{b}}T^{t\bar{j}}_{u\bar{t}} - \mathcal{A}\left(ab\right)T^{\bar{i}\bar{j}}_{\bar{t}\bar{a}}T^{t\bar{u}}_{u\bar{b}} - \mathcal{A}\left(\bar{i}\bar{j};\bar{a}\bar{b}\right)T^{\bar{u}}_{\bar{a}}T^{\bar{j}}_{\bar{t}}T^{t\bar{i}}_{u\bar{b}}, && (8.30)\\
M^{ij}_{ab} =& -\mathcal{A}\left(\bar{i}\bar{j};\bar{a}\bar{b}\right)\tau^{\bar{i}}_{\bar{a}}\left(T^{\bar{u}}_{\bar{t}}T^{t\bar{j}}_{u\bar{b}} + T^{t}_{u}T^{\bar{u}\bar{j}}_{\bar{t}\bar{b}}\right) \\
& -\mathcal{A}\left(\bar{i}\bar{j}\right)T^{t}_{u}T^{\bar{j}}_{\bar{t}}T^{\bar{u}\bar{i}}_{\bar{a}\bar{b}} - \mathcal{A}\left(\bar{a}\bar{b}\right)T^{t}_{u}T^{\bar{u}}_{\bar{b}}T^{\bar{i}\bar{j}}_{\bar{t}\bar{a}}, && (8.31)\\
M^{it}_{au} =& -T^{t\bar{i}}_{u\bar{a}}, && (8.32)\\
M^{ti}_{au} =& T^{t\bar{i}}_{u\bar{a}}, && (8.33)\\
M^{it}_{ua} =& T^{t\bar{i}}_{u\bar{a}}, && (8.34)\\
M^{ti}_{ua} =& -T^{t\bar{i}}_{u\bar{a}}, && (8.35)\\
M^{it}_{au} =& -\tau^{\bar{i}}_{\bar{a}}T^{t}_{u}, && (8.36)\\
M^{it}_{ua} =& +\tau^{\bar{i}}_{\bar{a}}T^{t}_{u}, && (8.37)\\
M^{ti}_{au} =& +\tau^{\bar{i}}_{\bar{a}}T^{t}_{u}, && (8.38)\\
M^{ti}_{ua} =& -\tau^{\bar{i}}_{\bar{a}}T^{t}_{u}. && (8.39)
\end{aligned}
$$

The all beta part of the doubles M tensor is obtained from the all alpha part analogously to the presented singles M tensor.

The alpha beta part is calculated as follows,

$$M^{i\bar{j}}_{u\bar{a}} = -\tau^{\bar{j}}_{\bar{a}} T^{t\bar{i}}_{u\bar{t}} - T^{j}_{u} T^{t\bar{i}}_{a\bar{t}} - T^{t}_{a} \tau^{j\bar{i}}_{u\bar{t}} - T^{\bar{i}}_{\bar{t}} T^{tj}_{au}, \tag{8.40}$$

$$M^{i\bar{j}}_{u\bar{a}} = T^{t}_{u} T^{j\bar{i}}_{a\bar{t}}, \tag{8.41}$$

$$M^{j\bar{i}}_{a\bar{t}} = -\tau^{j}_{a} T^{i\bar{u}}_{u\bar{t}} - T^{\bar{j}}_{\bar{t}} T^{i\bar{u}}_{u\bar{a}} - T^{\bar{u}}_{\bar{a}} \tau^{i\bar{j}}_{u\bar{t}} - T^{i}_{u} T^{\bar{j}\bar{u}}_{\bar{t}\bar{a}}, \tag{8.42}$$

$$M^{j\bar{i}}_{a\bar{t}} = T^{\bar{u}}_{\bar{t}} T^{i\bar{j}}_{u\bar{a}}, \tag{8.43}$$

$$M^{t\bar{i}}_{a\bar{b}} = \tau^{\bar{i}}_{\bar{b}} T^{t\bar{u}}_{u\bar{a}} + T^{t}_{b} T^{i\bar{u}}_{u\bar{a}} + T^{i}_{u} \tau^{t\bar{u}}_{b\bar{a}} + T^{\bar{u}}_{\bar{a}} T^{it}_{ub}, \tag{8.44}$$

$$M^{t\bar{i}}_{a\bar{b}} = -T^{t}_{u} T^{i\bar{u}}_{b\bar{a}}, \tag{8.45}$$

$$M^{i\bar{u}}_{b\bar{a}} = \tau^{i}_{b} T^{t\bar{u}}_{a\bar{t}} + T^{\bar{u}}_{\bar{b}} T^{t\bar{i}}_{a\bar{t}} + T^{\bar{i}}_{\bar{t}} \tau^{t\bar{u}}_{a\bar{b}} + T^{t}_{a} T^{\bar{i}\bar{u}}_{\bar{t}\bar{b}}, \tag{8.46}$$

$$M^{i\bar{u}}_{b\bar{a}} = -T^{\bar{u}}_{\bar{t}} T^{t\bar{i}}_{a\bar{b}}, \tag{8.47}$$

$$\begin{aligned} M^{i\bar{j}}_{a\bar{b}} = & -\tau^{\bar{j}}_{\bar{b}} (T^{\bar{u}}_{\bar{a}} T^{t\bar{i}}_{u\bar{t}} + T^{\bar{i}}_{\bar{t}} T^{t\bar{u}}_{u\bar{a}}) - \tau^{i}_{a} (T^{j\bar{u}}_{u\bar{t}} T^{t}_{b} + T^{t\bar{u}}_{b\bar{t}} T^{j}_{u}) \\ & + T^{\bar{i}}_{\bar{t}} T^{\bar{u}}_{\bar{a}} T^{tj}_{ub} + T^{t}_{b} T^{j}_{u} T^{\bar{u}\bar{i}}_{\bar{t}\bar{a}} - T^{j}_{u} T^{\bar{u}}_{\bar{a}} T^{t\bar{i}}_{b\bar{t}} - T^{t}_{b} T^{\bar{i}}_{\bar{t}} T^{j\bar{u}}_{u\bar{a}} \\ & - \tau^{j\bar{i}}_{u\bar{t}} \tau^{t\bar{u}}_{b\bar{a}} + T^{j\bar{u}}_{u\bar{t}} T^{t\bar{i}}_{b\bar{a}} + T^{t\bar{i}}_{u\bar{t}} T^{j\bar{u}}_{b\bar{a}} + T^{t\bar{u}}_{b\bar{t}} T^{j\bar{i}}_{u\bar{a}} \\ & + T^{t\bar{u}}_{u\bar{a}} T^{j\bar{i}}_{b\bar{t}} - T^{tj}_{ub} T^{\bar{u}\bar{i}}_{\bar{t}\bar{a}} - T^{t\bar{i}}_{u\bar{a}} T^{j\bar{u}}_{b\bar{t}} - T^{j\bar{u}}_{u\bar{a}} T^{t\bar{i}}_{b\bar{t}}, \end{aligned} \tag{8.48}$$

$$\begin{aligned} M^{i\bar{j}}_{a\bar{b}} = & -\tau^{\bar{j}}_{\bar{b}} T^{t}_{u} T^{\bar{u}\bar{i}}_{\bar{a}\bar{t}} - \tau^{i}_{a} T^{\bar{u}}_{\bar{t}} T^{jt}_{ub} + T^{t}_{u} T^{\bar{i}}_{\bar{t}} T^{j\bar{u}}_{b\bar{a}} + T^{t}_{u} T^{\bar{u}}_{\bar{a}} T^{j\bar{i}}_{b\bar{t}} \\ & + T^{\bar{u}}_{\bar{t}} T^{t}_{b} T^{j\bar{i}}_{u\bar{a}} + T^{\bar{u}}_{\bar{t}} T^{j}_{u} T^{t\bar{i}}_{b\bar{a}}, \end{aligned} \tag{8.49}$$

$$M^{t\bar{i}}_{u\bar{t}} = T^{i}_{u}, \tag{8.50}$$

$$M^{i\bar{u}}_{u\bar{t}} = T^{\bar{i}}_{\bar{t}}, \tag{8.51}$$

$$M^{t\bar{i}}_{u\bar{a}} = \tau^{it}_{ua}, \tag{8.52}$$

$$M^{i\bar{u}}_{a\bar{t}} = \tau^{\bar{i}\bar{u}}_{\bar{t}\bar{a}}, \tag{8.53}$$

$$M^{t\bar{i}}_{a\bar{t}} = \tau^{i\bar{u}}_{u\bar{a}}, \tag{8.54}$$

$$M^{i\bar{u}}_{u\bar{a}} = \tau^{t\bar{i}}_{a\bar{t}}, \tag{8.55}$$

$$M^{t\bar{u}}_{u\bar{a}} = -T^{t}_{a}, \tag{8.56}$$

$$M^{t\bar{u}}_{a\bar{t}} = -T^{\bar{u}}_{\bar{a}}, \tag{8.57}$$

$$M^{i\bar{j}}_{u\bar{t}} = -\tau^{j\bar{i}}_{u\bar{t}}, \tag{8.58}$$

$$M^{t\bar{u}}_{a\bar{b}} = -\tau^{t\bar{u}}_{b\bar{a}}. \tag{8.59}$$

The effective Hamiltonian W is just the all active part of the residuum,

$$W = R^{t\bar{u}}_{u\bar{t}}. \tag{8.60}$$

The all internal doubles $T^{t\bar{u}}_{u\bar{t}}$ coupled cluster amplitude is set to zero at the beginning of every iteration. At the end of every iteration the all internal doubles residuum $R^{t\bar{u}}_{u\bar{t}}$ is set to zero.
IAS contribution to the energy,

$$\Delta E_{\mathrm{IAS}} = -W^{t\bar{u}}_{u\bar{t}} T^{t}_{u} T^{\bar{u}}_{\bar{t}}. \tag{8.61}$$

Chapter 9

Tensor Decomposed Coupled Cluster (SVD-CC)

In this chapter we describe the tensor decomposed coupled cluster methods implemented in ElemCo.jl, following the SVD-DC-CCSDT approach of Ref. [11].

9.1 SVD-DC-CCSDT method

9.1.1 Overview

The SVD-DC-CCSDT method combines two key ideas:

1. The distinguishable cluster (DC) approximation applied to the CCSDT triples residual equations, which removes the expensive exchange terms while improving accuracy.

2. A singular value decomposition (SVD) of the triples amplitudes and residuals, which together with the density fitting (or Cholesky) factorization of the electron repulsion integrals reduces the scaling of the method from $\mathcal{O}(N^8)$ to $\mathcal{O}(N^6)$.

The singles contributions to the triples residuals are implemented using the similarity-transformed Hamiltonian,

$$\tilde{H}_N = e^{-\hat{T}_1}\hat{H}_N e^{\hat{T}_1}, \tag{9.1}$$

which implies dressing of the Fock matrices and the ERI factorization tensors v_p^{qL} with the singles amplitudes. The dressed quantities are denoted with a hat.

9.1.2 Triples amplitudes decomposition

The triples amplitudes T^{ijk}_{abc} are decomposed as

$$T^{ijk}_{abc} = T_{XYZ}\, U^{iX}_a U^{jY}_b U^{kZ}_c, \tag{9.2}$$

where T_{XYZ} is a symmetric three-index quantity in the SVD basis and U^{iX}_a are the transformation (compression) matrices connecting the SVD basis (X, Y, Z) with pairs of orbital indices (one occupied, one virtual). The amplitudes are symmetric with respect to the interchange of SVD indices, i.e., $T_{XYZ} = T_{YXZ} = T_{XZY} = \dots$

In the SVD-DC-CCSDT method the triples amplitudes are varied directly in the SVD basis, such that the residual R_{XYZ} in the SVD basis goes to zero. The U-transformation can be applied independently to each SVD index, so that in each diagram only the indices contracted with the Hamiltonian need to be transformed to the orbital basis, while the others remain in the SVD basis. The transformation matrices satisfy $U^{\dagger a}_{iX} U^{iY}_{a} = \delta^{Y}_{X}$.

The factorization of the electron repulsion integrals (cf. Eq. (2.7)) is also employed:

$$v^{rs}_{pq} \approx v^{rL}_{p} v^{sL}_{q}. \tag{9.3}$$

9.1.3 DC-CCSDT approximation for triples

The DC approximation to CCSDT modifies the triples residual equations by removing or rescaling certain exchange terms.

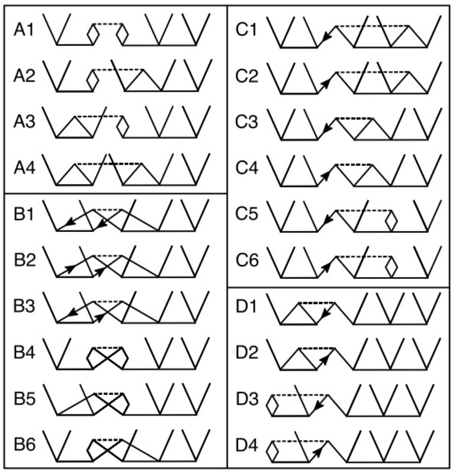

Figure 9.1: Spin-summed diagrams of $\hat{T}_2\hat{T}_3$ contributions to the triples residuals in CCSDT.

Classifying the spin-summed $\hat{T}_2\hat{T}_3$ diagrams contributing to the triples residuals into groups A, B, C, and D:

- Diagrams **A1–A4**: kept unchanged,
- Diagrams **B1–B6**: fully removed,
- Diagrams **C1–C6** and **D1–D4**: multiplied by $\frac{1}{2}$.

This approximation preserves size-extensivity, particle-hole symmetry, exactness for three-electron (or three-hole) systems, and invariance to rotations within the occupied and virtual spaces.

9.1.4 Residual equations in the SVD basis

The bra-projection for the singles and doubles residual equations uses the contravariant CSFs. For the triples, the contravariant CSF is

$$\widetilde{\Phi}^{ijk}_{abc} = \Phi^{ijk}_{abc} - \Phi^{ijk}_{cab}. \tag{9.4}$$

Due to the overlap properties of this choice the full triples residual has the structure

$$\bar{R}^{ijk}_{abc} = R^{ijk}_{abc} - R^{ijk}_{cab} = \mathcal{A}(abc, cab)\, R^{ijk}_{abc}, \tag{9.5}$$

where $\mathcal{A}(abc, cab)F(a,b,c) = F(a,b,c) - F(c,a,b)$. Since $R^{ijk}_{abc} = 0$ implies $R^{ijk}_{cab} = 0$, it is sufficient to solve $R^{ijk}_{abc} = 0$ with only half the terms.

Triples contributions to the singles residual

The triples contributions to the singles residual read:

$$R^i_a \leftarrow U^{iX}_a \left[T_{XYZ}\left(2A^{YL}A^{ZL} - B^{YZ}\right)\right] - U^{jY}_a\Big[2B^{iXL}_j V^L_{XY} - G^i_{jY}\Big], \tag{9.6}$$

with intermediates

$$\begin{aligned} A^{XL} &= B^{iXL}_i, & B^{jXL}_i &= v^{aL}_i U^{jX}_a, \\ B^{XY} &= B^{jXL}_i B^{iYL}_j, & w^{dX}_k &= v^{dL}_l B^{lXL}_k, \\ V^L_{XY} &= T_{XYZ}A^{ZL}, & G^i_{jY} &= T^i_{dYZ} w^{dZ}_j, \\ T^i_{aYZ} &= T_{XYZ}U^{iX}_a. \end{aligned} \tag{9.7}$$

Triples contributions to the doubles residual

$$\begin{aligned} R^{ij}_{ab} \leftarrow \mathcal{S}\,(ij,ab)\Big\{ &-\left(\bar{B}^{iLZ}_a - B^{iLZ}_a\right)V^{jL}_{bZ} + U^{jZ}_b\Big[2\left(\bar{B}^{iLY}_a - B^{iLY}_a\right)V^L_{YZ} \\ &+ T^i_{aYZ}\left(\hat{f}^c_k U^{kY}_c\right) - T_{XYZ}\left(U^{iY}_c\left(\bar{B}^{kLX}_a v^{cL}_k\right) - U^{kY}_a\left(B^{iLX}_c v^{cL}_k - \hat{f}^c_k U^{iX}_c\right)\right)\Big]\Big\}, \end{aligned} \tag{9.8}$$

with intermediates

$$\begin{aligned} B^{iLX}_a &= \hat{v}^{iL}_j U^{jX}_a, & \bar{B}^{iLX}_a &= \hat{v}^{bL}_a U^{iX}_b, \\ V^{jL}_{bZ} &= B^{jYL}_k T^k_{bYZ}, \end{aligned} \tag{9.9}$$

and the symmetrization operator $\mathcal{S}\,(ij,ab)\,X^{ij}_{ab} = X^{ij}_{ab} + X^{ji}_{ba}$.

Triples residual

The triples residual equations in the SVD basis read:

$$R_{XYZ} = Q_{XYZ} + Q_{YXZ} + Q_{XZY} + Q_{ZYX} + Q_{ZXY} + Q_{YZX} + q_{XYZ} + q_{YXZ} + q_{ZYX} \tag{9.10}$$

with the main intermediates

$$Q_{XYZ} = U^{\dagger b}_{jY}\Big[T^l_{bX}X^j_{lZ} - X^d_{bZ}T^j_{dX} + T^l_{bY'Z}\left(U^{jY'}_d\left(v^{dL}_l Y^L_X\right)\right)\Big] \tag{9.11}$$

and

$$\begin{aligned} q_{XYZ} = T_{X'YZ}\Big[U^{\dagger a}_{iX}\Big(&\left(\hat{f}^i_l + \frac{1}{2}v^{dL}_l V^{iL}_d\right)U^{lX'}_a - \left(\hat{f}^d_a - \frac{1}{2}v^{dL}_l V^{lL}_a\right)U^{iX'}_d \\ &+ \left(\hat{v}^{iL}_l \hat{v}^{dL}_a\right)U^{lX'}_d\Big) - 2Y^L_X A^{X'L}\Big] - T_{XY'Z'}\left(W^{Y'L}_Y W^{Z'L}_Z\right). \end{aligned} \tag{9.12}$$

The further intermediates used in these expressions are

$$X^d_{bZ} = \hat{v}^{dL}_b Y^L_Z - \hat{f}^d_l T^l_{bZ} - U^{lX}_b\left(v^{dL}_l \tilde{V}^L_{ZX}\right) + \left(\hat{v}^{di}_{lk}T^{lk}_{ba} - \hat{v}^{dc}_{la}T^{li}_{bc}\right)U^{\dagger a}_{iZ} - \hat{v}^{dc}_{lb}T^l_{cZ} + \frac{1}{2}T^k_{bYZ}w^{dY}_k, \tag{9.13}$$

$$X_{lZ}^{j} = \hat{v}_l^{jL} Y_Z^L + U_d^{jX} \left(v_l^{dL} \tilde{V}_{ZX}^L \right) + \left(\hat{v}_{la}^{dc} T_{dc}^{ji} - \hat{v}_{lk}^{di} T_{da}^{jk} \right) U_{iZ}^{\dagger a} - \hat{v}_{lk}^{dj} T_{dZ}^{k} - \frac{1}{2} G_{lZ}^{j}, \tag{9.14}$$

and

$$\begin{aligned} V_a^{iL} &= v_k^{cL} \widetilde{T}_{ac}^{ik}, & Y_X^L &= U_{iX}^{\dagger a} \left(\hat{v}_a^{iL} + V_a^{iL} \right), \\ T_{aX}^{i} &= U_{jX}^{\dagger b} T_{ab}^{ij}, & \widetilde{T}_{ab}^{ij} &= 2T_{ab}^{ij} - T_{ba}^{ij}, \\ \hat{v}_{la}^{dc} &= v_l^{dL} \hat{v}_a^{cL}, & \hat{v}_{lk}^{di} &= v_l^{dL} \hat{v}_k^{iL}, \\ W_Y^{Y'L} &= U_{jY}^{\dagger b} \left(\bar{B}_b^{jLY'} - B_b^{jLY'} \right), & \tilde{V}_{ZX}^{L} &= V_{ZX}^{L} - \frac{1}{2} V_{cX}^{kL} U_{kZ}^{\dagger c}. \end{aligned} \tag{9.15}$$

None of the contractions above involve more than six different indices. This means that the tensor decompositions reduce the scaling of the DC-CCSDT method from $\mathcal{O}(N^8)$ to $\mathcal{O}(N^6)$, provided the number of SVD basis functions grows linearly with the system size. The most expensive terms in the SVD-DC-CCSDT residual equations scale as $N_o^2 N_v N_X^2 N_L$, where N_o and N_v are the numbers of occupied and virtual orbitals, and N_X and N_L are the numbers of SVD and density-fitting basis functions.

Projected intermediates

An additional approximation can be introduced by calculating the V_{aX}^{iL} intermediate in an SVD basis, which reduces the scaling of the most expensive terms to $N_o^2 N_v N_X^3$:

$$\begin{aligned} V_{aX}^{iL} &\approx \bar{V}_{X\tilde{Z}}^{L} \tilde{U}_a^{i\tilde{Z}}, \\ \bar{V}_{X\tilde{Z}}^{L} &= \left(\left(\tilde{U}_{j\tilde{Z}}^{\dagger b} U_b^{kY} \right) T_{cXY}^{j} \right) v_k^{cL}. \end{aligned} \tag{9.16}$$

The $\tilde{Z}$ SVD space covers both the triples SVD space (for the $\tilde{V}_{XZ}^L$ intermediate) and the doubles space (for the $\tilde{U}_a^{i\tilde{Z}}$ projection in the doubles residual equations). It is constructed by expanding the triples SVD space by important orthogonal contributions from the doubles SVD space.

9.1.5 Construction of the SVD basis

The construction of the singular vectors U_a^{iX} must also scale as $\mathcal{O}(N^6)$. This is achieved using a two-step procedure.

Step 1: Doubles SVD basis

First, the relevant block of an approximate two-particle density matrix is obtained from converged CCSD doubles amplitudes:

$$\mathcal{D}_{ib}^{aj} = T_{ik}^{\dagger ac} T_{bc}^{jk}, \tag{9.17}$$

and eigendecomposed:

$$\mathcal{D}_{ib}^{aj} \approx \bar{U}_{i\tilde{\tilde{X}}}^{\dagger a} \bar{D}_{\tilde{\tilde{X}}} \bar{U}_b^{j\tilde{\tilde{X}}}. \tag{9.18}$$

The SVD basis $\{\tilde{\tilde{X}}\}$ is truncated by disregarding eigenvectors with eigenvalues smaller than a threshold: $\bar{D}_{\tilde{\tilde{X}}} < \varepsilon_{T2}$.

The basis is canonicalized by diagonalizing the Fock matrix in the SVD basis:

$$f_{\tilde{\bar{X}}}^{\tilde{\bar{Y}}} = \bar{U}^{\dagger a}_{i\tilde{\bar{X}}}(\epsilon_a - \epsilon_i)\bar{U}^{i\tilde{\bar{Y}}}_a. \tag{9.19}$$

In the basis of its eigenvectors $U^{\bar{X}}_{\tilde{\bar{X}}}$ the Fock matrix is diagonal, $f^{\bar{X}}_{\bar{Y}} = \epsilon_{\bar{X}}\delta^{\bar{X}}_{\bar{Y}}$, and the complete transformation to the $\bar{X}$-basis is given by

$$\bar{U}^{i\bar{X}}_a = \bar{U}^{i\tilde{\bar{X}}}_a \bar{U}^{\bar{X}}_{\tilde{\bar{X}}}. \tag{9.20}$$

Step 2: Triples SVD basis via CC3-type density matrix

Having the transformation matrices to the doubles SVD basis, a partially decomposed CC3-type triples residual is computed:

$$R^i_{a\bar{X}\bar{Y}} = \left(X^{jL}_{b\bar{X}}\bar{U}^{\dagger b}_{j\bar{Y}}\right)\hat{v}^{iL}_a + V^d_{a\bar{X}}T^i_{d\bar{Y}} - V^i_{l\bar{X}}T^l_{a\bar{Y}} + E^d_{l\bar{X}\bar{Y}}T^{il}_{ad}, \tag{9.21}$$

with intermediates

$$\begin{aligned} X^{jL}_{b\bar{X}} &= T^j_{c\bar{X}}\hat{v}^{cL}_b - T^l_{b\bar{X}}\hat{v}^{jL}_l, & T^i_{a\bar{X}} &= \bar{U}^{\dagger b}_{j\bar{X}}T^{ij}_{ab}, \\ E^d_{l\bar{X}\bar{Y}} &= V^d_{a\bar{X}}\bar{U}^{\dagger a}_{l\bar{Y}} - V^i_{l\bar{X}}\bar{U}^{\dagger d}_{i\bar{Y}}, & W^L_{\bar{X}} &= \hat{v}^{kL}_c\bar{U}^{\dagger c}_{k\bar{X}}, \\ V^d_{a\bar{X}} &= \hat{v}^{dL}_a W^L_{\bar{X}}, & V^j_{l\bar{X}} &= \hat{v}^{jL}_l W^L_{\bar{X}}. \end{aligned} \tag{9.22}$$

The last term of the residual scales as N^6, while all other terms scale as N^5. Since the SVD basis is canonicalized, the partially transformed triples amplitudes are readily available:

$$T^i_{a\bar{X}\bar{Y}} = -\frac{R^i_{a\bar{X}\bar{Y}} + R^i_{a\bar{Y}\bar{X}}}{\epsilon_{\bar{X}} + \epsilon_{\bar{Y}} + \epsilon_a - \epsilon_i}. \tag{9.23}$$

The triples two-particle density matrix is then computed as

$$D^{aj}_{ib} = T^{\dagger a\bar{X}\bar{Y}}_i T^j_{b\bar{X}\bar{Y}} \tag{9.24}$$

at N^6 cost.

The triples amplitudes in the SVD basis implicitly include nonphysical components with triply repeated occupied orbital indices (e.g., T^{iii}_{abc}), which would cancel in case of a full SVD basis. Their contributions to the density matrix can be explicitly isolated and removed:

$$\begin{aligned} \Delta D^{aj}_{ib} &= (T^{\dagger a}_{i\bar{X}\bar{Y}}\mathcal{U}^{\bar{X}}_{i\bar{X}'}\mathcal{U}^{\bar{Y}}_{i\bar{Y}'})T^{j\bar{X}'\bar{Y}'}_b, \\ \mathcal{U}^{\bar{X}}_{i\bar{X}'} &= \bar{U}^{\dagger a}_{i\bar{X}'}\bar{U}^{i\bar{X}}_a, \\ \widetilde{D}^{aj}_{ib} &= D^{aj}_{ib} - \left(1 - \frac{1}{2}\delta^i_j\right)\mathcal{S}\left(ij, ab\right)\Delta D^{aj}_{ib}. \end{aligned} \tag{9.25}$$

Finally, the actual SVD basis for the triples in DC-CCSDT is obtained by diagonalizing the corrected triples two-electron density matrix:

$$\widetilde{D}^{aj}_{ib} \approx U^{\dagger a}_{i\tilde{X}}D_{\tilde{X}}U^{j\tilde{X}}_b. \tag{9.26}$$

Eigenvectors with eigenvalues smaller than ε_{T3} are discarded. The SVD basis is again canonicalized by diagonalizing the Fock matrix,

$$f^{\tilde{Y}}_{\tilde{X}} = U^{\dagger a}_{i\tilde{X}}(\epsilon_a - \epsilon_i)U^{i\tilde{Y}}_a, \tag{9.27}$$

and the final SVD transformation matrices are

$$U^{iX}_a = U^{i\tilde{X}}_a U^X_{\tilde{X}}. \tag{9.28}$$

Additional SVD basis for the projection approximation

The additional SVD basis for the projection approximation is constructed by half-transforming the contravariant doubles amplitudes $\widetilde{T}^{ij}_{ab}$ to the SVD basis, projecting the part not covered by the triples SVD space:

$$\Delta\widetilde{T}^{ij}_{ab} = \widetilde{T}^{ij}_{ab} - U^{iX}_{a} U^{\dagger c}_{kX} \widetilde{T}^{kj}_{cb}, \tag{9.29}$$

computing SVD of $\Delta\widetilde{T}^{ij}_{ab}$, and selecting the most important left vectors $\Delta U^{i\bar{X}}_{a}$, which are then added to the triples SVD basis. A symmetric orthogonalization of the combined space is performed at the end.

9.1.6 Amplitude update and convergence

The eigenvalues ϵ_X of the Fock matrix in the SVD basis are employed in the amplitude update, carried out directly in the SVD basis:

$$\Delta T_{XYZ} = -\frac{R_{XYZ}}{\epsilon_X + \epsilon_Y + \epsilon_Z}. \tag{9.30}$$

To accelerate convergence, DIIS is employed for singles, doubles, and triples amplitude vectors.

9.1.7 Truncation thresholds

The truncation of the SVD spaces is governed by two tolerances: ε_{T3} for the triples and ε_{T2} for the auxiliary doubles SVD. These are linked by a weight factor w_{T2}:

$$\varepsilon_{T2} = w_{T2}\,\varepsilon_{T3}. \tag{9.31}$$

The choice of ε_{T2} has very little effect on the results provided it is not larger than ε_{T3}.

A hybrid approach with an SVD-error correction at the CCSD(T) level can be used to accelerate convergence with the SVD truncation threshold:

$$E_{\text{DC-CCSDT}} \approx E_{\text{SVD-DC-CCSDT}} + \delta_{\text{SVD}}, \tag{9.32}$$

with

$$\delta_{\text{SVD}} = E_{\text{CCSD(T)}} - E_{\text{SVD-CCSD(T)}}. \tag{9.33}$$

9.1.8 Correlation energy

The correlation energy is calculated using the singles and doubles amplitudes:

$$E = \left(2T^{i}_{a}T^{j}_{b} - T^{i}_{b}T^{j}_{a}\right) v^{ab}_{ij} + 2T^{i}_{a} f^{a}_{i} + (2T^{ij}_{ab} - T^{ij}_{ba}) v^{ab}_{ij}. \tag{9.34}$$

9.1.9 Scaling summary

- **Full DC-CCSDT**: $\mathcal{O}(N^8)$.
- **SVD-DC-CCSDT**: $\mathcal{O}(N^6)$, provided the number of SVD basis functions grows linearly with system size (confirmed by tests on alkane chains).

- The most expensive terms in SVD-DC-CCSDT scale as $N_o^2 N_v N_X^2 N_L$. Assuming $N_v \approx N_X$, these terms are only a factor of N_L/N_v more expensive than the most expensive term in CCSD.
- The 4-external ladder diagram of the triples is reduced to N^5.
- The SVD-error and the size of the SVD bases scale linearly with system size, confirming size extensivity.

Chapter 10

Full Configuration Interaction and Selected CI Methods

10.1 Introduction

Full Configuration Interaction (FCI) provides the exact solution to the time-independent Schrödinger equation within a given orbital basis set. While FCI is the ultimate benchmark for quantum chemistry calculations, its exponential scaling with system size limits its applicability to small systems. This chapter describes the implementation of FCI and selected CI methods in ElemCo.jl, including Heat-bath CI (HCI), multi-state calculations, P-space enhanced Davidson solver, and reduced density matrix calculations.

10.2 Full Configuration Interaction

10.2.1 Determinantal Representation

A Slater determinant $|\Phi_I\rangle$ is specified by occupation numbers for each spin-orbital. In the Knowles-Handy algorithm[12], determinants are represented separately for α and β spins:

$$|\Phi^I\rangle = |\alpha^I\rangle \otimes |\beta^I\rangle \tag{10.1}$$

where $|\alpha^I\rangle$ and $|\beta^I\rangle$ are strings of occupied spin-orbitals.

The FCI wave function is expanded as:

$$|\Psi\rangle = c_I|\Phi^I\rangle \tag{10.2}$$

where c_I are the CI coefficients to be determined.

10.2.2 Slater-Condon Rules

Matrix elements $\langle\Phi_I|\hat{H}|\Phi^J\rangle$ can be evaluated using Slater-Condon rules:

Identical determinants ($|\Phi^I\rangle = |\Phi^J\rangle$):

$$\langle\Phi_I|\hat{H}|\Phi^I\rangle = \sum_{i\in I} h_i^i + \frac{1}{2}\sum_{i,j\in I}\left[v_{ij}^{ij} - v_{ij}^{ji}\right] \tag{10.3}$$

Single excitation ($|\Phi^J\rangle = \hat{a}_a^\dagger \hat{a}_i |\Phi^I\rangle$):

$$\langle \Phi_I | \hat{H} | \Phi^J \rangle = \mathrm{sgn}_{ia} \left[h_a^i + \sum_{j \in I} \left[v_{aj}^{ij} - v_{aj}^{ji} \right] \right] \tag{10.4}$$

where sgn_{ia} is the fermionic phase factor from orbital reordering.

Double excitation ($|\Phi^J\rangle = \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i |\Phi^I\rangle$):

$$\langle \Phi_I | \hat{H} | \Phi^J \rangle = \mathrm{sgn}_{iajb} \left[v_{ab}^{ij} - v_{ba}^{ij} \right] \tag{10.5}$$

Higher excitations: Matrix elements vanish for determinants differing by more than two orbitals.

10.2.3 String-Based Algorithm

The Knowles-Handy algorithm[12] evaluates the matrix-vector product $\mathbf{v}_{\text{out}} = \hat{H}\mathbf{c}_{\text{in}}$ efficiently using string-based techniques and resolution of identity:

$$v_I = H_I^J c_J = \langle \alpha_I \beta_I | \hat{H} | \alpha^J \beta^J \rangle c_J \tag{10.6}$$

For the two-electron part of the Hamiltonian, we have:

$$\langle I | \frac{1}{2} v_{pq}^{rs} \left(\hat{E}_r^p \hat{E}_s^q - \hat{E}_s^p \delta_r^q \right) | J \rangle c_J, = \frac{1}{2} v_{pq}^{rs} \langle I | \hat{E}_r^p | K \rangle \langle K | \hat{E}_s^q | J \rangle c_J - \cdots, \tag{10.7}$$

The Hamiltonian is decomposed into one-electron and two-electron parts:

$$\hat{H} = \hat{H}_1^{(\alpha)} + \hat{H}_1^{(\beta)} + \hat{H}_2^{(\alpha\alpha)} + \hat{H}_2^{(\beta\beta)} + \hat{H}_2^{(\alpha\beta)} \tag{10.8}$$

The one-electron part is absorbed into the two-electron integrals. For remaining components, string substitutions are performed:

- **Same-spin two-electron**: Substitution in one string, sum over other string
- **Opposite-spin two-electron**: Substitution in both strings simultaneously

10.3 Davidson Iterative Diagonalization

10.3.1 Davidson Algorithm

The Davidson method finds the lowest eigenvalues of large matrices without explicit matrix construction. Given initial guess vectors $\{\mathbf{u}^1, \ldots, \mathbf{u}^k\}$, the algorithm iteratively builds a subspace and solves:

$$\mathbf{H}_{\text{sub}} \mathbf{c} = E \mathbf{c}, \tag{10.9}$$

where $[\mathbf{H}_{\text{sub}}]_i^j = \langle \mathbf{u}_i | \hat{H} | \mathbf{u}^j \rangle$.

10.3.2 Subspace Expansion

The residual vector for eigenstate $\mathfrak{n}$ is:

$$\mathbf{r}^{\mathfrak{n}} = \mathbf{H}\mathbf{x}^{\mathfrak{n}} - E^{\mathfrak{n}}\mathbf{x}^{\mathfrak{n}}, \tag{10.10}$$

where $\mathbf{x}^{\mathfrak{n}} = c_i^{\mathfrak{n}}\mathbf{u}^i$ is the approximate eigenvector.

A correction vector $\mathbf{u}^{\mathfrak{n}} \to \mathbf{u}^{k+1}$ is computed and added to the subspace. In ElemCo.jl, we use either the diagonal preconditioner,

$$u_I^{\mathfrak{n}} = \frac{r_I^{\mathfrak{n}}}{E^{\mathfrak{n}} - H_I^I}, \tag{10.11}$$

or the P-space enhanced preconditioner (see Sec. 10.3.3 and Sec. 10.3.5).

10.3.3 Jacobi-Davidson Preconditioner

For improved convergence, especially for excited states, we implement the Jacobi-Davidson correction. The correction vector $\mathbf{u}^{\mathfrak{n}}$ solves:

$$(\mathbf{I} - \mathbf{x}^{\mathfrak{n}}\mathbf{x}_{\mathfrak{n}}^{\dagger})(\mathbf{H} - E^{\mathfrak{n}})(\mathbf{I} - \mathbf{x}^{\mathfrak{n}}\mathbf{x}_{\mathfrak{n}}^{\dagger})\mathbf{u}^{\mathfrak{n}} = -\mathbf{r}^{\mathfrak{n}\perp}, \tag{10.12}$$

where $\mathbf{x}^{\mathfrak{n}}$ is the normalized Ritz vector and $\mathbf{r}^{\mathfrak{n}\perp}$ is the residual orthogonalized to $\mathbf{x}^{\mathfrak{n}}$.

In our implementation, we use a P-Q space decomposition where P is a selected subspace of important determinants and Q is its complement. For the Q space we use the simple diagonal preconditioner, Eq. (10.11), while for the P space we solve the Jacobi-Davidson equation exactly, including the coupling to Q space. The correction equation becomes:

$$\left[(\mathbf{I}_{\{P\}} - \mathbf{x}^{\mathfrak{n}}_{\{P\}}\mathbf{x}^{\dagger}_{\mathfrak{n}\{P\}})(\mathbf{H}_{\{P\}} - E^{\mathfrak{n}})(\mathbf{I}_{\{P\}} - \mathbf{x}^{\mathfrak{n}}_{\{P\}}\mathbf{x}^{\dagger}_{\mathfrak{n}\{P\}}) + \alpha\mathbf{x}^{\mathfrak{n}}_{\{P\}}\mathbf{x}^{\dagger}_{\mathfrak{n}\{P\}}\right]\mathbf{u}^{\mathfrak{n}}_{\{P\}} = -\mathbf{r}^{\mathfrak{n}\perp}_{\{P\}} + \mathbf{s}_{\{P\}}, \tag{10.13}$$

with coupling terms:

$$\alpha = \mathbf{x}^{\dagger}_{\mathfrak{n}\{Q\}}(\mathbf{H}_{\{Q\}} - E^{\mathfrak{n}})\mathbf{x}^{\mathfrak{n}}_{\{Q\}} \tag{10.14}$$

$$\mathbf{s}^{\mathfrak{n}}_{\{P\}} = (\mathbf{I}_{\{P\}} - \mathbf{x}^{\mathfrak{n}}_{\{P\}}\mathbf{x}^{\dagger}_{\mathfrak{n}\{P\}})\left(\mathbf{H}_{\{P\}} - E^{\mathfrak{n}}\right)\mathbf{x}^{\mathfrak{n}}_{\{P\}}\left(\mathbf{x}^{\dagger}_{\mathfrak{n}\{Q\}}\mathbf{u}^{\mathfrak{n}}_{\{Q\}}\right) + \beta\mathbf{x}^{\mathfrak{n}}_{\{P\}} \tag{10.15}$$

$$\beta = \mathbf{x}^{\dagger}_{\mathfrak{n}\{Q\}}(\mathbf{H}_{\{Q\}} - E^{\mathfrak{n}})(\mathbf{I}_{\{Q\}} - \mathbf{x}^{\mathfrak{n}}_{\{Q\}}\mathbf{x}^{\dagger}_{\mathfrak{n}\{Q\}})\mathbf{u}^{\mathfrak{n}}_{\{Q\}}. \tag{10.16}$$

Since $\mathbf{H}_Q$ is diagonal, α and β are computed efficiently using only diagonal elements.

The equation is solved using a direct linear solver (LU or QR decomposition) since the P space is small.

10.3.4 Multi-State Davidson

For computing multiple states simultaneously, the algorithm is extended to maintain orthogonality between states. The subspace is expanded for all states:

$$\mathbf{H}_{\text{sub}}\mathbf{C} = \mathbf{C}\mathbf{E}, \tag{10.17}$$

where $\mathbf{E} = \text{diag}(E^1, E^2, \ldots, E^{\mathfrak{n}})$ and columns of $\mathbf{C}$ contain eigenvector coefficients.

Dual convergence criteria: Both energy convergence ($|\Delta E^{\mathfrak{n}}| < \epsilon_E$) and residual convergence ($\|\mathbf{r}^{\mathfrak{n}}\| < \epsilon_R$) must be satisfied for all states to ensure correct excited state energies.

10.3.5 P-Space Selection

The P-space is a subset of important determinants used to generate high-quality initial guess vectors and preconditioners. Determinants are selected by:

Excitation level: Include all determinants up to n-fold excitations from the Hartree-Fock reference.

Energy criterion: Select determinants with diagonal elements satisfying:

$$H_I^I - E_{\text{HF}} < \Delta E_{\text{threshold}} \tag{10.18}$$

Hybrid selection: Combine excitation level and energy criteria.

HCI-based selection: Use Heat-bath CI (see Sec. 10.4) to select the most important determinants.

10.4 Heat-bath Configuration Interaction (HCI)/CIPSI

10.4.1 Algorithm Overview

Heat-bath CI (HCI)[13] is a selected CI method that efficiently selects important determinants using selection inspired by heat-bath sampling. An extension towards CIPSI[14] is also implemented.

The algorithm consists of three phases plus an extension towards CIPSI:

Phase I (Setup): Pre-compute sorted lists of double excitation matrix elements:

$$\text{For each } (p, q): \quad \{(r, s, H_{rs\leftarrow pq})\} \text{ sorted by } |H_{rs\leftarrow pq}| \text{ (decreasing)} \tag{10.19}$$

Phase II: Variational calculation in the current selected space:

$$\mathbf{H}_{\text{sel}}\mathbf{c} = E_{\text{var}}\mathbf{c} \tag{10.20}$$

Phase III (Heat-Bath Selection): Generate candidate determinants connected to the variational space by $|H_J^I c_I| > \epsilon_h$:

For each determinant $|\Phi_I\rangle$ in the variational space:

- Generate single excitations with $|f_i^a c_I| > \epsilon_h$
- Use pre-sorted lists to generate only double excitations with $|v_{ij}^{ab} c_I| > \epsilon_h$
- Adaptive threshold: $\epsilon_{\text{adaptive}} = \epsilon_h / |c_I|$

Phase IV (Perturbative selection à la CIPSI): Select determinants corresponding to highest perturbative amplitudes:

$$^{pert}c_J^2 \approx \frac{\left|\sum_{I\in\{|H_J^I c_I|>\epsilon_h\}} H_J^I c_I\right|^2}{(E_{\text{var}} - H_J^J)^2} > \epsilon_p^2 \tag{10.21}$$

The sum runs only over determinants that have a large contribution to the J determinant according to the heat-bath criterion, and therefore can be calculated during the Phase III, essentially at no extra cost.

Note that in Phase III, the candidates are generated only for the new determinants, however, the $H_J^I c_I$ values are accumulated for all determinants in the variational space.

10.4.2 Efficient Excitation Generation

The key to HCI performance is the setup phase, which pre-computes and sorts excitation lists. For restricted Hartree-Fock (RHF), store:

$$\text{double\_excitations}[(p,q)] = \{(r,s,H_{rs\leftarrow pq})\}_{\text{sorted}}, \tag{10.22}$$

where $H_{rs\leftarrow pq} = v_{pq}^{rs} - v_{pq}^{sr}$ is the antisymmetrized matrix element, and $H_{rs\leftarrow pq} = v_{pq}^{rs}$ for p,q of opposite spin.

For unrestricted Hartree-Fock (UHF), store separately:

- α-α excitations: $v_{pq}^{rs} - v_{pq}^{sr}$
- β-β excitations: $v_{pq}^{rs} - v_{pq}^{sr}$
- α-β mixed excitations: v_{pq}^{rs}

10.4.3 Multi-State HCI/CIPSI

For multi-state calculations, use **state-maximum selection**:

$$^{pert,max}c_J^2 = \max_{\mathfrak{n}=1,\ldots,\mathfrak{N}_{\text{states}}} \left[\frac{\left|\sum_{I\in\text{var}\epsilon_h} H_J^I c_I^{\mathfrak{n}}\right|^2}{(E_{\text{var}}^{\mathfrak{n}} - H_J^J)^2} \right] \tag{10.23}$$

This ensures determinants important for *any* state are included, preventing bias toward the ground state.

The initial guess is generated by diagonalizing a small-space Hamiltonian:

$$N_{\text{small}} = \max\left(100, \sqrt{N_{\text{target}}}, 5N_{\text{states}}\right) \tag{10.24}$$

with determinants selected using hybrid method (energy + excitation level), and diagonalized to get initial eigenvectors for all states. This prevents missing excited states due to poor initialization.

10.4.4 Hamiltonian Matrix Construction

The Hamiltonian matrix in the selected space is constructed using the Slater-Condon rules (see Sec. 10.2) and stored in a sparse format (for each row, store only non-zero elements and their column indices).

The connections between determinants are identified using three helper lists (similar to Ref. [15]):

- **Same alpha list**: For each alpha string, store all indices of determinants sharing that alpha string. This list is used to find beta single and double excitations efficiently.
- **Same beta list**: For each beta string, store all indices of determinants sharing that beta string. This list is used to find alpha single and double excitations efficiently.
- **Single alpha excitation list**: For each alpha string, store all alpha strings reachable by a single excitation. This list is used to find mixed alpha-beta double excitations efficiently.

Using these lists, a list of connected determinants i for each determinant j is generated (for $i < j$) and used to compute and store the Hamiltonian matrix elements.

10.4.5 Second-Order Energy Correction

The PT2 correction recovers dynamical correlation from determinants outside the variational space[13]:

$$\Delta E_{\mathrm{PT2}} = \sum_{K \notin \mathrm{var}} \frac{\left(\sum_{I\in \mathrm{var}} H_K^I c_I\right)^2}{E_{\mathrm{var}} - H_K^K}, \tag{10.25}$$

where K runs over all determinants connected to the variational space but not included in it.

To compute PT2 efficiently, adaptive thresholding is employed. For each variational determinant $|\Phi^I\rangle$,

$$\epsilon_{\mathrm{adaptive}} = \frac{\epsilon_{pt2}}{|c_I|}, \tag{10.26}$$

and only excitations are generated with $|H_K^I| > \epsilon_{\mathrm{adaptive}}$ using pre-sorted lists from setup.

In order to reduce memory usage, an additional layer of thresholding is applied during PT2 calculation. This involves discarding excitations with small contributions before they are fully generated. For this purpose, the instantaneous PT2 amplitudes are calculated as $H_K^I c_I/(E_{\mathrm{var}} - H_K^K)$, and excitations with amplitudes below a second threshold ϵ_{pt2}^c are not added to the list of candidates for PT2 correction. If the K determinant is already present in the list, its amplitude is updated accordingly, otherwise an energy contribution of this instantaneous excitation is added to the PT2 energy directly and also to the uncertainty estimate of the PT2 energy.

In practice, the calculation corresponds to the Phase III + Phase IV of the HCI selection with a different (tight) threshold ϵ_{pt2}.

10.5 Reduced Density Matrices

10.5.1 One-Particle Density Matrix

The one-particle reduced spin-resolved density matrix (1-RDM) is:

$$\gamma_q^p = \langle \Psi | \hat{a}_p^\dagger \hat{a}_q | \Psi \rangle = c^{I*} c_J \langle \Phi_I | \hat{a}_p^\dagger \hat{a}_q | \Phi^J \rangle \tag{10.27}$$

For separate spin components (p, q and $\bar{p}, \bar{q}$ correspond to α and β orbitals, respectively):

$$\begin{aligned} \gamma_q^p &= c^{I*} c_J \langle \alpha_I \beta_I | \hat{a}_p^\dagger \hat{a}_q | \alpha^J \beta^J \rangle \\ \gamma_{\bar{q}}^{\bar{p}} &= c^{I*} c_J \langle \alpha_I \beta_I | \hat{a}_{\bar{p}}^\dagger \hat{a}_{\bar{q}} | \alpha^J \beta^J \rangle \end{aligned} \tag{10.28}$$

Properties:

- Trace: $\mathrm{tr}(\gamma^{(\alpha)}) = N_\alpha$, $\mathrm{tr}(\gamma^{(\beta)}) = N_\beta$
- Hermitian: $\gamma_{pq} = \gamma_{qp}^*$
- Eigenvalues: Natural orbital occupations $0 \le n_i \le 2$

10.5.2 Transition Density Matrix

For different states $|\Psi_{\mathtt{m}}\rangle$ and $|\Psi_{\mathtt{n}}\rangle$:

$$ {}^{\mathtt{n}}_{\mathtt{m}}\gamma^{p}_{q} = \langle \Psi_{\mathtt{m}} | \hat{a}^{\dagger}_{p} \hat{a}_{q} | \Psi^{\mathtt{n}} \rangle \tag{10.29} $$

Used for computing transition properties (e.g., oscillator strengths, transition dipole moments).

10.5.3 Two-Particle Density Matrix

The two-particle reduced (spin-resolved) density matrix (2-RDM) is:

$$ \Gamma^{pq}_{rs} = \langle \Psi | \hat{a}^{\dagger}_{p} \hat{a}^{\dagger}_{q} \hat{a}_{s} \hat{a}_{r} | \Psi \rangle \tag{10.30} $$

Energy verification:

$$ E = h^{q}_{p} \gamma^{p}_{q} + \frac{1}{2} v^{rs}_{pq} \Gamma^{pq}_{rs} + E_{\text{nuc}} \tag{10.31} $$

10.5.4 Natural Orbitals

Diagonalize the 1-RDM:

$$ \gamma \phi_i = n_i \phi_i \tag{10.32} $$

Natural orbital occupations n_i indicate importance of orbitals. Used for:

- Active space selection
- Identifying multireference character

Chapter 11

Orbital localization and region fragments

11.1 Localized occupied orbitals

The region workflow starts from localized occupied orbitals. In the default implementation, the valence occupied space is localized with the intrinsic bond orbital (IBO) functional

$$\mathcal{L}_{\mathrm{IBO}} = \sum_i \sum_A \left(q_A^i\right)^p , \tag{11.1}$$

with IAO charges

$$q_A^i = \sum_{\mu \in A} \left|\langle \mathrm{IAO}_\mu | \phi_i \rangle\right|^2 , \tag{11.2}$$

where A labels atoms, i labels occupied orbitals, and $p = 4$ by default. The working IAOs are obtained from the occupied orbitals C_{occ} and the minimal-basis overlap blocks

$$P_{12} = S^{-1} S_{12}, \qquad B = S_{21} C_{\mathrm{occ}}, \qquad D = S_{22}^{-1} B. \tag{11.3}$$

Following the revised formulation from supplementary materials of Ref. [16], ElemCo.jl first builds the proto-IAOs

$$A_{\mathrm{proto}} = P_{12} + \left(C_{\mathrm{occ}} - P_{12} D \left(B^\dagger D\right)^{-1}\right) B^\dagger, \tag{11.4}$$

and then performs a metric Löwdin orthogonalization,

$$C_{\mathrm{IAO}} = A_{\mathrm{proto}} \left(A_{\mathrm{proto}}^\dagger S A_{\mathrm{proto}}\right)^{-1/2} . \tag{11.5}$$

Only non-ghost minimal-basis functions are retained in this construction. The localized occupied orbitals are related to the input occupied orbitals through a rotation matrix

$$|\tilde{\phi}_i\rangle = \sum_j |\phi_j\rangle R_{ji}. \tag{11.6}$$

Let $\mathcal{C}_{\mathrm{arg}}$ denote the explicit center list passed to `@region`, and let $\mathcal{C}_{\mathrm{incl}}^{\mathrm{opt}}$ and $\mathcal{C}_{\mathrm{excl}}^{\mathrm{opt}}$ denote `region.inclusive_centers` and `region.exclusive_centers`, respectively. The effective center sets are

$$\mathcal{C}_{\text{incl}} = \begin{cases} \mathcal{C}_{\text{incl}}^{\text{opt}} \cup \mathcal{C}_{\text{arg}}, & \texttt{region.mode = :inclusive}, \\ \mathcal{C}_{\text{incl}}^{\text{opt}}, & \texttt{region.mode = :exclusive}, \end{cases} \tag{11.7}$$

$$\mathcal{C}_{\text{excl}} = \begin{cases} \mathcal{C}_{\text{excl}}^{\text{opt}}, & \texttt{region.mode = :inclusive}, \\ \mathcal{C}_{\text{excl}}^{\text{opt}} \cup \mathcal{C}_{\text{arg}}, & \texttt{region.mode = :exclusive}. \end{cases} \tag{11.8}$$

An occupied orbital is retained when it satisfies either the inclusive or the exclusive rule:

$$i \in \mathcal{F}_{\text{occ}}^{\text{incl}} \iff \exists A \in \mathcal{C}_{\text{incl}} : q_A^i \geq \tau_{\text{occ}}, \tag{11.9}$$

$$i \in \mathcal{F}_{\text{occ}}^{\text{excl}} \iff \left(\exists A \in \mathcal{C}_{\text{excl}} : q_A^i \geq \tau_{\text{occ}}\right) \wedge \left(\forall B,\ q_B^i \geq \tau_{\text{occ}} \Rightarrow B \in \mathcal{C}_{\text{excl}}\right), \tag{11.10}$$

where τ_{occ} = `region.occ_charge_thr`. The selected occupied orbitals are written as fragment-inactive orbitals in the tagged dump.

11.2 Fragment virtual orbitals from support-atom OPAOs

In the non-pi workflow, the fragment virtual space is built from support-atom projected atomic orbitals. If $\mathcal{C}_{\text{all}} = \mathcal{C}_{\text{incl}} \cup \mathcal{C}_{\text{excl}}$, the support set is defined as

$$\mathcal{S} = \mathcal{C}_{\text{all}} \cup \left\{ A \,\middle|\, \exists i \in \mathcal{F}_{\text{occ}},\ q_A^i \geq \tau_{\text{atom}} \right\}, \tag{11.11}$$

where τ_{atom} = `region.atom_charge_thr`.

Let C_v denote the canonical virtual orbitals and let $S_{\text{AO},\mathcal{S}}$ denote the AO overlap columns corresponding to AO functions on support atoms. The fragment PAOs are formed as

$$C_{\text{PAO}} = C_v C_v^\dagger S_{\text{AO},\mathcal{S}}. \tag{11.12}$$

Their overlap matrix is

$$S_{\text{PAO}} = C_{\text{PAO}}^\dagger S C_{\text{PAO}}. \tag{11.13}$$

A locality-preserving factor M is built such that

$$M^\dagger S_{\text{PAO}} M = I. \tag{11.14}$$

Near-redundant directions of S_{PAO} — those with eigenvalue below `opaofac` · `redthr` · $\lambda_{\max}$ (`loc.opaofac` times the AO basis redundancy threshold `scf.redthr`) — are dropped, so M has fewer columns than C_{PAO} when the support PAOs are linearly dependent in the virtual space. The retained PAOs are selected by a rank-revealing column-pivoted QR of the kept eigenvectors of S_{PAO} and orthogonalized by a symmetric Löwdin transformation (with one refinement step), keeping the OPAOs local. This yields orthogonal PAOs

$$C_{\text{OPAO}} = C_{\text{PAO}} M. \tag{11.15}$$

The corresponding virtual rotation is obtained from

$$R_{\text{virt}} = C_v^\dagger S C_{\text{OPAO}}. \tag{11.16}$$

11.3 Local pi axes and target orbitals

For `region.pi = :occupied` or `:both`, one local target p'_z orbital is constructed for each selected atom. The bonding graph is built from the distance criterion

$$r_{AB} < 1.3\,(R_A + R_B)\,, \tag{11.17}$$

where R_A and R_B are element-specific covalent radii.

For each selected atom A, a local plane neighborhood $\mathcal{P}_A$ is assembled from bonded neighbors first, with nearest-atom fallbacks if the bonded graph is insufficient to define a plane. The code then constructs the symmetric matrix

$$G_A = \sum_{B\in\mathcal{P}_A} (\mathbf{R}_B - \mathbf{R}_A)\,(\mathbf{R}_B - \mathbf{R}_A)^\dagger\,, \tag{11.18}$$

and takes the normalized eigenvector corresponding to the smallest eigenvalue as the local plane normal $\mathbf{n}_A$.

If $|p_x^A\rangle$, $|p_y^A\rangle$, and $|p_z^A\rangle$ denote the highest-valence p-type IAOs on atom A, the target local pi orbital is

$$|p'_{z,A}\rangle = n_{A,x}|p_x^A\rangle + n_{A,y}|p_y^A\rangle + n_{A,z}|p_z^A\rangle. \tag{11.19}$$

11.4 Pi-space projector

Let P_Z be the matrix whose columns are the target orbitals $|p'_{z,A}\rangle$ and let C_{blk} denote either the occupied or the virtual block under consideration. The projected overlaps are

$$X = P_Z^\dagger S C_{\text{blk}}, \qquad S_\pi = P_Z^\dagger S P_Z. \tag{11.20}$$

ElemCo.jl diagonalizes the projector-overlap matrix

$$O = X^\dagger S_\pi^{-1} X. \tag{11.21}$$

If

$$OU = U\Lambda, \tag{11.22}$$

the rotated block is formed with eigenvectors ordered by descending eigenvalues, so the leading orbitals are those with the largest pi-projector overlap.

The automatic total number of local π electrons is

$$n_{\pi,e} = \begin{cases} \texttt{region.pi\_electrons}, & \text{if this option is set non-negative,} \\ \sum\limits_{A\in\mathcal{C}_{\text{all}}} n_{\pi,e}(A), & \text{otherwise,} \end{cases} \tag{11.23}$$

where the fallback atomic contributions $n_{\pi,e}(A)$ use an element- and coordination-based main-group p-block rule.

For restricted references, the default occupied and virtual π counts are

$$n_{\pi,\text{occ}} = \frac{n_{\pi,e}}{2}, \qquad n_{\pi,\text{virt}} = N_{\text{centers}} - n_{\pi,\text{occ}}. \tag{11.24}$$

These defaults can be overridden by `region.pi_occupied` and `region.pi_virtual`, in which case only the leading occupied and virtual projector-ordered orbitals are retained. For unrestricted references, the alpha and beta occupied counts are split using the spin population difference $m_s^{(2)} = N_\alpha - N_\beta$,

$$n_{\pi,\text{occ}}^\alpha = \frac{n_{\pi,e} + m_s^{(2)}}{2}, \qquad n_{\pi,\text{occ}}^\beta = \frac{n_{\pi,e} - m_s^{(2)}}{2}. \tag{11.25}$$

11.5 Pseudo-canonicalization

If `region.pseudo = true`, the selected fragment occupied and fragment virtual blocks are semicanonicalized by diagonalizing the corresponding Fock subblocks. For a fragment block C_f this amounts to diagonalizing

$$F_f = C_f^\dagger F_{\mathrm{AO}} C_f, \tag{11.26}$$

and rotating the block with the eigenvectors of F_f.

Chapter 12

Automatically generated UCCSDT and UDC-CCSDT

Unrestricted implementations of CCSDT and DC-CCSDT[17, 18, 19] were generated with version 1.0.1 of the Quantwo program[20]. The Quantwo inputs are listed below.

- UCCSDT Quantwo input file:

```
prog,spinintegr=0,nobrafac=1,explspin=1,algo=2
output,level=1,maxlenline=70

\beq
<\Phi^{a}_{i}| \op H (1 + \op T_2 + \op T_3) |0>_C
\eeq
\beq
<\Phi^{ab}_{ij}| \op H (1 + \op T_2 + \half \op T_2 \op T_2 + \op T_3) |0>_C
\eeq
\beq
<\Phi^{abc}_{ijk}| \op H (\op T_2 + \op T_3 + \half \op T_2 \op T_2 + \op T_2 \op T_3) |0>_C
\eeq
```

- UDC-CCSDT Quantwo input file:

```
%singles and doubles amplitude equations from UCCSDT
%we only modify the triples amplitude equation

prog,spinintegr=0,nobrafac=1,explspin=1,algo=2
output,level=1,maxlenline=70

\beq
<\Phi^{abc}_{ijk}| \op H (\op T_2 + \op T_3 + \frac{1}{2} \op T_2 \op T_2
+ \op T_2 \op T_3) |0>_C
+ (1 - \Perm{IJ}{JI} - \Perm{IK}{KI})(1 - \Perm{AB}{BA} - \Perm{AC}{CA})
(\sum_{LMDE} \tnsr \intg{LE}{MD} \tnsr T^{IL}_{AD} \tnsr T^{MJK}_{EBC})
- \frac{1}{2}(1 - \Perm{KI}{IK} - \Perm{KJ}{JK})
\sum_{LMDE} \tnsr \intg{LD}{ME} \tnsr T^{IJ}_{DE} \tnsr T^{LMK}_{ABC}
- \frac{1}{2}(1 - \Perm{CA}{AC} - \Perm{CB}{BC})
\sum_{LMDE} \tnsr \intg{LD}{ME} \tnsr T^{LM}_{AB} \tnsr T^{IJK}_{DEC}
+ \frac{1}{2}(1 - \Perm{IJ}{JI} - \Perm{IK}{KI})
\sum_{LMDE} \tnsr \intg{LD}{ME} \tnsr T^{LI}_{DE} \tnsr T^{MJK}_{ABC}
+ \frac{1}{2}(1 - \Perm{AB}{BA} - \Perm{AC}{CA})
\sum_{LMDE} \tnsr \intg{LD}{ME} \tnsr T^{LM}_{DA} \tnsr T^{IJK}_{EBC}
+ \frac{1}{2}(1 - \Perm{KI}{IK} - \Perm{KJ}{JK})(1 - \Perm{AB}{BA} - \Perm{AC}{CA})
\sum_{LMDE} \tnsr \intg{LD}{ME} \tnsr T^{IJ}_{AD} \tnsr T^{LMK}_{BEC}
+ \frac{1}{2}(1 - \Perm{IJ}{JI} - \Perm{IK}{KI})(1 - \Perm{CA}{AC} - \Perm{CB}{BC})
\sum_{LMDE} \tnsr \intg{LD}{ME} \tnsr T^{IL}_{AB} \tnsr T^{JMK}_{DEC}
\eeq
```

The program generates TensorOperations code. The generated code used by ElemCo.jl is located in the `src/algo` directory.

Bibliography

[1] Knowles, P. J. & Handy, N. C. A determinant based full configuration interaction program. *Comput. Phys. Commun.* **54**, 75 (1989).

[2] Sun, Q. Libcint: An efficient general integral library for gaussian basis functions. *J. Comput. Chem.* **36**, 1664 (2015).

[3] Yarkony, D. R. Comment on the use of the augmented matrix in mcscf theory. *Chem. Phys. Lett.* **77**, 634–635 (1981).

[4] Meier, U. & Staemmler, V. An efficient first-order casscf method based on the renormalized fock-operator technique. *Theor. Chem. Acc.* **76**, 95–111 (1989).

[5] Kreplin, D. A., Knowles, P. J. & Werner, H.-J. MCSCF optimization revisited. II. Combined first- and second-order orbital optimization for large molecules. *J. Chem. Phys.* **152**, 074102 (2020).

[6] Kats, D. & Manby, F. R. Sparse tensor framework for implementation of general local correlation methods. *J. Chem. Phys.* **138**, 144101 (2013).

[7] Hampel, C., Peterson, K. A. & Werner, H.-J. A comparison of the efficiency and accuracy of the quadratic configuration interaction (QCISD), coupled cluster (CCSD), and Brueckner coupled cluster (BCCD) methods. *Chem. Phys. Lett.* **190**, 1 (1992).

[8] Knowles, P. J., Hampel, C. & Werner, H.-J. Coupled cluster theory for high spin, open shell reference wave functions. *J. Chem. Phys.* **99**, 5219 (1993).

[9] Knowles, P. J., Hampel, C. & Werner, H.-J. Erratum: "Coupled cluster theory for high spin, open shell reference wave functions" [J. Chem. Phys. 99, 5219 (1993)]. *J. Chem. Phys.* **112**, 3106 (2000).

[10] Szalay, P. G. & Bartlett, R. J. Analytic energy gradients for the two-determinant coupled cluster method with application to singlet excited states of butadiene and ozone. *J. Chem. Phys.* **101**, 4936 (1994).

[11] Rickert, C., Usvyat, D. & Kats, D. Tensor decomposed distinguishable cluster. I. Triples decomposition. *J. Chem. Phys.* **163** (2025).

[12] Knowles, P. J. & Handy, N. C. A new determinant-based full configuration interaction method. *Chem. Phys. Lett.* **111**, 315–321 (1984).

[13] Holmes, A. A., Tubman, N. M. & Umrigar, C. J. Heat-Bath Configuration Interaction: An Efficient Selected Configuration Interaction Algorithm Inspired by Heat-Bath Sampling. *J. Chem. Theory Comput.* **12**, 3674–3680 (2016).

[14] Huron, B., Malrieu, J. P. & Rancurel, P. Iterative perturbation calculations of ground and excited state energies from multiconfigurational zeroth-order wavefunctions. *J. Chem. Phys.* **58**, 5745–5759 (1973).

[15] Sharma, S., Holmes, A. A., Jeanmairet, G., Alavi, A. & Umrigar, C. J. Semistochastic Heat-Bath Configuration Interaction Method: Selected Configuration Interaction with Semistochastic Perturbation Theory. *J. Chem. Theory Comput.* **13**, 1595–1604 (2017).

[16] Senjean, B., Sen, S., Repisky, M., Knizia, G. & Visscher, L. Generalization of Intrinsic Orbitals to Kramers-Paired Quaternion Spinors, Molecular Fragments, and Valence Virtual Spinors. *J. Chem. Theory Comput.* **17**, 1337–1354 (2021).

[17] Kats, D. & Köhn, A. On the distinguishable cluster approximation for triple excitations. *J. Chem. Phys.* **150**, 151101 (2019).

[18] Rishi, V. & Valeev, E. F. Can the distinguishable cluster approximation be improved systematically by including connected triples? *J. Chem. Phys.* **151**, 064102 (2019).

[19] Schraivogel, T. & Kats, D. Accuracy of the distinguishable cluster approximation for triple excitations for open-shell molecules and excited states. *J. Chem. Phys.* **155**, 064101 (2021).

[20] Kats, D. & Schraivogel, T. Quantwo: second-quantization program (2024). `github.com/fkfest/quantwo` [Accessed: 2024-02-22].